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Climate change impact across the Southland region

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Purpose

The global climate system is changing and will continue to change for decades to come. These changes will have implications not only for New Zealand's climate and weather systems but also for freshwater availability for downstream users and for hazard exposure (inland and coastal). Due to the nature of climate change, trends will vary across the country, throughout the century, and among different climate change scenarios. This report addresses potential impacts of climate change on climate, hydrology, marine and coastal processes across the Southland region using downscaled Global Climate Model (GCM) outputs from 1971–2099 under different global warming scenarios. The combination of six GCMs and three emissions scenarios allows consideration of a plausible range of future trajectories of greenhouse gas emissions and climatic responses.

Methods

The climate projections used in this report are derived from the 6th Coupled Model Intercomparison Project (CMIP6), which incorporates several significant scientific and modelling advances. The projections are based on six GCMs which are different from those previously used for New Zealand studies that were derived from the 5th Coupled Model Intercomparison Project (CMIP5). Climate change projections are generated for three emissions scenarios: SSP2-4.5, SSP3-7.0 and SSP5-8.5. Dynamical downscaling was used to produce the CMIP6 NZ climate projections using CSIRO's Conformal Cubic Atmospheric Model (CCAM) to at hourly climate simulations, for all climate variables of interest, available at 12 km spatial resolution across New Zealand.

The hydrological modelling builds on the Deep South National Science Challenge project 'Climate change impact on national hydrology' that coupled New Zealand's CMIP5 climate projections with the TopNet hydrological model, Earth Sciences New Zealand's uncalibrated, nationally parameterised surface water hydrological model. The modelling methodology follows recommendations provided as part of the Ministry for the Environment project 'Pilot CMIP5-CMIP6 hydrological projections intercomparison study'. The TopNet model has been implemented as part of the New Zealand Water Modelling (NZWaM) framework and is routinely used for surface water hydrological modelling applications in New Zealand.

Future projections of New Zealand's Earth System Model were used to quantify sea surface temperatures, marine heatwaves and ocean acidification. This fully-coupled Earth System Model has a finer spatial resolution around New Zealand compared to other ocean models in the CMIP5/6 archives, enabling better representation of coastal circulation and coastal processes.

Two tide-gauge records were analysed: the Port of Bluff gauge operated by South Port and archived by LINZ, and the Stead Street Bridge gauge operated by Environment Southland. Both records span more than 30 years, providing a robust basis for assessing mean sea-level trends, tidal characteristics, and extreme storm-tide levels. All datasets were quality-assured to identify gaps, check datum consistency, and remove any artefacts or offset changes prior to analysis.

Future sea-level rise projections are based on five Shared Socioeconomic Pathway (SSP) scenarios aligned with Ministry for the Environment guidance and the NZSeaRise programme. These projections indicate a wide range of possible outcomes, with higher-emissions scenarios leading to substantially greater sea-level rise by 2150. The timing of when key sea-level thresholds are reached varies across scenarios, reinforcing the need for adaptive, staged planning approaches.

Intercomparison of the CMIP5-CMIP6 atmospheric changes

ESNZ has previously conducted an intercomparison across New Zealand comparing projected changes across a range of climate variables for two scenarios producing similar levels of CO_2 by the end of the century. The scenarios considered were RCP4.5 (CMIP5) versus SSP2-4.5 (CMIP6) and RCP8.5 (CMIP5) versus SSP3-7.0 (CMIP6)[1]. The key messages relevant to the Southland region are as follows:

- **Air temperature:** CMIP6 projects generally larger increases than those under CMIP5. However, the number of hot days (>25°C) predicted is lower under CMIP6.
- **Precipitation:** CMIP6 projects generally smaller overall changes than CMIP5 but with large seasonal differences in autumn and winter.
- **Dry days:** The number of days for which precipitation is less than 1 mm is projected to increase under CMIP6 compared to CMIP5. Heavy rainfall extremes (i.e., 99th percentile calculated on the wet days) are projected to be larger under SSP2-4.5 (CMIP6) than RCP 4.5 (CMIP5), and smaller under SSP3-7.0 (CMIP6) than RCP8.5 (CMIP5) by the end of the century.
- **Wind speed:** Average and strong wind speeds are projected to change less under CMIP6 than CMIP5. Work on wind speed projections is ongoing, and it is currently advised to use caution when using wind projections.

Results

It is not possible to attribute the modelled differences between two time periods (in this report, mid-century and end of century) solely to climate change, as natural climate variability may add to, or subtract from, the climate change effect. The potential impacts of climate change are presented as averages across the six model projections, which reduces the underlying natural variability to some extent. With these caveats in mind, the potential effects of changing climate over this century across the Southland region and Freshwater Management Units (FMUs) are summarised below across the different domains for each metric considered.

Climate domain

The projected mean temperature changes across the Southland region increase with time and emissions scenario. Future annual average warming spans a wide range: 0.9–1.4°C by 2050, and 1.8–3.6°C by 2090, largely dependent on scenario. Seasonally, most warming occurs in summer by mid-century and during autumn-winter and summer by the end of century. Changes in maximum and minimum temperature follow similar patterns, with the largest change in the

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[1] The mismatch in RCP-SSP scenario for high emission scenario (unlike low-emission scenarios RCP4.5 and SSP2-4.5) is explained by the comparison of global annual mean CO2 concentration by 2100 between the different scenarios. G lobal annual mean CO2 concentration is projected to reach 840 ppm by 2100 under RCP8.5, 870 ppm under SSP3-7.0 and 1130 ppm under SSP5-8.5

maximum temperature projected during summer (up to 3.9 °C for SSP5-8.5 by the end of century), while the largest change in the minimum temperature is projected during winter (up to 3.6°C for SSP5-8.5 by the end of century). Diurnal range (i.e., difference between daily minimum and maximum temperature) is expected to increase with time and emission scenario, with the largest increase projected for summer and autumn.

Changes in extreme temperatures mirror the average annual signal. The average number of hot days is expected to rise from 3–5 days by 2050 to 6-19 days by 2090. Heatwave days (consecutive days >25°C) are projected to increase — particularly at higher elevations — across all scenarios by up to 8 days by 2050 and up to 36 days by 2090. Conversely, frost days are expected to decline by up to 20 days by mid-century, and by 25–45 days by the end of the century. Growing degree days will increase with time and emissions scenario, with the largest seasonal rise during summer.

Projected changes in rainfall show a marked seasonality and variability across the Southland region. Annual rainfall is expected to increase slightly by mid-century (5%), rising to 8–13% by the end of the century, with larger increases in western and eastern areas. One exception is the Garvie Mountains region (headwaters of the Waikaia River) where annual rainfall is projected to decrease by up to 10% across all scenarios. Seasonally the largest increases are projected during winter and spring (up to 40%), while summer and autumn precipitation is more variable with increase/decrease (between -20% and + 20%) depending on elevation, location, time and emission scenarios.

By mid-century, the number of wet days may increase slightly (up to 1 day) but decrease by up to 2.5 days by the end of the century. Spatially, increases in wet days occur mainly in the headwaters of the three Waiau hydro-electric generation lakes and coastal Southland. Decreases are concentrated in the eastern Waiau and north of the Hokonui syncline.

The number days where the total precipitation exceeds 10 mm is projected to increase across Southland at all time slices and emission scenarios, except for some areas in the lower Fiordland (Cameron/Princess Mountain ranges) and upper Mataura (Waikaia catchment) where small decreases (up to eight days) are expected for both mid-century and end of century.

The number of very wet days (i.e., days where the total precipitation exceeds 25 mm) is projected to increase across Southland at all time slices and emission scenarios, except for a small area in the eastern Waiau catchment (northern part of the Livingstone/Thomson Mountain range) and lower Fiordland (Cameron/Princess Mountain ranges) where a small decrease is expected by mid-century.

The number of heavy wet days (i.e., days where the total precipitation exceeds 50 mm) is also projected to increase across Southland under all scenarios, with the largest increase in northern Fiordland (north of the Cameron-Princess-Kaherekoau Mountain ranges). This increase is smaller than that for very wet days.

Projected changes in maximum annual 1-day rainfall (Rx1d) rise with time and emissions scenario, reflecting changes in very wet and heavy wet days. Rx1d is expected to increase by an average of 11 mm by mid-century and 23 mm by the end of the century, with local increases of up to 80 mm in the Waiau headwaters.

By mid-century, decreases in annual maximum 5-day rainfall (Rx5d) are projected for the Thomson and Ailsa Mountain ranges (up to 25 mm) while other areas, particularly Fiordland, may see increases of up to 50 mm. At the end of the century, Rx5d is projected to increase across the region, with Fiordland experiencing rises of up to 75–100 mm.

By mid-century the number of dry days may decrease slightly (up to 1 day), mainly in low-elevation areas, except in the three Waiau hydro-electric generation lakes. The number of dry days is expected to increase by up to 2 days by century's end, particularly north of the Hokonui syncline, while Waiau headwaters may see decreases.

Relative humidity is projected to slightly decrease with time and emission scenarios by up to 1%.

Solar radiation is projected to rise slightly (up to 1 $W.m^{-2}$) except under SSP 2-4.5 where a decrease is expected. Seasonally, solar radiation is expected to increase in summer and autumn (by up to 7 $W.m^{-2}$ in summer by end of century) and decrease in winter and spring (by up to 4 $W.m^{-2}$ in winter by end of century). Spatially, the largest decrease occurs in Fiordland's Darran Range, while the largest increase is across Mataura and Oreti Freshwater Management Units (FMUs).

Cloud cover is expected to decrease slightly (by up to 2%), mainly from October to April, with some minor increases expected for July/August. Fiordland headwaters may see increases mid-century, while Mataura–Oreti and eastern Waiau experience the largest decreases by the end of the century.

We note that work is ongoing on the wind speed projections, therefore caution is required in using the following results:

Annual average wind speed may rise slightly (up to 4%) across Southland under all scenarios. Seasonally, wind speed is expected to increase in winter and spring (up to 12% in winter under SSP5-8.5) and decrease for summer and autumn (up to 5 % under SSP3-7.0) by the end of the century. Spatially, annual average wind speed may decrease in Fiordland (north of the Franklin Range) and north-east Mataura (Old Man Range), while the largest increases occur north of the Hokonui syncline for the Mataura and Oreti, for most of the Aparima and most of the Waiau River downstream of Manapouri and south of Takitimu mountain ranges.

The number of calm days (i.e., mean wind speed <1.5 $m.s^{-1}$) is expected to remain relatively unchanged, with minor seasonal shifts: slight decreases in June–September across scenarios and slight increases by the end of the century for February–March.

Windy days (i.e., mean wind speed >10 $m\ s^{-1}$) are expected to increase (up to 10 days annually) especially during June–October (up to 3 days per month by the end of the century). Changes in windy days are spatially variable; northern Fiordland may experience reductions (up to 20 days/yr), while most of the Mataura-Oreti-Aparima and Waituna may experience increases (up to 20 days/yr). Strong wind changes (99th percentile) are likely to follow similar patterns.

Southland's prevailing wind direction - typically spanning the westerly quarter (from northwest through to southwest) - is projected to remain unchanged through time and warming.

Changes in meteorological drought (assessed using Potential Evaporation Deficit or PED) indicate Southland will experience increased drought: limited under SSP2-4.5 (up to 7 mm) and greater under higher emission scenarios (up to 23 mm). Spatially, the headwaters of the Waikaia River may see the largest increase (up to 150 mm under SSP5-8.5), with smaller changes elsewhere. Conversely, the probability of annual PED accumulation >200 mm reflects similar trends: the Waikaia and upper Mataura may exceed 200 mm by at least 50%, while coastal Oreti and Waituna catchments may see decreases (up to 20%).

Changes in rainfall intensity correlate with temperature changes: mid-century rainfall intensity increases range from 4.8% (2-year ARI, 1-hour event) to 18% (100-year ARI, 1-hour event); by century's end the range is 16% to 46% for the same events.

Probable Maximum Precipitation (PMP) is projected to change linearly with mean annual air temperature at a rate of 8.34% per °C, regardless of scenario or storm duration.

Cryosphere domain

Changes in snow volume reflect changes in precipitation and temperature. Decreases in snow volume are expected to reach at least 50% across Southland mountain ranges by mid-century for SSP3-7.0 and SSP5-8.5, with smaller reductions in the lower-elevation range of the Waiau and the Garvie mountain range under SSP2-4.5. By the end of the century and regardless of the level of warming, snow volume is projected to decrease by at least 50% across Southland.

Changes in the number of snow days are projected to remain largely unchanged by the 2050s across emission scenarios, although the northern headwater mountain ranges are projected to experience a reduction of up to 30 snow days. Similar spatial patterns are projected for the end of the century with greater reductions in snow days under higher-emission scenarios. The Garvie Mountains (Mataura) are projected to experience snow day reductions of at least 90 days.

Snow line is expected to rise across Southland mountain ranges over time and under all emission scenarios. By mid-century, the snow line is expected to increase by up to 300 m across Southland regardless of scenario. By the end of the century, snow line is expected to increase by approximately 400–670 m depending on the level of warming. While snow may persist on the highest mountain ranges, it is projected that lower mountain ranges will have fewer than 90 days of snow cover by the end of the century.

Hydrology domain

The bias-corrected hydrological modelling chain is judged appropriate for use in this project. The hydrological bias correction was undertaken using a simple method that aims to match the long-term average riverine and cryospheric metrics across Southland rivers. This method allowed the bias-corrected model to adequately reproduce the mean flow metrics for all natural flow catchments. The bias corrected models also matched the mid- to lower-end values of the high flow metrics (95th percentile of non-exceedance) and the mid- to high-end values of the low flow metrics (25th percentile of the annual flow time series) for the natural flow catchments. However, the bias corrected models were less able to reproduce the more extreme flows, i.e., the lowest values for the low flow metrics, or the highest values for the high flow metrics. This indicates that further work is required to ensure the adequacy of the bias correction methods for these more extreme flow conditions.

Mean annual flood is projected to increase across nearly all Southland River reaches (at least 95%), with the greatest increases by the end of the century under SSP5-8.5 (up to 42%). While the magnitude of mean annual flood is projected to decrease for a small pocket within each

FMU, flood magnitude is projected to decrease for the headwaters of the Waikaia catchment (Mataura FMU).

Changes in the magnitude of the Q5 (i.e., river flow magnitude exceeded five percent of the time) and of the Q25 (i.e., river flow magnitude exceeded twenty-five percent of the time) reflect changes in mean annual flood albeit with a reduced range of magnitude (up to 20% for changes in Q5 and up to 9% for changes in Q25), suggesting a possible shift in the distribution of high flows under warming conditions.

Changes in the average annual number of Fre3 events are projected to vary across Southland rivers depending on location, time and emission scenario. By the end of the century under SSP5-8.5, the number of annual Fre3 events is projected to increase by between one and two events across all FMUs, except the Waiau FMU for which the increase is limited to one additional event. Spatially, decreases are projected for up to 11% of river reaches across Southland with decreases of up to five events in some areas of the Aparima and Mataura FMUs.

Changes in the magnitude of mean discharge reflect changes in average annual precipitation. The magnitude of the mean flow is projected to increase across nearly all Southland river reaches (at least 90%), with the greatest increases by the end of the century under SSP5-8.5 greenhouse gas concentration scenarios (average increase of 15% across Southland rivers). However, the magnitude of mean flow is projected to decrease for several reaches within each FMUs (up to 16% in the Waiau FMU and 23% in the Mataura FMU).

Changes in annual flows were considered in terms of changes in mean annual flows (more sensitive to extremes) and median annual flows. Across most Southland rivers, mean annual flow is projected to increase, with the greatest increases by the end of the century under SSP5-8.5 (average increase of 15% across Southland rivers). A notable difference occurs in the lower Mataura River, whereby the end of the century under very high greenhouse gas concentration scenarios, median annual flows are projected to increase by 10%, while mean annual flows are projected to increase only by up to five percent. The magnitude of mean annual flow is projected to decrease in some areas within each FMU (e.g., up to 17% of the reaches in the Mataura and Waiau FMUs).

The magnitude of seasonal average flows is projected to increase across nearly all Southland rivers in every season except summer, with the greatest increases by the end of the century under SSP5-8.5 (average increase of at least 30% across Southland rivers). In contrast, the direction of change in summer flows is projected to vary with time and scenario, while the magnitude of change is limited to a range of -5 to 5%. Regardless of time and scenario, summer average flows are predicted to decrease in the Waiau and Fiordland FMUs, and across the Mataura FMU by the end of century under SSP3-7.0 and mid-century under SSP5-8.5. Summer flows across the Aparima and Oreti FMUs are projected to increase regardless of time and scenarios. Overall, changes in seasonal average flows suggest a possible larger shift in Southland rivers' hydrological regimes under warming conditions, with spatial differences identified across FMUs.

Changes in the magnitude of the Q75 (i.e., river flow magnitude exceeded seventy-five percent of the time) and Q95 (i.e., river flow magnitude exceeded ninety-five percent of the time), shows a different spatial pattern to the mean flow. Both metrics are projected to decrease by the end of the century under high (SSP3-7.0) and very high (SSP5-8.5) greenhouse gas scenarios, with average decrease up to 6% across Southland rivers. However, while Q75 is projected to increase across the Fiordland FMU, and to a lesser extent the Aparima and Waituna FMUs, Q95 is projected to increase only for the Fiordland FMU. Overall, changes in low flow metrics associated to changes in seasonal average flows suggest a possible larger shift in Southland rivers' hydrological regimes under warming conditions, with spatial differences identified across FMUs.

Changes in the magnitude of mean annual 7-day low flow vary across Southland rivers in both direction and magnitude. At FMU scale, the largest decrease in mean annual 7-day low flow is predicted across the Waiau FMU by mid-century under SSP3-7.0 (-11.1%), while the largest increase is predicted across the Aparima FMU by the end of the century under SSP2-4.5 (+ 50%). Spatially, mean annual 7-day low flow is expected to decrease by at least 23% for most reaches located in the Oreti and Mataura main stems by mid-century century under SSP3-7.0. Mean annual 7-day low flow is projected to decrease in parts of every FMU (up to more than 60% across all FMUs, except for the Waituna FMU). These changes point to significant shifts in low flow hydrological regimes across Southland rivers under emission scenarios.

Changes in the magnitude of flow reliability vary across Southland rivers in both direction and magnitude. At FMU scale, the largest decrease in flow reliability is predicted across the Aparima, Waituna and Mataura FMUs by end of century under SSP2-4.5 (up to at least -0.05), while the largest increase is predicted across the Waiau FMU by the end of the century under all emission scenarios (+ 0.02). Spatially, flow reliability is projected to decrease across at least 69% of the reaches in every FMU, with the largest decrease projected in the Aparima-Oreti and Mataura main stems (up to -0.08) by the end of century under SSP5-8.5. Changes in flow reliability point to river water being less consistently available to meet environmental or water-use needs under future warming, reflecting a shift towards greater variability and uncertainty in water regimes.

Changes in the magnitude of water supply reliability vary across Southland rivers in both direction and magnitude. At FMU scale, the largest decrease in water supply reliability is predicted across the Aparima, Waituna and Mataura FMUs by end of century under SSP2-4.5 (up to at least -0.05), while the largest increase is predicted across the Waiau FMU by the end of the century under SSP2-4.5 and SSP5-8.5 (+ 0.01). Spatially, water supply reliability is projected to decrease for most FMU river main stems (up to -0.1) by end of century under SSP5-8.5. Changes in water supply reliability extend changes in flow reliability and indicate river water will be less consistently available to meet water demand during the irrigation season, likely increasing the frequency of water-use shortages under future warming.

Changes in the magnitude of land surface recharge vary across Southland shallow aquifers in both direction and magnitude. At annual and FMU scale, land surface recharge is projected to decrease across the Oreti FMU with warming (up to -6%), with the largest increase in land surface recharge predicted across all other FMUs under all emission scenarios (up to 12.6% for the Aparima by end of century under SSP5-8.5). At seasonal scale, land surface recharge is projected to increase during autumn and winter across all FMUs through time and emission scenarios, except for small pockets within the Oreti and Waiau FMUs. In summer, however, land surface recharge is projected to decrease with warming across most shallow aquifers in the Oreti (except the Five Rivers aquifer), Waituna and lower Mataura FMUs (by up to 50% for some aquifers under SSP5-8.5). In spring, decreases in land surface recharge are limited to parts of the lower Oreti and Waiau aquifers, specifically under SSP5-8.5. Overall, decreases in spring and summer land surface recharge points to intensifying risks of hydrological and agricultural drought, water scarcity (e.g., access to groundwater resource for drinking water supply in the lower Mataura), and ecosystem stress across the Oreti-Waituna and lower Mataura FMUs under future warming.

Magnitude of the maximum daily and hourly annual peak discharge is generally projected to increase with time, warming and return period across Southland rivers, noting that for some locations the largest rises occur under SSP3-7.0 rather than the SSP5-8.5 scenario. These results mirror the changes in precipitation extremes.

Magnitude in peak daily discharge associated with 12hr and 24 hr duration PMP (referred as 12hr PMF and 24 hr PMF) is projected to increase with warming. Under a 1°C warming scenario, most FMUs, except the Mataura FMU, are projected to experience an increase between 10 and 20% in the magnitude of the PMFs (independently of the event duration). Spatial differences in changes to the 12hr and 24 hr PMF magnitude are more pronounced for warming greater than 2°C, with the largest changes predicted for the Mataura FMU because of the timing of the PMP event (winter PMP). While changes in PMF across different storm durations generally follow the same pattern as rainfall depth changes predicted by the HIRDS tool, changes in PMF are predicted to not increase with storm duration at some locations. This may be due to the methodology used to predict climate-change adjustments to PMP rainfall depths across duration. Further investigations are needed to understand these changes and better characterise the impacts of rarer rainfall events.

Stream temperature is projected to increase uniformly across Southland rivers with time and warming, at both annual and seasonal time scale. Spatially, the largest increases are projected under SSP5-8.5 at the outlets of the main river stems (up 4°C), while the lowest increases are projected in the upper reaches of FMUs (up to 1°C). These changes directly reflect changes in air temperature at both annual and seasonal time scale and signal greater stress on freshwater ecosystems.

Intercomparison of the CMIP5-CMIP6 changes in hydrological metric projection

Intercomparison of the magnitude of the projected change in three hydrological metrics (mean annual flood, annual and seasonal average discharge and water supply reliability) was carried out between the CMIP5 derived assessment and this assessment. As per the atmospheric analysis, the intercomparison was completed for two scenarios producing similar levels of CO_2 by the end of the century. The scenarios considered were RCP4.5 (CMIP5) versus SSP2-4.5 (CMIP6) and RCP8.5 (CMIP5) versus SSP3-7.0 (CMIP6). While the CMIP5 assessment was developed using a non-bias-corrected modelling chain, the CMIP6 assessment uses a bias-corrected modelling chain. The bias correction methodology does not affect the intercomparison of changes in river flow metrics. The key messages relevant to the Southland region are:

- Change in the magnitude of mean annual flood under CMIP6 are generally smaller than those projected under CMIP5. However, the spatial distribution of these changes differs, especially under the lower-emission scenario, mean annual flood projected to decrease up to 50% under RCP4.5 (CMIP5) for the upper Oreti FMU by the end of the century, while increases of up to 20% are projected under SSP2-4.5 (CMIP6).

- Changes in annual average discharge remain similar under the lower-emission scenario for the Mataura FMU. However, while little change is projected for the remaining Southland main rivers under RCP 4.5 (CMIP5), increases of up to 20% are projected under SSP2-4.5 (CMIP6). Under higher emission scenarios, similar magnitudes of change are projected under RCP8.5 (CMIP5) and SSP5-8.5 (CMIP6), except for the Mataura FMU, where little change or a decrease is projected under CMIP6.
- While differences in the magnitude of change are noted for seasonal average flow, with smaller changes projected under CMIP6, differences in spatial distribution and direction of change are identified depending on the emission scenario, as follows:
  - Winter flows are projected to decrease for the Waikawa catchment under CMIP6, while winter flows are projected to increase under CMIP5. Winter flows are projected to increase in lower Mataura and Oreti FMU under SSP2-4.5 (CMIP6), with little change projected under RCP4.5 (CMIP5) for those locations.
  - Changes in summer discharge are consistent between the two assessments for the Mataura FMU. However, differences in the direction of the change are projected for the Aparima, Oreti and Waiau FMUs particularly under higher-emission scenarios.
  - Changes in spring discharge are consistent between the two assessments with differences only in the magnitude of change.
  - Changes in autumn discharge are not spatially consistent between the two assessments, particularly for the Aparima, Oreti and Mataura FMUs, with increases in autumn discharge projected under RCP4.5 (CMIP5), while decreases are projected under SSP2-4.5 (CMIP6).
- The magnitude of absolute change in water supply reliability is consistent across both assessments. However, differences in the direction of the change are identified for both emissions scenarios, with water supply reliability projected to decrease across most rivers under RCP8.5 (CMIP5) while increasing under SSP5-8.5 (except for the lower Oreti FMU).

Sea-level domain

Tidal analysis showed differences in the tidal characteristics at Bluff and Stead Street, with Bluff experiencing a larger tidal range. High-tide exceedance curves were used to identify how often high tides might reach critical elevations, and these were then adjusted to reflect projected sea-level rise. Additionally, storm-tide levels—caused by the combined effects of tide, storm surge, and wave setup—were statistically modelled for both locations. Bluff was found to be more exposed to extreme sea levels under storm-tide conditions, which are expected to become more frequent as sea levels rise.

Historical observations from Bluff and Stead Street show a clear and ongoing rise in mean seas level, consistent with trends observed elsewhere in New Zealand. While year to year variability remains high, recent decaded show higher average sea levels than earlier periods, increasing the frequency of coastal flooding during high tides and storm events.

Future projections, based on IPCC AR6 scenarios and data from the NZSeaRise programme, indicate that sea levels will continue to rise under all emission pathways. These projections include estimates of when critical sea-level thresholds (e.g., 0.2 m, 0.6 m, and 1.0 m above 2020 levels) are likely to be reached, both with and without adjustments for local vertical land movement—information that is essential for future risk and infrastructure planning.

This sea-level analysis highlights the increasing exposure of coastal communities and infrastructure to flooding and erosion hazards. It provides important information to inform planning, infrastructure investment, and risk management decisions across Southland. The modelling shows that Southland property exposure to coastal flooding (with a 100-year ARI) will increase anywhere from 20% to 76% over time, under different emission scenarios and if land subsidence over the next 100 years. The results underline the need for proactive adaptation measures that consider not only global sea-level rise but also local land movement and extreme events.

Marine domain

Sea surface temperatures (SSTs) are projected to increase over time and under all emission scenarios due to anthropogenic greenhouse gas emissions. By the end of the century, SSTs are projected to increase by up to 1°C under the low-emission stabilisation scenario (SSP1-2.6) and by more than 3°C under SSP3-7.0.

Linked to SST change, marine heatwaves are expected to increase over time and under all emission scenarios. By the end of the century, marine heatwave intensity is projected to increase by up to 0.5°C under SSP1-2.6 and exceed 2 °C under SSP3-7.0. Conversely, the frequency of marine heatwaves is expected to increase from 15–30 days per year (under current climate conditions) to 120–200 days per year under SSP1-2.6 and up to 365 days per year under SSP3-7.0.

The below table is provided in this Executive Summary to assist readers interested in the key findings for each Freshwater Management Unit (FMU).

FMU	Summary of change
Mataura	<u>Climate domain</u> By the end of century, the average temperatures are expected to increase by 1.8 to 3.6°C depending on the emission scenario. Projected changes in seasonal temperature are elevation-dependent during summer and autumn in higher emission scenario, and more uniform during winter and spring season. Similar changes are projected for the minimum temperature. Hot days are expected to increase by approximately 11 to 30 days, with the largest increase for the upper Mataura between an area delineated by Gore-Waikaka and Lumsden townships, including the Waikaia River valley (up to 60 days).

FMU	Summary of change
	The number of cold nights (taken as the number of cold days) is projected to decrease by between 26 to 42 days (depending on the scenario). Coastal Mataura (south of Gore township) may experience up to 50 fewer cold days, while high elevations (Garvie and Umbrella mountain ranges) may have up to 100 fewer cold days. Heatwave frequency is expected to increase with emission scenario and elevation with increase ranging from 20 to 60 days in coastal Mataura up to 80 days in Mataura's mountain ranges. By the end of century, the annual precipitation is projected to increase with emission scenario between7.5 to 10 percent. Spatial decrease in precipitation (up to 10%) is expected independently of emission scenario in the Garvie Mountains. The area of largest increase in precipitation develops from an area between Gore - Waikaia and Cattle Flats townships under SSP3-7.0, to south of a line Cattle Flats-Waikaia townships under SSP 5-8.5. From a seasonal point of view, winter then spring and summer seasons are projected to experience the largest increases (ranging from 5 to 20%), while autumn precipitation increases moderately (up to 2%) under emission scenarios SSP3-7.0 and SSP5-8.5. Number of wet days are projected to decrease up to 9 days with emission scenario and elevation, except under SSP 2-4.5 under which it is projected to increase up to 3 days across lower elevation terrain. However, number of wet and very wet days and rain days above 10 mm are projected to increase independently of emission scenario and elevation by up to 3 to 6 days. Conversely, the number of dry days is projected to increase by up to 3 days independent of emission scenario and elevation, except for north of Gore where the number of dry days is expected up to 6 days, with larger changes projected for the Old Man range (up to 9 days). Rx1d magnitude is projected to increase up to 20 mm independently of elevation and emission scenario. Conversely, Rx5d is projected to follow similar spatial pattern with increase up to 50 mm. Relative humidity is projected to slightly decrease with emission scenario across the FMU by up to 3%, while solar radiation is expected to increase by up to 5% (except for the Garvie Mountains where solar radiation is expected to decrease by up to 10%). Wind speed is projected to increase with emission scenario and elevation by up to 10%, except for the Garvie Mountains where wind speed is projected to decrease by up to 10% independently of the emission scenario. Meteorological drought frequency is projected to decrease independent of emission scenario by up to 40 days south of a line from Lumsden-Gore township, with decrease up to 20 days for the remainder of the FMU except for the Garvie and Umbrella mountains which are projected to experience an increase by up to 40 days of meteorological drought frequency. Change in rainfall depth follows change in annual average temperature. Rainfall depth is projected to increase from 8% (for 2-year ARI of 120 hrs storm duration) up to 49% (for a 100-year ARI of 1hr storm duration). Similar patterns are projected for the change in PMP depth independently of storm duration. PMP depth is projected to increase with warming from 15% up to 30%.

FMU	Summary of change
	Hydrology domain By the end of the century high flow metrics (mean annual flood, Q5 and Q25) are expected to increase by 4% to 40% by the end of century depending on emission scenario level and high flow metric for most of the river reaches in the Mataura FMU. However, some decreases are projected in isolated pockets and to a larger extent in the headwaters of the Waikaia River. Annual average number of Fre3 events are projected to slightly increase with emission scenario up to an additional 1.2 event per year for most of the river reaches in the Mataura FMU. From a season point of view, winter and summer are projected to experience the moderate increase up to 0.3 event, with smaller increase projected for the remaining seasons. Average flow metrics (mean flow and mean annual flows) are expected to increase by 8% to 14% by the end of century, for most of the river reaches in the Mataura FMU, with the largest increase projected under low emission scenario (SSP2-4.5). Some decreases are projected in isolated pockets and to a larger extent in the headwaters of the Waikaia River, associated with the change in precipitation. Seasonal flow is projected to increase with emission scenario, for most of the river reaches in the Mataura FMU, during winter-spring and autumn ranging from 0.4% to 21.4%, with smallest increase projected under SSP3-7.0. Summer flows are projected to be more variable, with largest increase (up to 2.6%) under SSP2-4.5 and largest decrease (up to 12.2%) under SSP3-7.0. Annual Q25 is projected to increase with emission scenario, for most river reaches in the Mataura FMU, up to 14% under SSP2-4.5 and SSP5-8.5, with smallest increase projected under SSP3-7.0 (up to 7.5%) and SSP5-8.5 (up to 11.4%). Q75 and Q95 are projected to moderately increase up to 4.3% under SSP2-4.5, while decreasing (between 10 and 12% for Q75 and 17% for Q95) with increased emission scenarios. 7-day mean annual low flow magnitude is projected to increase, for most river reaches in the Mataura FMU, under SSP2-4.5 (up to 39.2%) and SSP5-8.5 (up to 9.8%). 7-day mean annual low flow magnitude is projected to moderately decrease (up to 2.2%) under SSP3-7.0. Flow reliability and water supply reliability are projected to decrease independent of emission scenario with the largest decrease (up to 0.04) projected under SSP2-4.5. Annual average groundwater recharge is projected to increase across emission scenario, except for isolated pockets of the Mataura shallow aquifers, with largest increase projected under SSP2-4.5 (up to 8.5%) and lowest increase under SSP3-7.0 (up to 1.9%). Seasonally, winter recharge is projected to increase with emission scenario (up to 23.9%), while summer recharge is more variable. Summer recharge is projected to slightly increase under SSP2-4.5 and SSP5-8.5 (up to 2.7%) and to decrease up to 20.4% under SSP3-7.0. Magnitude of the maximum daily and hourly annual peak discharge is projected to increase with time-emission scenario and return period across the FMU. However, the magnitude of changes varies by location and emission scenario. Change in PMF magnitude associated with 12hr and 24 hr PMP duration event is increasing with warming and is impacted by the timing of the PMP event (summer/winter event) and onset cryospheric contribution. Summer PMF is projected to increase from 10% up to 60%, while change in winter PMF magnitude is amplified (compared to summer PMF) due to change in precipitation phase with warming and added contribution of snow melt to flood generation mechanism.

FMU	Summary of change
	Coastal domain By the end of century SLR increase with emission scenario and position along the coastline, with larger SLR (including LVM) projected with increased emission scenario. A 50 cm SLR increment could be reached between 2050 under SSP5-8.5 and 2081 under SSP2-4.5.
Oreti	Climate domain By the end of century, average temperatures are expected to increase by between 1.8 to 3.5 °C depending on the emission scenario. Changes in seasonal temperature are elevation dependant during summer and autumn season, and more uniform during winter and spring season. Similar changes are projected for the minimum temperature. Hot days are expected to increase by between 11 to 32 days, with the largest increase for the mid Oreti in an area located between Winton-Lumsden and Mossburn townships (up to 60 days). Number of cold nights (taken as the number of cold days) is expected to decrease with emission scenario and elevation by between 26 to 42 days across the Oreti. Coastal Oreti is projected to experience a reduction of the number of cold days by up to 50 days independently of emission scenario, increasing with elevation (Eyre Mountain range) up to 100 days. Heatwave frequency is expected to increase with emission scenario and elevation, with increases ranging from 20 to 60 days in coastal Oreti, and up to 80 days in the Oreti River's mountain ranges. Annual average precipitation is expected to increase with emission scenario, with increases ranging from 10 to 14 percent across the Oreti. The largest increases (up to 20%) are located south of Lumsden township. Precipitation in the Oreti mountain ranges is expected to increase by up 10% independently of emission scenario. From a seasonal point of view, winter then spring and summer seasons are projected to experience the largest increase 11 to 25% with increased emission scenarios, while autumn precipitation increases moderately (up to 5%) under emission scenarios SSP3-7.0 and SSP5-8.5. Number of wet days is projected to decrease up to 6 days with emission scenario and elevation, except under SSP 2-4.5 under which it is projected to increase up to 3 days across lower elevation terrain. However, number of wet and very wet days are projected to increase independently of emission scenario and elevation by up to 5 to 6 days. Conversely, the number of dry days is projected to increase by up to 3 days independent of emission scenario and elevation, except for north of the Hokonui syncline where number of dry days are expected up to 6 days. Rx1d magnitude is projected to increase up to 20 mm independently of elevation and emission scenario. Conversely, Rx5d is projected to follow similar spatial pattern with increases up to 50 mm. By the end of century relative humidity is projected to decrease slightly with emission scenario across the FMU by up to 3%, while solar radiation is expected to increase by up to 5%. Wind speed is projected to increase with emission scenario and elevation by up to 10% north of Dipton township.

FMU	Summary of change
	Meteorological drought frequency is projected to decrease independent of emission scenario by up to 40 days, except for south of the Eyre mountains for which a decrease is projected by up to 20 days. Change in rainfall depth follows changes in annual average temperature. Rainfall depth is projected to increase from 8% (for 2-year ARI of 120 hrs storm duration) up to 48% (for a 100-year ARI of 1hr storm duration). Similar patterns are projected for the change in PMP depth independently of storm duration. PMP depth is projected to increase with warming from 15% up to 29%. <u>Hydrology domain</u> By the end of century, high flow metrics (mean annual flood, Q5 and Q25) are expected to increase by 4.1% to 50% depending on emission scenario level and high flow metric for most of the river reaches in the Oreti FMU. Little change in the annual average number of Fre3 events are projected with emission scenario (up by 1.3 event per year) for most of the river reaches in the Oreti FMU. From a season point of view, winter and summer are projected to experience the moderate increase up to 0.5 event, with similar decrease projected for the remaining seasons. Average flow metrics (mean flow and mean annual flows) are expected to increase with emission scenario by 14% to 21% by the end of century, for most of the river reaches in the Oreti FMU, with the smallest increase projected under high emission scenario (SSP3-7.0). Seasonal flow is projected to increase with emission scenario, for most of the river reaches in the Oreti FMU, ranging from 7.8% to 24%, with the largest increases (larger than 16%) projected during winter and spring and the smallest increase projected in summer under SSP3-7.0 (7.8%). Annual Q25 is projected to increase with emission scenario, for most river reaches in the Oreti FMU, up to 21% under SSP5-8.5, with the smallest increase (up to 14.7%) projected under SSP3-7.0. Changes in Q75 vary with time and emission scenarios. Q75 is projected to slightly increase (1.0%) under SSP2-4.5, and to decrease up to 20% with higher emission scenarios. Q95 is projected to decrease with emission scenario by 3.1% to 20.8% by the end of the century, for most reaches in the Oreti FMU. Changes in 7-day mean annual low flow magnitude are more variable to those in Q75. 7-day mean annual low flow is projected to increase, for most river reaches in the Oreti FMU, under SSP2-4.5 (up to 37.4%) and SSP3-7.0 (up to 6%), but to decrease up to 4% under SSP5-8.5. Flow reliability and water supply reliability are projected to decrease independent of emission scenario with the largest decrease (up to 0.03) projected under SSP2-4.5. Annual average groundwater recharge is generally projected to decrease with emission scenario, except for isolated pockets of the Oreti shallow aquifers, with largest decrease projected under SSP3-7.0 (up to 6.2%) and increase under SSP2-4.5 (up to 1.0%). Seasonally, winter recharge is projected to increase across emission scenario (largest increase up to 23.3% under SSP2-4.5), while summer recharge is projected to decrease with emission scenario (up to 22.6% under SSP3-7.0 and SSP5-8.5).

FMU	Summary of change
	Magnitude of the maximum daily and hourly annual peak discharge is projected to increase with time-emission scenario and return period across the FMU. However, the magnitude of changes varies by location and emission scenario. Change in PMF magnitude associated with 12hr and 24 hr PMP duration event is increasing with warming. PMF is projected to increase from 9% (under +1.0°C warming) up to 34% (under +3.0°C warming) ranging from 9% to 49% depending on GCM and timing of the PMP event. <u>Coastal domain</u> By the end of century SLR increase with emission scenario and position along the coastline, with larger SLR (including LVM) projected with increased emission scenario. A 50 cm SLR increment could be reached between 2056 under SSP5-8.5 and 2104 under SSP2-4.5.
Aparima	<u>Climate domain</u> By the end of century, average temperatures are expected to increase between 1.8 to 3.5°C depending on the emission scenario. Changes in seasonal temperature are elevation dependant during summer and autumn, and more uniform during winter and spring season. Similar changes are projected for the minimum temperature. Hot days are expected to increase between 11 to 32 days, with the largest increase (up to 60 days) for the mid-Aparima for an area south of the Hokonui syncline to Otautau township. Number of cold nights (taken as the number of cold days) is expected to decrease with emission scenario and elevation, with decreases ranging from 28 to 46 days across the Aparima, except for the Takitimu range for which the number of cold days is projected to increase up to 10 days. Heatwave frequency is expected to increase with emission scenario and elevation with increases ranging from 20 to 60 days for most of the Aparima, and up to 80 days in the Takitimu mountain range. By the end of century, annual average precipitation increases with emission scenario from 10 to 14 percent across the Aparima. Spatially, the largest increases are located south of Wreys Bush township. From a seasonal point of view, winter then spring and summer seasons are projected to experience the largest increases (11-24%), while autumn precipitation increases up to 10% but decrease with increased emission scenarios. Number of wet days is projected to decrease by up to 6 days with emission scenario and elevation, except under SSP2-4.5 under which it is projected to increase up to 3 days across lower elevation terrain. However, the number of wet and very wet days is projected to increase independently of emission scenario and elevation by up to 5 to 6 days. Conversely, the number of dry days is projected to increase by up to 3 days independent of emission scenario and elevation, except for the Takitimu and Eyre mountain ranges where number of dry days are expected to increase up to 6 days. Rx1d magnitude is projected to increase by up to 20 mm, independent of elevation and emission scenario. Rx5d is projected to follow similar spatial pattern with increase up to 50 mm. By the end of century relative humidity is projected to decrease slightly with emission scenario across the FMU by up to 3%, while solar radiation is expected to increase by up to 5%. Wind speed is projected to increase with emission scenario and elevation by up to 10% for most of the FMU.

FMU	Summary of change
	Meteorological drought frequency is projected to decrease independent of emission scenario by up to 40 days, except for the Takitimu Mountains for which a decrease up to 20 days is projected. Change in rainfall depth is following change follows change in annual average temperature. Rainfall depth is projected to increase from 8% (for 2-year ARI of 120 hrs storm duration) up to 48% (for a 100-year ARI of 1hr storm duration). Similar patterns are projected for the change in PMP depth independently of storm duration. PMP depth is projected to increase with warming from 15% up to 29%. Hydrology domain By the end of century high flow metrics (mean annual flood, Q5 and Q25) are expected to increase, for most of the river reaches in the Aparima FMU, by 6% to 68% depending on emission scenario and high flow metric. Annual average number of Fre3 events are projected to slightly increase with emission scenario up to an additional 1.6 event per year for most of the river reaches in the Aparima FMU. From a season point of view, winter and summer are projected to experience moderate increase up to 0.5event, while smaller changes are projected for the remaining seasons. Average flow metrics (mean flow and mean annual flows) are expected to increase with emission scenario by 12% to 21% by the end of century, for most of the river reaches in the Aparima FMU, with the lowest increase projected under high emission scenario (SSP3-7.0). Seasonal flow is projected to increase with emission scenario, for most of the river reaches in the Aparima FMU, up to 28%, with the smallest increase projected under SSP3-7.0. Annual Q25 is projected to increase with emission scenario, for most river reaches in the Aparima FMU, up to 17% under SSP2-4.5 and SSP5-8.5, with the smallest increase (up to 10%) projected under SSP3-7.0. Q75 is projected to increase under SSP2-4.5 (up to 14.6%) and SSP5-8.5 (up to 4.7%) but decrease (by 2.7%) under SSP3-7.0. Changes in 7-day mean annual low flow magnitude are more variables than those in Q75. 7-day mean annual low flow is projected to increase, for most river reaches in the Aparima FMU, under SSP2-4.5 and SSP5-8.5 (up to 51.9%), but to decrease up to 7.8% under SSP3-7.0. Q95 is projected to increase under SSP2-4.5 (up to 12.6%) and decrease with increased warming (by 6.1% under SSP3-7.0). Flow reliability and water supply reliability are projected to decrease independent of emission scenario with the largest decrease (up to 0.05) projected under SSP2-4.5. Annual average groundwater recharge is projected to increase across emission scenario, except for isolated pockets of the Aparima shallow aquifers, with largest increase projected under SSP5-8.5 (up to 12.7%) and lowest increase under SSP3-7.0 (up to 4.6%). Seasonally, winter recharge is projected to increase with emission scenario (up to 25%), while summer recharge is more variable. Summer recharge is projected to increase under SSP2-4.5 and SSP5-8.5 (up to 25.5% under SSP5-8.5) and to decrease for lower emission scenario (up to 2.2 % under SSP3-7.0). Magnitude of the maximum daily and hourly annual peak discharge is projected to increase with time-warming and return period across the FMU. However, the magnitude of changes varies by location and emission scenario. Change in PMF magnitude associated with 12hr and 24 hr PMP duration event is increasing with warming. PMF is projected to increase from 16% (under +1.0°C warming) up to 60% (under +3.0°C warming and ranging from 11% to more than 100% depending on location, GCM and timing of the PMP event.

FMU	Summary of change
	<u>Coastal domain</u> By the end of century SLR increase with emission scenario and position along the coastline, with larger SLR (including LVM) projected with increased emission scenario. A 50 cm SLR increment could be reached between 2059 under SSP5-8.5 and 2094 under SSP2-4.5.
Waituna	<u>Climate domain</u> By the end of century, average and seasonal temperatures are expected to increase between 1.7 to 3.4°C depending on the emission scenario. Similar changes are projected for the minimum temperature. Hot days are expected to increase between 10 to 32 days. Cold nights (taken as the number of cold days) is expected to decrease by between 12 to 16 days depending on emission scenario. Heatwave frequency is expected to increase with emission scenario between 20 days up to 60 days under largest emission scenario. Annual average precipitation is projected to increase with emission scenario, with increases from 11 to 18 %. From a seasonal point of view, winter then spring and summer seasons are projected to experience the largest increase (6 to 30%) with increased emission scenarios, while autumn precipitation increases are moderate (up to 10%) but decrease with increased emission scenarios. Number of wet days is projected to decrease up to 3 days with emission scenario and elevation, except under SSP2-4.5 under which it is projected to increase up to 3 days. However, number of wet and very wet days are projected to increase independently of emission scenario by up to 5 to 6 days. Conversely, the number of dry days is projected to increase by up to 3 days independent across the Waituna. Rx1d magnitude is projected to increase up to 20 mm. Conversely, Rx5d is projected to follow similar spatial pattern with increase up to 50 mm. By the end of century relative humidity is projected to slightly decrease with emission scenario across the FMU by up to 3%, while solar radiation is expected to increase by up to 5%. Wind speed is projected to increase with emission scenario and elevation by up to 5%. Meteorological drought frequency is projected to decrease independent of emission scenario by up to 40 days. Change in rainfall depth is following change in annual average temperature with warming. Rainfall depth is projected to increase from 8% (for 2-year ARI of 120 hrs storm duration) up to 46% (for a 100-year ARI of 1hr storm duration). Similar patterns are projected for the change in PMP depth independently of storm duration. PMP depth is projected to increase with warming from 14% up to 28%.

FMU	Summary of change
	<u>Hydrology domain</u> By the end fo century high flow metrics (mean annual flood, Q5 and Q25) are expected to increase by 17% to 60% by the end of century depending on emission scenario level and high flow metric for most of the river reaches in the Waituna FMU. Annual average number of Fre3 events are projected to slightly increase with emission scenario up to an additional 1.4 event per year for most of the river reaches in the Waituna FMU. From a season point of view, summer is projected to experience the moderate increase up to 0.4 event, with smaller increase projected for the remaining seasons, exept for winter (up to 0.2 event independently of emission scenario). Average flow metrics (mean flow and mean annual flows) are expected to between 20% and 25% by the end of century, for most of the river reaches in the Waituna FMU, with the largest increase projected under very high emission scenario (SSP5-8.5). Seasonal flow is projected to increase with emission scenario, for most of the river reaches in the Waituna FMU, up to 18% under SSP2-4.5 and SSP3-7.0, to 30% projected under SSP5-8.5. Q75 is projected to increase under SSP2-4.5 (up to 12.7%) and SSP5-8.5 (up to 4.6%),but decrease (by 8.4%) under SSP3-7.0. Annual Q25 is projected to increase with emission scenario, for most river reaches in the Waituna FMU, up to 25% under SSP2-4.5 and SSP5-8.5, with the smallest increase (up to 17%) projected under SSP3-7.0. 7-day mean annual low flow magnitude is projected to increase with emission scenario, for most river reaches in the Waituna FMU. The largest increase in 7-day mean annual low flow magnitude is projected under SSP2-4.5 and SSP5-8.5 (up to 24.5 %) and the lowest increase projected under SSP3-7.0 (up to 16.7 %). Q95 is projected to increase under SSP2-4.5 (up to 10.1%) and decrease with increased warming (by 13.3% under SSP3-7.0). Flow reliability and water supply reliability are projected to decrease independent of emission scenario with the largest decrease (up to 0.08) projected under SSP2-4.5 reducing to a decrease up to 0.02 under SSp5-8.5. Annual average groundwater recharge is projected to increase across emission scenario, except for isolated pockets of the Waituna shallow aquifers, with largest increase projected under SSP2-4.5 and SSP5-8.5 (up to 17%) and lowest increase under SSP3-7.0 (up to 10.6%). Seasonally, winter recharge is projected to increase with emission scenario (up to 31.7%), while summer recharge is more variable. Summer recharge is projected to increase under SSP2-4.5 and SSP5-8.5 (up to around 12%) and to decrease (up to 17.9%) under SSP3-7.0. <u>Coastal domain</u> By the end fo century SLR increase with emission scenario and position along the coastline, with larger SLR (including LVM) projected with increased emission scenario. A 50 cm SLR increment could be reached between 2055 under SSP 5-8.5 and 2083 under SSP 2-4.5.

FMU	Summary of change
Waiau	<u>Climate domain</u> By the end of century average temperatures are expected to increase between 1.8 to 3.5°C depending on the emission scenario. Changes in seasonal temperature are elevation dependant during summer and autumn season, and more uniform during winter and spring season. Similar changes are projected for the minimum temperature. Hot days are expected to increase relatively uniformly between 5 to 16 days depending on emission scenario level, except for the Thomson Mountain range projected to experience lesser increase. Number of cold nights (taken as the number of cold days) is expected to decrease with emission scenario and elevation by between 35 to 60 days. Coastal Waiau and the Waiau River valley are projected to experience reduction on cold days by up to 50 days independently of emission scenario, while higher elevation (especially headwaters of the three hydro-electric generation lakes) are projected to experience lager change up to 125 days. Heatwave frequency is expected to increase with emission scenario and elevation with increase ranging from 20 to 40 days for most of the Waiau, and up to 80 days in all Waiau mountain ranges. Annual average precipitation is projected to increase with emission scenario, with increases ranging from 7 up to 12% across the Waiau. Spatially, the largest increases are localised in the headwaters of the Upukerora and Whitestone Rivers, headwater of Lake Te Anau and Manapouri and coastal Waiau (up to 20%) under the largest emission scenario. From a seasonal point of view, winter and spring seasons are projected to experience similarly the largest increase ranging from 11 to 26%, while autumn precipitations increase is minimal under increased emission scenario. Number of wet days is projected to decrease up to 9 days with emission scenario and elevation, except under SSP2-4.5 under which it is projected to increase up to 3 days including the western headwaters of the three hydro-electricity generation lakes. However, number of and very wet days are projected to increase by up to 5 to 6 days across the Waiau, except in the headwaters of the three hydro-generation lakes for which increase in number of wet days is projected to be up to 9 to 10 days. Conversely, the number of dry days are projected to increase by up to 3 days independent of emission scenario and elevation, except for the Livingstone – Thompson and Takitimu mountain ranges where number of dry days are expected up to 6 days. Rx1d magnitude is projected to increase with elevation from 20 mm at the coast to up to 80 mm in the headwaters of the three hydro-generation lakes. Conversely, Rx5d is projected to follow similar spatial pattern with increase up to 75 mm in the headwaters of the three hydro-generation lakes. By the end of century relative humidity is projected to slightly decrease with emission scenario across the FMU by up to 3%, and up to 3% south of the Takitimu Mountains. Solar radiation is expected to increase by up to 5% east of the three hydro-generation lakes. Wind speed is projected to increase with emission scenario independently of elevation by up to 5%, with an area delimited by Kepler to Takitimu mountains to Monowai township for which wind speed is projected to decrease by up to 10% under SSP 5-8.5.

FMU	Summary of change
	Meteorological drought frequency is projected to decrease independent of emission scenario by up to 40 days east of the Waiau River and the lower reaches of the Whitestone and Upukerora rivers. Similar patterns are projected for the headwaters of the three hydro-generation lakes. Change in rainfall depth is following change in annual average temperature with warming. Rainfall depth is projected to increase from 8% (for 2-year ARI of 120 hrs storm duration) up to 49% (for a 100-year ARI of 1hr storm duration). Similar patterns are projected for the change in PMP depth independently of storm duration. PMP depth is projected to increase with warming from 15% up to 29%. <u>Hydrology domain</u> By the end fo century high flow metrics (mean annual flood, Q5 and Q25) are expected to increase by 5% to 45% by the end of century depending on emission scenario level and high flow metric for most of the river reaches in the Waiau FMU. Annual average number of Fre3 events are projected to slightly increase with emission scenario up to an additional 0.9 event per year for most of the river reaches in the Waiau FMU. From a season point of view, most of the changes are projected to occu in winter, while little change projected for the remaining seasons. Average flow metrics (mean flow and mean annual flows) are expected to increase with emission scenario by 9% to 15.4% by the end of century, for most of the river reaches in the Waiau FMU, with the smallest increase projected under high emission scenario (SSP3-7.0). Seasonal flow is projected to increase with emission scenario, for most of the river reaches in the Waiau FMU, during winter-spring and autumn. Change winter flows is ranging from 27% to 40% during winter, with the largest increase projected under SSP5-8.5, while lower increases projected for autumn and spring (largest increase up to 12% under SSP2-4.5). Summer flows are projected to little change to large decrease across emission scenarios, with largest decrease (up to 11%) projected under SSP3-7.0. Annual Q25 is projected to increase with emission scenario, for most river reaches in the Waiau FMU, up to 13.8% under SSP5-8.5, with the smallest increase (up to 8.8%) projected under SSP3-7.0. Changes in Q75 and Q95 are variable with emission scenario. Q75 and Q95 are projected to increase under SSP2-4.5 (up to 5% for both Q75 and Q95), and to decrease with higher emission scenarios, with the largest decrease up to 5.5%, projected for Q75 under SSP3-7.0 and for Q95 under SSP5-8.5. Changes in 7-day mean annual low flow magnitude are similar to those in Q75 and Q95. 7-day mean annual low flow is projected to increase, for most river reaches in the Waiau FMU, under SSP2-4.5 (up to 12.7%), but to decrease up to 11% under SSP3-7.0 and SSP5-8.5. Projected changes in flow reliability and water supply reliability are variables. Flow reliability is projected to increase with emission scenario (up to 0.02 under SSP5-8.5), while water supply reliability is projected to decrease (up to 0.01) under SSP2-4.5 and remain unchanged with increased emission scenario. Annual average groundwater recharge is projected to increase across emission scenario, except for isolated pockets of Waiau shallow aquifers, with largest increase projected under SSP5-8.5 (up to 6.1) and lowest increase under SSP3-7.0 (up to 1.7%). Seasonally, winter recharge is projected to increase with emission scenario (up to 32%), while summer recharge is more variable. Summer recharge is projected to increase under SSP5-8.5 (up to 1.5%) and to decrease for lower emission scenario (up to 12.9% under SSP3-7.0).

FMU	Summary of change
	Magnitude of the maximum daily and hourly annual peak discharge is projected to increase with time-emission scenario and return period across the FMU. However, the magnitude of changes varies by location and emission scenario. Change in PMF magnitude associated with 12hr and 24 hr PMP duration event is increasing with warming and is impacted by timing of the PMP event (summer/winter event) and onset cryospheric contribution. Summer PMF is projected to increase from 10% up to 40%, while change in winter PMF magnitude is amplified (compared to summer PMF) due to change in precipitation phase with warming and added contribution of snow melt to flood generation mechanism.
Fiordland	<u>Climate domain</u> By the end of century, average temperatures are expected to increase between 1.7 to 3.3 °C depending on the emission scenario. Changes in seasonal temperature are elevation dependant for summer and autumn season, and more uniform during winter and spring season. Similar changes are projected for the minimum temperature. Hot days are expected to increase relatively uniformly by between 1 to 5 days depending on emission scenario level and elevation, with coastal Fiordland experiencing less change. Number of cold nights (taken as the number of cold days) are expected to decrease with emission scenario and elevation, with increases ranging from 22 to 35 days across Fiordland. Coastal Fiordland is projected to experience reduction of cold days by up to 50 days independently of emission scenario, while higher elevations are projected to experience lager changes up to 100 days. Heatwave frequency is expected to increase with warming independent of elevation with increases ranging from 20 to 60 days. Annual average precipitation is projected to increase with emission scenario and elevation, with increases ranging from 7% up to 13%. From a seasonal point of view, winter then spring seasons are projected to experience the largest increase ranging from 11 to 26 %, while summer and autumn precipitation increases are moderate (up to 5%) and the lowest under SSP3-7.0. Number of wet days are projected to decrease up to 6 days with emission scenario and elevation, except under SSP 2-4.5 under which southern Fiordland is projected to increase up to 3 days. However, number of wet and very wet days are projected to increase by up to 12 to 15 days closer to the Fiordland coast, depending on emission scenario, and north of the Franklin Mountains. Conversely, the number of dry days is projected to increase by up to 3 days independent of emission scenario and elevation. Rx1d magnitude is projected to increase with elevation and distance to coast from 20 mm in coastal Fiordland to up to 80 mm in Fiordland headwater catchments. Rx5d is projected to follow similar trend with increase up to 100 mm north of the Franklin Mountains. By the end of the century, relative humidity is projected to slightly decrease with emission scenario across the FMU by up to 2%. Solar radiation is expected to increase by up to 5% south of the Cameron and Kaherekoau mountain ranges while decreasing by up to 10% north of the Franklin Mountain range.

FMU	Summary of change
	Meteorological drought frequency is projected to decrease independent of emission scenario by up to 40 days west and north of the Cameron and Kaherekoau Mountains with decrease in meteorological drought frequency projected for low elevation coastal Fiordland. Change in rainfall depth is following change in annual average temperature with warming. Rainfall depth is projected to increase from 8% (for 2-year ARI of 120 hrs storm duration) up to 45% (for a 100-year ARI 1hr storm duration). Similar patterns are projected for the change in PMP depth independently of storm duration. PMP depth is projected to increase with warming from 14% up to 27%. <u>Hydrology domain</u> By the end of century high flow metrics (mean annual flood, Q5 and Q25) are expected to increase by 6% to 29% by the end of century depending on emission scenario level and high flow metric for most of the river reaches in the Fiordland FMU. Annual average number of Fre3 events are projected to slightly increase with emission scenario up to an additional 1.4 event per year for most of the river reaches in the Fiordland FMU. From a season point of view, winter is projected to experience a moderate increase up to 0.8 event, while little change is projected during summer (-0.1 event). Average flow metrics (mean flow and mean annual flows) are expected to increase with emission scenario by 8% to 12% by the end of century, for most of the river reaches in the Fiordland FMU. Seasonal flow is projected to increase with emission scenario, for most of the river reaches in the Fiordland FMU, during winter and spring ranging from 13% to 35%. Changes in autumn flow are depending on emission scenario, with increase by up to 8% under SSP2-4.5 and decrease by 2% under SSP3-7.0. Summer flows are projected to decrease across emission scenarios, with largest decrease (by up to 8%) projected under SSP3-7.0. Annual Q25 is projected to increase with emission scenario, for most river reaches in the Fiordland FMU, up to 14.8% under SSP5-8.5. Changes in Q75 and Q95 are projected to increase with emission scenario. Changes in Q75 and Q95 are projected up to 12%, under SSP2-4.5, with the smallest increase, up to 6%, projected under SSP3-7.0. Changes in 7-day mean annual low flow magnitude are more variables than those in Q75 and Q95. 7-day mean annual low flow is projected to increase, for most river reaches in the Fiordland FMU, under SSP2-4.5 and SSP3-7.0 (up to 14.1%), but to decrease up to 2.2 % under SSP5-8.5. Projected changes in flow reliability and water supply reliability are variables. Those are projected to decrease under SSP2-4.5 and SSP3-7.0 (up to 0.02) and increase (up to 0.01) projected under SSP5-8.5. <u>Coastal domain</u> By the end fo century SLR increase with emission scenario and position along the coastline, with larger SLR (including LVM) projected with increased emission scenario. A 50 cm SLR increment could be reached between 2058 under SSP5-8.5 and 2092 under SSP2-4.5

1 Introduction

The climate is changing and it is accepted internationally that further changes will result from increasing amounts of greenhouse gases in the atmosphere. In addition, climate will also vary from year to year and decade to decade due to natural processes such as El Niño. Climate change effects over the next decades are predictable with some level of certainty, and will vary from place to place throughout New Zealand (Ministry for the Environment, MfE 2018).

Environment Southland, in collaboration with Invercargill City Council, Southland District Council and Gore District Council (the Councils), commissioned Earth Sciences New Zealand (ESNZ) to produce an update of the 2018 regional assessment of the impacts of climate change for Southland (Zammit et al. 2018) that was completed using downscaled climate projections derived from the 5th Coupled Model Intercomparison Project (CMIP5), (IPCC 2014). The updated assessment combines recently released national climate model projections with regionally specific information and modelling to produce regionally specific projections across climate, hydrological, coastal and ocean domains. The potential impacts of changing climate over this century are generated and summarised across Southland region, for riverine and coastal areas, and for specific requested locations (depending on the domain considered). Analyses across the domains are provided through summary maps describing the differences between the historical period 1995–2014 and two future periods: mid-century (2031–2050) and late-century (2081–2099).

Climate domain

The national climate projections are based on a six Global Circulation Model (GCM) ensemble included within the 6th Coupled Model Intercomparison Project (CMIP6) (IPCC 2023), which incorporates several significant science and modelling advances. Those GCMs were validated, downscaled for New Zealand, and evaluated over the period 1980–2015 (with the period 1995–2014 being referred as the historical period) compared to global observational reanalysis data, and based on a large number of climate variables and indicators (Gibson et al. 2024a). The updated assessment aims to provide regional information about numerous variables for multiple spatial and temporal criteria within the Southland region to assess impacts of climate change which can be used to support strategic planning and adaptation by councils and the community. The work will also summarise the expected impacts of climate change on specific issues, sectors, or environments.

The potential impacts of a changing climate over this century are generated and summarised across Southland region, for riverine and coastal areas and for specific requested locations (depending on the domain considered). Climate projections associated with three emission scenarios were assessed: SSP2-4.5, SSP3-7.0 and SSP5-8.5. Dynamical downscaling was used to produce the CMIP6 NZ climate projections, which was undertaken using CSIRO's Conformal Cubic Atmospheric Model (CCAM) (Gibson et al. 2024a) to produce hourly climate simulations, for all climate variables of interest, available at 12 km spatial resolution across New Zealand.

For operational reasons, analysis of associated high intensity rainfall information continues to use CMIP5 data. Analysis results, performed predominantly at daily timesteps, are presented as absolute (usually for the historical period) and relative change in the parameters and variables of interest and are available in GIS format. Supporting datasets are also provided. Analysis will be performed predominantly at daily timesteps with the analysis results presented as monthly, seasonal, annual, or probability summaries. Analysis will be performed for the Southland region and its component Freshwater Management Units, catchments, or town/port (as specified in the RFP).

Changes across the climate domain are provided as netcdf gridded information to Environment Southland for current warming level (historical period as absolute value) and change across SSP2-4.5, SSP3-7.0 and SSP5-8.5 scenarios by mid-century (2031–2050) and late-century (2081–2099).

Hydrological domain

The hydrological projections are generated by coupling the CMIP6 climate projections with the TopNet hydrological model (Clark et al. 2008). The TopNet model has been implemented as part of the New Zealand Modelling (NZWaM) framework and is routinely used for surface water hydrological modelling applications in New Zealand. The coupling methodology (referred as the modelling chain) is adapted from the Deep South National Science Challenge project 'Climate change impact on national hydrology' (Collins 2020). It follows recommendations provided as part of the Ministry for the Environment project 'Pilot CMIP5-CMIP6 hydrological projections intercomparison study' (Zammit et al. 2024).

Changes across the hydrological domain are provided as csv dataset as well as ArcGIS-Pro layouts to Environment Southland for current warming level (historical period as absolute value) and change across SSP2-4.5, SSP3-7.0 and SSP5-8.5 scenarios by mid-century (2031–2050) and late-century (2081–2099).

Sea-level

Understanding how sea level is changing is critical for managing the risks to Southland's coastal communities, infrastructure, and ecosystems. This component of the climate change impact assessment focuses on evaluating past and future changes in sea level and how these interact with tides and storm events to influence flood and erosion risk.

The analysis combines measured tide gauge records with the latest national projections from the NZSeaRise programme. These projections consider global climate scenarios alongside local factors such as vertical land movement, which is occurring in parts of Southland and contributes to increased relative sea-level rise. Two long-term sea-level monitoring sites—Bluff and Stead Street—provided the basis for trend analysis and extreme sea-level modelling.

The study assessed not only gradual sea-level rise but also how future higher sea levels will amplify the impact of high tides and storm-tide events. By modelling storm-tide extremes and considering projected sea-level increments, the report helps identify when key risk thresholds might be exceeded and what this means for coastal exposure across the region.

This work supports strategic planning by providing a robust, locally relevant understanding of sea-level rise and its potential impacts under a changing climate.

Ocean domain

ESNZ assessed future projections of New Zealand's Earth System model to quantify sea surface temperatures, marine heatwaves and ocean acidification (Behrens et al. 2022). This fully coupled Earth System model has a finer spatial resolution around New Zealand compared to other ocean models used as part of the CMIP5/6 archives and therefore captures coastal circulation and coastal processes better. ESNZ assessed the relevant local variations to apply to the three New Zealand-wide sea-level rise (SLR) scenarios to 2120 that ESNZ (formerly NIWA) developed for the MfE Coastal Hazards and Climate Change Guidance (Ministry for the Environment 2017).

The report associated with the "Climate change impact across the Southland region-IPCC 6th Assessment Report update" report is composed of the following four volumes:

- Volume 1 (this report) introduces the climate change projections (Section 2), the methodology used to estimate climate change impact across the different domains (Section 3), a brief description of Southland's present climate (Section4) and presents a summary of the potential impact of climate change across different industry sectors in Southland (Section 5).
- Volume 2 presents the results of the expected changes across the atmospheric domain.
- Volume 3 presents the modelling chain bias correction, the expected changes across the Cryosphere and Hydrology domains.
- Volume 4 presents the results of the expected changes across the Coastal and Ocean domains.

2 Introduction to climate change and natural variability

2.1 Global climate modelling

Assessing possible changes for our future climate due to anthropogenic activity is difficult because climate projections depend strongly on estimates for future greenhouse gas concentrations. Those concentrations depend on global greenhouse gas emissions that are driven by factors such as economic activity, population changes, technological advances and policies for sustainable resource use. In addition, for a specific future trajectory of global greenhouse gas emissions, different climate model simulations produced somewhat different results for future climate change.

Coupled global atmosphere-ocean general circulation models are used to generate climate change projections for prescribed scenarios of future greenhouse gas concentrations. The CMIP6 project (Eyring et al. 2016) comprises model results from approximately 100 climate models across 49 different modelling groups (Hausfather 2019, WCRP 2021). Some of the notable improvements to CMIP6 models, compared with CMIP5 models, include higher atmosphere and ocean resolution, and the inclusion of new and more complex processes such as interactive chemistry, active land processes, ice sheets and permafrost (Simpkins 2017).

2.2 Downscaling of global climate model data

In regions characterised by complex and coastal terrain, such as New Zealand, the raw output from GCMs can contain large biases. This generally means that GCMs should not be used directly in downstream climate impact studies, especially for local and regional applications. Well-known issues include that GCMs struggle to capture orographic precipitation and extremes over New Zealand, the intensity of extreme events such as tropical cyclones, as well as the impact of elevation and coastal processes on temperature variability (Gibson et al. 2024a).

Due to these issues, downscaling[2] is a valuable tool for better capturing smaller scale processes that impact climate while enhancing the spatial resolution of projections. The Conformal Cubic Atmospheric Model (CCAM) is used as the primary model for dynamical downscaling of selected CMIP6 GCMs over New Zealand. CCAM has been extensively used for downscaling over Australia in CMIP3 (Perkins et al. 2014), CMIP5 (Evans et al. 2021), and CMIP6 (Chapman et al. 2023, Grose et al. 2023) climate projections.

When compared to NIWA's previous CMIP5 downscaling (MfE 2018), the current CMIP6 downscaling has been driven by updated CMIP6 GCMs and scenarios, different regional models run at higher resolution, and a modified bias correction methodology has been applied. Gibson et al. (2023, 2024a) extensively documents the CCAM model, grid configuration, physics settings, and experiment design utilised for the CMIP6 downscaling effort. Considering the improvements outlined above, CMIP6 data are more suitable for use than corresponding CMIP5 data.

[2] Deriving local climate information from larger-scale model or observation dataset. See glossary for further details.

2.3 Climate change scenarios

Assessing possible changes for our future climate due to human activity is challenging because climate projections strongly depend on estimates for future greenhouse gas concentrations. In turn, those concentrations depend on global greenhouse gas emissions that are driven by factors such as economic activity, population changes, technological advances and policies for mitigation and sustainable resource use. This range of uncertainty has been dealt with by the Intergovernmental Panel on Climate Change (IPCC) through consideration of 'scenarios' that describe concentrations of greenhouse gases in the atmosphere. The range of scenarios are associated with possible economic, political, and social developments during the twenty-first century.

In the IPCC Sixth Assessment Report, which is based on CMIP6 models, the scenarios are called Shared Socio-economic Pathways (SSPs). This is a new range of pathways compared to the IPCC Fifth Assessment Report, which used Representative Concentration Pathways (RCPs). In contrast to RCPs — which focus only on changes in atmospheric composition —SSP emission scenarios originate from a wide array of socioeconomic drivers (Bodeker et al. 2022). These include economic, political, and technological developments, as well as population growth.

The updated projections for New Zealand use SSP scenarios. These SSP scenarios start in 2015, and are abbreviated as SSP1-1.9, SSP1-2.6, SSP2-4.5, SSP3-7.0 and SSP5-8.5, in order of increasing greenhouse gas emissions. These SSPs represent narratives characterised as 'sustainability' (SSP1), 'middle of the road' (SSP2), 'regional rivalry' (SSP3) and 'fossil-fuel intensive development' (SSP5). SSPs also specify the end-of-21st-century radiative forcing reached e.g., the '4.5' in SSP2-4.5 assumes a radiative forcing of 4.5 W m2 in 2100. The following definition of each SSP is adapted from IPCC (2021):

- SSP1-1.9 and SSP1-2.6 are scenarios with very low and low greenhouse gas emissions, with CO2 emissions declining to net zero around or after 2050, followed by varying levels of net negative CO_2 emissions.
- SSP2-4.5 is an intermediate greenhouse gas emissions scenario, with CO_2 emissions remaining around current levels until the middle of the century.
- SSP3-7.0 and SSP5-8.5 are high and very high greenhouse gas emissions scenarios, with CO_2 emissions that roughly double from current levels by 2100 and 2050, respectively.

Bodeker et al. (2022) note that while some of the RCP and SSP scenarios reach the same radiative forcing by 2100 (e.g., RCP4.5 and SSP2-4.5), the greenhouse gas emissions and concentration pathways taken to reach that radiative forcing are not necessarily the same (Figure 2-1, Figure 2-2). However, the premise of both RCPs and SSPs is the same; they present scenarios that cover the range of possible future development of anthropogenic drivers of climate change (IPCC 2021). Note, neither the IPCC nor scenario developers have placed likelihoods on any scenario.

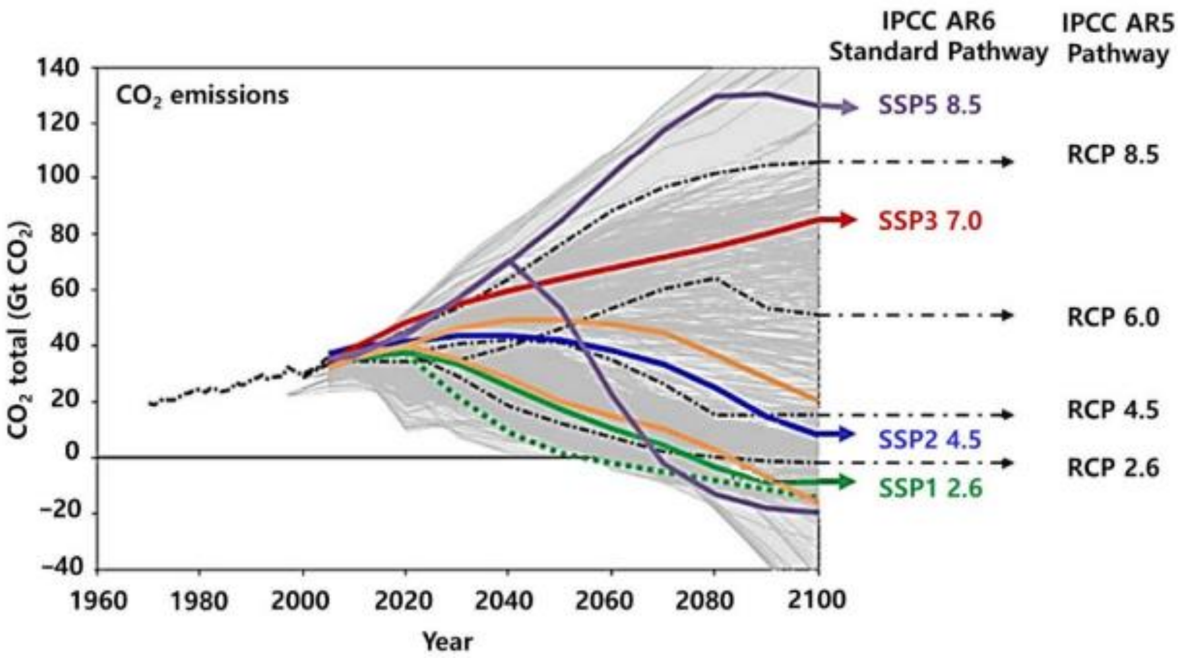

Figure 2-1: Global CO_2 emissions for the 21st century scenarios. Sourced from An et al. (2022).

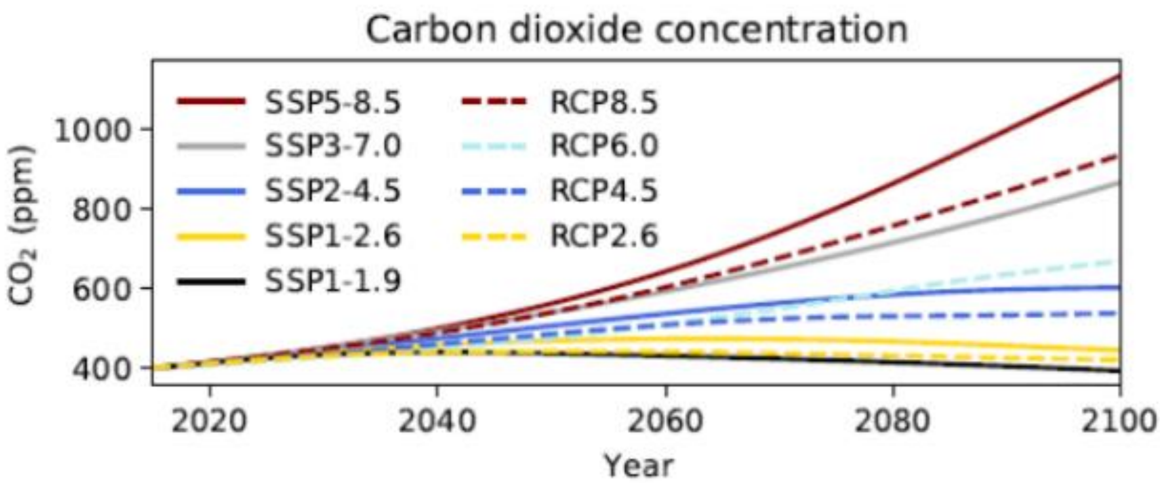

Figure 2-2: Global annual mean CO_2 concentrations for the 21st century scenarios. Sourced from Bodeker et al. (2022).

2.4 New Zealand climate change

Published information about the expected impacts of climate change on New Zealand is summarised and assessed in the Australasia chapter of the IPCC Working Group II Risk Assessment report (Lawrence et al. 2023) as well as a report published by the Ministry for the Environment (Bodeker et al. 2022), and recent publications on NIWA's CMIP6 projections (Broadbent et al. 2025, Gibson et al. 2025). Key findings from these publications include:

- The regional climate is changing. The Australasia region continues to demonstrate long-term trends toward higher surface air and sea surface temperatures, more hot extremes and fewer cold extremes, and changed rainfall patterns.

- The region has exhibited warming to the present and is virtually certain to continue to do so. New Zealand mean annual temperature has increased, on average, by 0.11°C per decade over the period 1909–2024 (Figure 2-3). Annual temperature changes have emerged above natural variability.

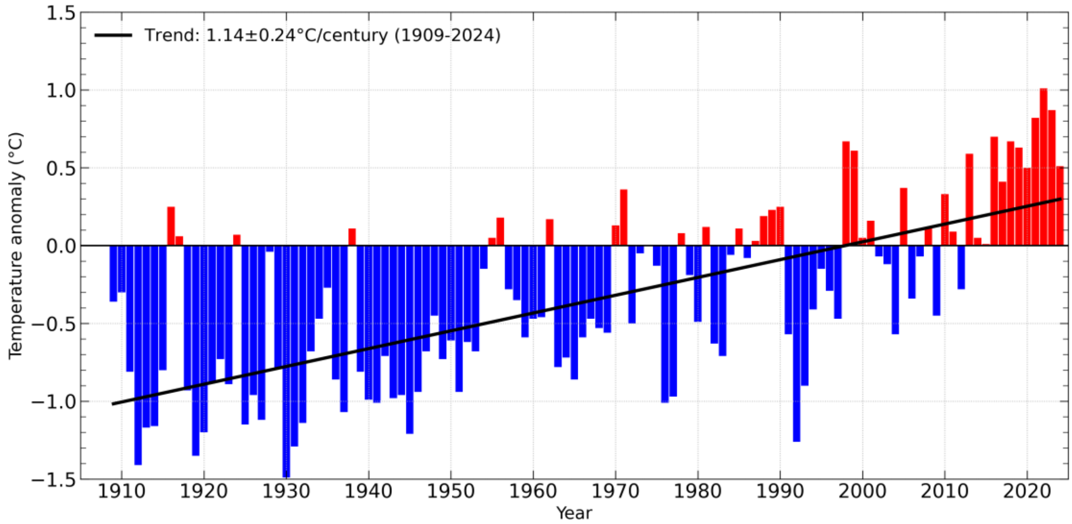

Figure 2-3: Air temperature anomalies (in C) over land, relative to a 1995–2014 baseline. Air temperature anomalies calculated from the New Zealand seven-station.

- Warming is projected to continue through the 21st century along with other changes in climate. Warming is expected to be associated with rising snow lines, decrease in snow depth, more frequent hot extremes, less frequent cold extremes, and increasing heavy rainfall related to flood risk for most areas across New Zealand. In particular:
  - Enhanced warming in the East Australian Current region of the Tasman Sea is projected.
  - An increase in marine heatwaves and ocean acidity is projected.
  - Fire weather is projected to increase in many parts of New Zealand. Regional sea-level rise will very likely exceed the historical rate, consistent with global mean trends.
  - Different spatial pattern of projected changes to annual maximum compared to minimum temperatures.
  - Precipitation is projected to increase in southern New Zealand. Precipitation changes are seasonally and orientation dependant (projected increase in winter and spring rainfall in the west and south, with less rainfall in the east and north, and more summer rainfall in the east of both islands, with less rainfall in the west and central North Island).
  - A general increase in annual maximum precipitation even in some areas of decreased annual precipitation.

- Future effects of the El Niño Southern Oscillation (ENSO) cycle upon New Zealand climate are not expected to change significantly this century. The future of trends in the Southern Annular Mode (SAM) and the location of the mid-latitude jet stream depends on greenhouse gas mitigation action and how quickly the ozone hole recovers.

- Without adaptation, further climate-related changes are projected to have substantial impacts on water resources, coastal ecosystems, infrastructure, health, agriculture, and biodiversity.

Additional information about recent New Zealand climate change can be found in Gibson et al. (2024b).

2.5 Natural factors causing fluctuation in climate patterns over New Zealand

Much of the material in this report focuses on the projected impact on the climate of, and oceans surrounding Southland over the coming century of increases in global anthropogenic greenhouse gas concentrations. However, natural variations will also continue to occur. Much of the variation in New Zealand's climate is random and lasts for only a short period, but longer term, quasi-cyclic variations in climate can be attributed to different factors. Three large-scale oscillations that influence climate in New Zealand are the El Niño-Southern Oscillation, the Interdecadal Pacific Oscillation, and the Southern Annular Mode (Ministry for the Environment, 2008). Those involved in (or planning for) climate-sensitive activities in Southland will need to cope with the sum of both anthropogenic climate change and natural climate variability.

2.5.1 The effect of El Nino and La Nina

El Niño-Southern Oscillation (ENSO) is a natural mode of climate variability that has wide-ranging impacts around the Pacific basin (Murphy and Timbal 2014). ENSO involves a movement of warm ocean water from one side of the equatorial Pacific to the other, changing atmospheric circulation patterns in the tropics and subtropics, with corresponding shifts for rainfall and mean sea level across the Pacific.

During El Niño, easterly trade winds weaken and warm water 'spills' eastward across the equatorial Pacific, accompanied by higher rainfall than normal in the central-east Pacific. La Niña produces opposite effects and is typified by an intensification of easterly trade winds, and retention of warm ocean waters over the western Pacific. ENSO events occur on average 3 to 7 years apart, typically becoming established in April or May and persisting for about a year thereafter (Figure 2-4).

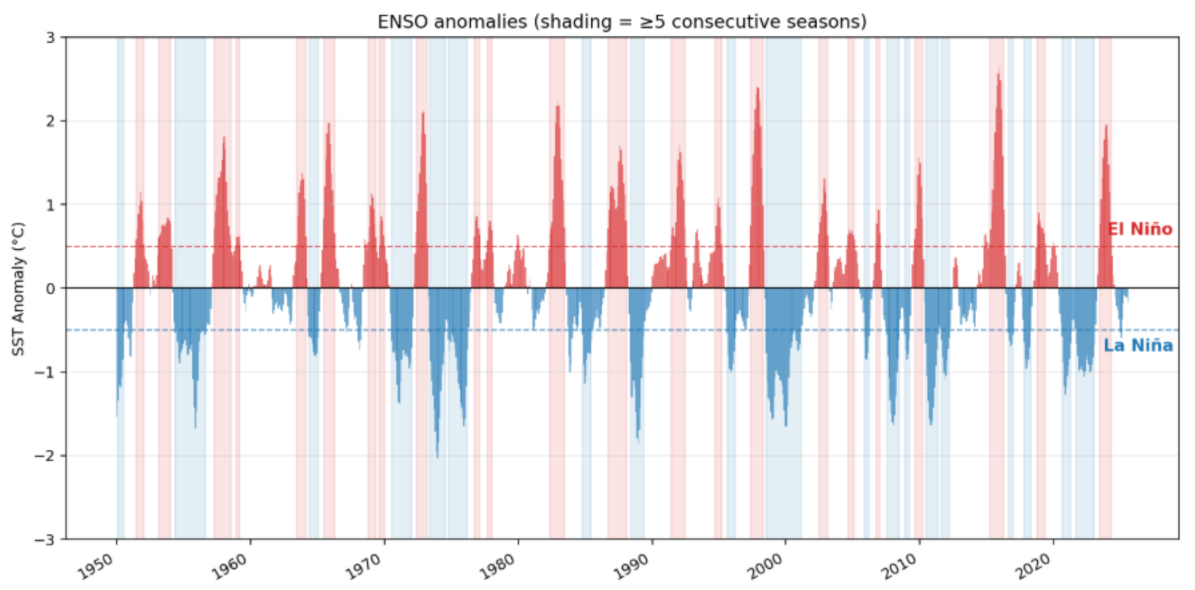

Figure 2-4: Time series of NINO3.4 sea surface temperature anomalies from 1950-2025 with reference period being 1991–2020. Values >0.5 correspond with El Niño, values <-0.5 correspond with La Niña and values between [-0.5,0.5] are considered as weak SSTs anomalies. Data sourced from https://www.ncei.noaa.gov/access/monitoring/enso/sst.

During El Niño events, the weakened trade winds cause New Zealand to experience on average a stronger than normal south-westerly airflow. This generally brings lower seasonal temperatures to the country and drier than normal conditions to the north and east of New Zealand, and wetter than usual conditions for parts of Southland (Hennessy et al. 2007) (Figure 2-4). Mean sea level around New Zealand can be several centimetres (up to 12 cm) lower than "normal" during peak El Niño events (Ministry for the Environment 2017). During La Niña conditions, the strengthened trade winds usually cause New Zealand to experience more north-easterly airflow than normal, higher-than-normal temperatures (especially during summer), and drier conditions for much of the South Island, including Southland (Figure 2-5). Mean sea level is generally higher than normal.

According to IPCC (2023), ENSO is highly likely to remain the dominant mode of natural climate variability in the 21st century (Bodeker et al. 2022), and rainfall variability relating to ENSO is likely to increase. While there is uncertainty in whether there will be meaningful change in ENSO variability over the next 50 to 100 years, there is evidence that the intensity of events may increase.

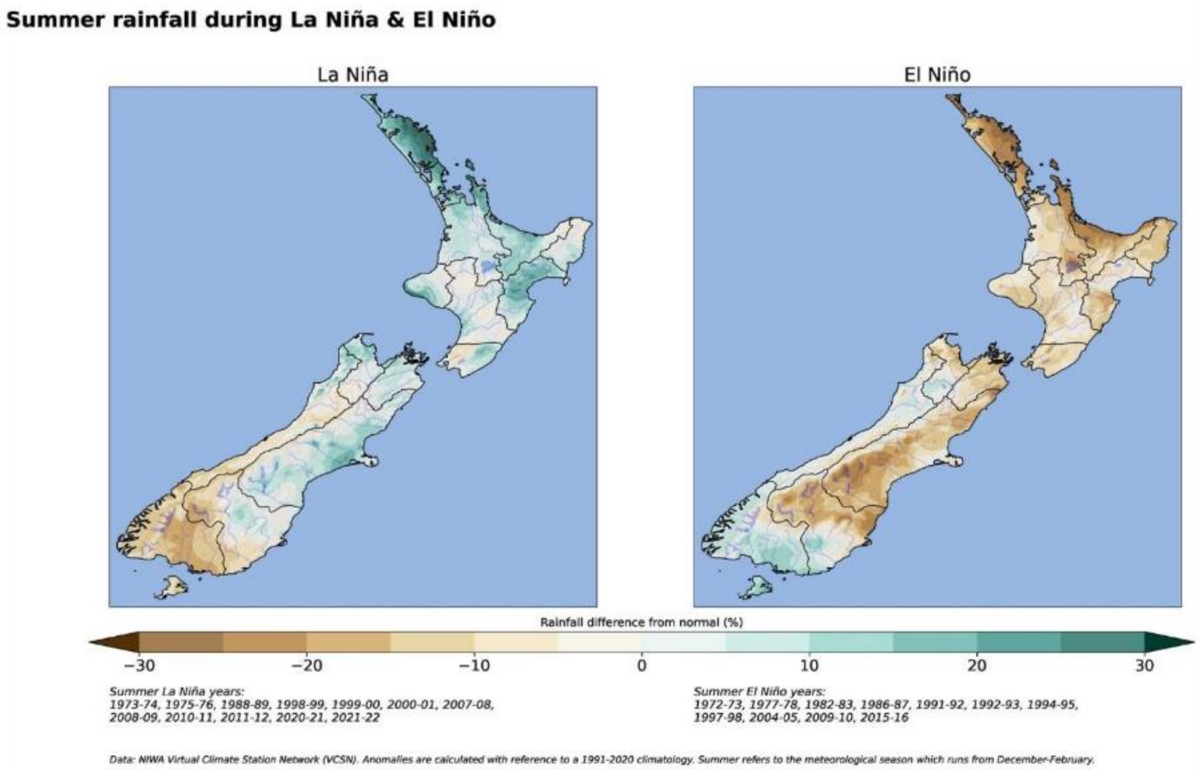

Figure 2-5: Average summer percentage of normal rainfall during El Niño (left) and La Niña (right) in New Zealand. Sourced from ESNZ.

2.5.2 The effects of the Interdecadal Pacific Oscillation

The Interdecadal Pacific Oscillation (IPO) is a large-scale, long-period oscillation that influences climate variability over the Pacific Basin including New Zealand (Salinger et al. 2001). The IPO operates at a multi-decadal scale, with phases lasting around 20 to 30 years. During the positive phase of the IPO, sea surface temperatures around New Zealand tend to be lower, and westerly winds stronger, resulting in wetter conditions for Southland (Figure 2-6). Mean sea level around New Zealand tends to be lower than "normal" or trends in sea-level rise reduced. The opposite occurs in the negative IPO phase (e.g. the Pacific has been in this phase since ~ 1989[3]). The IPO can modify New Zealand's connection to ENSO, and it also positively reinforces (dampens) the impacts of El Niño during IPO positive and negative phases, and of La Niña during the opposite IPO phases.

[3] IPO time series sourced from psl.noaa.gov/data/timeseries/IPOTPI/tpi.timeseries.ersstv5.data

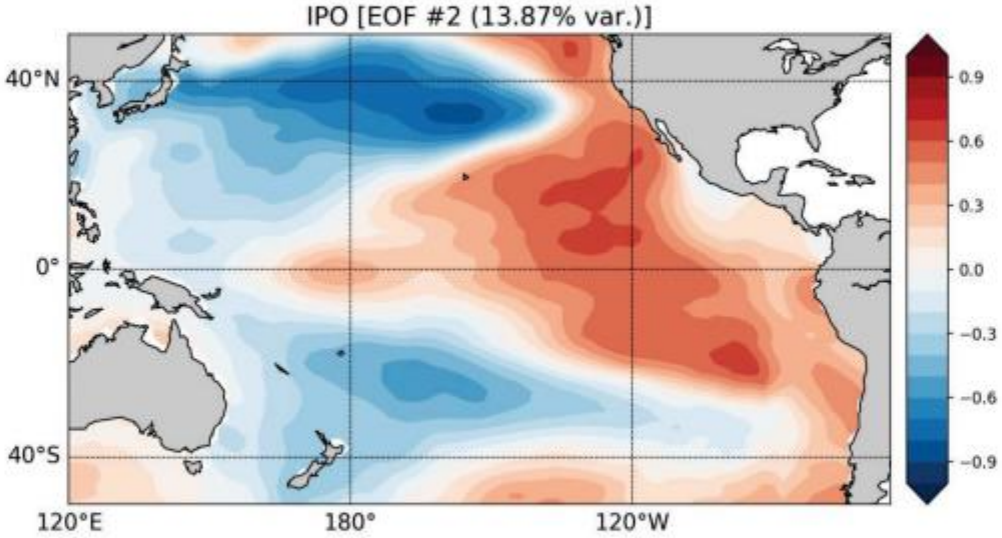

Figure 2-6: SST anomaly spatial pattern (Empirical Orthogonal Function, or EOF) associated with the positive phase of the Interdecadal Pacific Oscillation. The pattern shown, with positive SST anomalies in the eastern tropical Pacific, is IPO+. The IPO- phase has anomalies of opposite sign everywhere. Data source: ERSST version 5 dataset.

2.5.3 The effects of the Southern Annular mode

The Southern Annular Mode (SAM) represents the variability of circumpolar atmospheric jets that encircle the Southern Hemisphere and extend out to the latitudes of New Zealand. It is the leading mode of variability in the Southern hemisphere atmospheric circulation on month-to month and interannual time scales. The SAM is often modulated by ENSO, and both phenomena affect New Zealand's climate in terms of westerly wind strength and storm occurrence (Renwick and Thompson 2006). In its positive phase, the SAM is associated with relatively light winds and more settled weather over New Zealand, with stronger westerly winds further south towards Antarctica. In contrast, the negative phase of the SAM is associated with unsettled weather and stronger westerly winds over New Zealand, whereas wind and storms decrease towards Antarctica. (Figure 2-6, Figure 2-7). Kidston et al. (2009) describe the impacts of the SAM on the climate of southern New Zealand (including Southland):

- When the SAM is positive:
  - More easterly winds than normal during summer.
  - More northwesterly winds than normal during winter.
  - Reduced summer rainfall throughout Southland.
  - Increased winter rainfall for Fiordland, reduced winter rainfall for eastern and southern parts of Southland.
  - Higher than average monthly mean maximum temperatures in summer and winter.
- When the SAM is negative:
  - More southwesterly winds than normal.

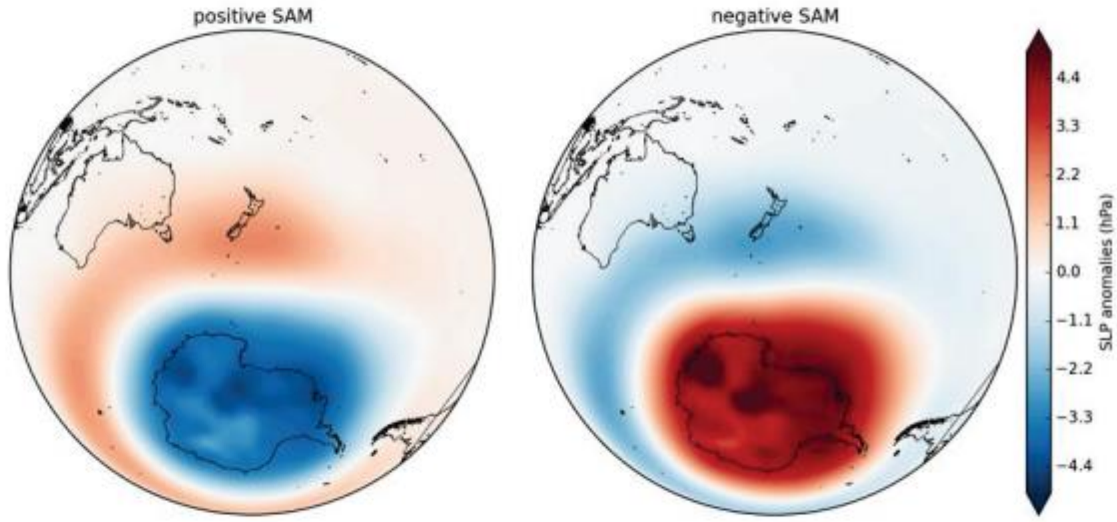

Figure 2-7: Pattern of the pressure variations associated with the positive (left) and negative (right) phases of the SAM. Blue shading indicates below-average pressures and red shading indicates above average pressures. Monthly composites were made using the PC associated with the first EOF of Southern Hemisphere monthly geopotential anomalies at 850 hPa from the NCEP / NCAR reanalysis (Section 2.3.3) using a threshold of +/- 1 std. Data source: http://www.esrl.noaa.gov/psd/data/gridded/data.ncep.reanalysis.html.

2.5.4 Interactions between natural climate cycles and climate changes

El Niño-Southern Oscillation

ENSO is highly likely to remain the dominant mode of natural climate variability in the 21st century, and that rainfall variability relating to ENSO is likely to increase (IPCC 2013a, Huang and Xie 2015). However, there is uncertainty about future changes to the amplitude and spatial pattern of ENSO. According to Cai et al. (2014), there may be an increase in 'extreme' El Niño events (like the 1982/83, 1997/98 and 2016/17 El Niño events) with increasing concentrations of greenhouse gases in the atmosphere, due to faster warming over the eastern equatorial Pacific Ocean. Bodeker et al. (2022) note there is no systematic signal in changes in Southern Hemisphere teleconnections associated with ENSO. Therefore, those authors suggest that the typical average effects of ENSO upon New Zealand seen in recent decades will continue into the future.

Interdecadal Pacific Oscillation

The IPO influences global mean temperatures through its influence on Pacific sea surface temperatures (Meehl et al. 2013). When the IPO is in its negative phase, Pacific SSTs are cooler than usual, which has led to observed hiatuses in global warming (for example, the shift from the positive to negative IPO in the early 2000s). In contrast, a positive IPO exhibits above normal SSTs in the Pacific, leading to an acceleration in global mean temperatures (e.g., the shift from the negative to positive IPO in the 1970s). The future behaviour of the IPO is uncertain but Meehl et al. (2013) suggests that hiatus periods (associated with negative IPO periods) may become slightly longer.

Southern Annular Mode

Two drivers are expected to have competing influences on the future trends in the SAM:

1. The ongoing reduction of ozone-depleting substances and associated recovery of the ozone hole is expected to drive more occurrences of the negative SAM, particularly in summer (Bodeker et al. 2022).
2. Increasing concentrations of greenhouse gases will encourage more occurrences of the positive SAM throughout the year (Bodeker et al. 2022).

By 2100, the net result for future trend in the SAM is therefore likely to be relatively little change from present (Banerjee et al. 2020, Simpson et al. 2025). However, uncertainty in how future greenhouse gas emissions will trade off against the recovery of the ozone hole leads to uncertainty in how the SAM will influence New Zealand's future climate. According to Bodeker et al. (2022), if higher emissions scenarios eventuate we would expect a SAM influence on New Zealand of more frequent warmer and drier conditions in many regions (but wetter conditions in the eastern North Island, and the northeast South Island in summer). Conversely, if low emissions scenarios eventuate then the expected SAM effect on New Zealand in summertime is for more frequent unsettled periods with more precipitation in many regions (but drier in the eastern North Island and northeast South Island), and cooler conditions.

Much of the material presented hereafter focuses on the projected impact on the climate and oceans of, and surrounding, the Southland region over the coming century of increases in global anthropogenic greenhouse gas concentrations. But natural variations, associated with El Niño, La Niña, the Interdecadal Pacific Oscillation, the Southern Annular Mode, and "climate noise", will also continue to occur.

An example of this for temperature (from an overall New Zealand perspective) is shown in Figure 2-8. This figure shows annual temperature anomalies relative to the 1986–2005 base period used throughout this report. The solid black line on the left-hand side represents NIWA's 7-station temperature anomalies (i.e., the average over Auckland, Masterton, Wellington, Nelson, Hokitika, Lincoln, and Dunedin), and the dashed black line represents the 1909–2014 trend of 0.92 °C/century extrapolated to 2100. All the other line plots and shading refer to the air temperature averaged over the region 33–48 °S, 160–190 °W, and thus encompasses air temperature over the surrounding seas as well as land air temperatures over New Zealand.

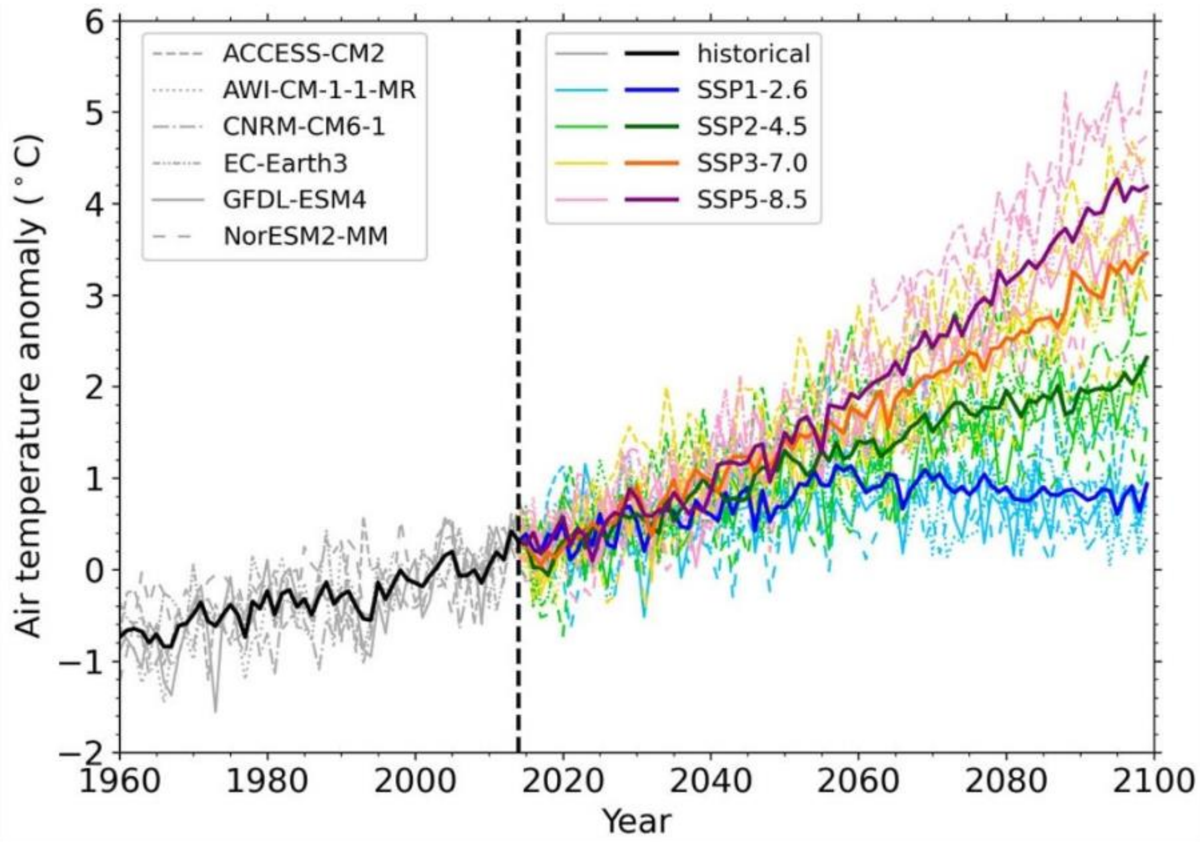

Figure 2-8: Evolution of national (land-based) annual air temperature anomalies projected by the downscaled models out to year 2100 under various SSPs. Anomalies are computed relative to the 1995–2014 base period. Source (NIWA): Updated national climate projections for New Zealand | Earth Sciences New Zealand | NIWA.

As discussed in Section 2.5, the IPO and the El Niño/La Niña cycle influence New Zealand sea level. So, the sea levels we experience over the coming century will also result from the sum of anthropogenic trend and natural variability.

In essence, natural variability will add to the human-induced trends at times, while at others it may offset part of the anthropogenic effect. Those involved in (or planning for) climate-sensitive activities in the Southland region will need to cope with the sum of both anthropogenic change and natural variability.

3 Methodology

3.1 Climate change impact assessment

3.1.1 Measuring climate change effect

To measure the effect of climate change on the chosen metrics, downscaled CMIP6 climate projections from the baseline period [1990s (1995–2014)] are compared to the mid-century [2050s (2041–2060)] and end of century [2090s (2081–2099)] using the IPCC recommended estimation method (IPCC 2023). The magnitude of the effect is determined by the difference between the metrics or thresholds calculated over the baseline and future periods as follows:

- For climate, ocean, coastal climate change impact assessment, analysis is reported using year defined as the period between 1 January and 31 December of each calendar year.
- Change in magnitude associated with extreme value analysis is reported using year defined as the period between 1 January and 31 December of each calendar year.
- Change in hydrological variable magnitude, is reported using hydrological year definition (period between 1 June and 31 May) with the numbered year indicating the calendar year of the hydrological year. As a result, the change in hydrological variable magnitude by the end of century covers the period 2081–2098, instead of 2081–2099.
- Change in cryospheric variable magnitude, is reported using snow year definition (period between 1 April and 31 March) with the number year indicating the calendar year of the snow year. As a result, the change in cryospheric variable magnitude by the end of century covers the period 2081–2098, instead of 2081–2099.
- Change in stream water temperature, is reported using water year definition (period between 1 October and 30 September) with the number year indicating the calendar year of the water year. For modelling purpose, seasons were defined as half years with Summer being the period October to March of the water year, and Winter being the period April to September of the water year.

3.1.2 Climate change impact assessment

A brief description of each variable to be reported is provided hereafter, as well as current limitations associated with the analysis. With the exception of SLR, all "changes" refer to differences between the historical period 1995–2014 and two future periods (2041–2060 and 2081–2099), noting the exceptions to the end year of the 2090s period listed above. Table 3-1 presents a description of the analysis carried out and reported across regional (Southland), Freshwater Management Unit (FMU) and catchment scales.

The change in these variables between the baseline and projection periods is represented in either percentage or absolute terms depending on the variable in question. The results for the six GCMs are combined into either a multi-model average (for climate variables) or a multi-model median (for hydrological variables), and the results of climate change scenarios are kept separate. This approach provides annual and seasonal maps for each statistic, representing average/median changes in that statistic between baseline and mid-century and baseline and late century. Changes in hydrological statistics are reported for the lower end of river reaches (for riverine characteristics) or as watershed average characteristic (for watershed scale hydrological and cryospheric characteristics) while changes in climate statistics are reported on at the scale (12 km) of the atmospheric model across the Southland region.

Table 3-1: Variable and methods used to report climate change impact across Southland, Freshwater Management Unit and catchment scales.

Variable	Description
Precipitation	Average precipitation: change in the annual-average and seasonal-average precipitation for each time slice.
	Maximum annual daily: change in the annual-average maximum daily precipitation for each time slice.
	Maximum annual 5 days: change in the annual-average maximum 5 days precipitation for each time slice.
	Days with rain above threshold: change in the annual average number of day above threshold for each time slice.
	Intensity: change of intensity is provided as per specific weather event analysis.
	Dry day projection is characterised by the change in annual average number of days where total daily precipitation is less than 1 mm/day.
	Wet day projection is characterised by the change in annual average number of days where total daily precipitation is equal to or larger than 1 mm/day.
	Heavy rain day projection is characterised by the change in annual average number of days where total precipitation is equal to or larger than 50 mm/day.
Temperature	Average temperature: change in the annual-average and seasonal-average daily temperature for each time slice.
	Average minimum temperature: change in the annual-average and seasonal-average mean minimum for each time slice.
	Number of hot days: Change in the annual number of days where temperature is 25°C or above.
	Number of cold days: Change in the annual number of days where minimum temperature is 0°C or below.
	Heatwave days: For this analysis a heatwave is defined as at least three consecutive days when the temperature is 25°C or above. The estimation of the distribution of heatwave (annual distribution) is carried out over the year July to June in order not to break up any heatwaves that cross the December-January period.
	Growing degree-days: change in the mean annual accumulative number of degrees Celsius exceeding a threshold over the period 1 July to 30 June, with a temperature threshold representing a threshold temperature for plant growth taken as 10°C.

Variable	Description
Other climate variables	Relative Humidity: change in the annual-average, seasonal-average and monthly average relative humidity for each time slice. Solar Radiation: change in the annual-average and monthly average shortwave and longwave solar radiation for each time slice. Cloud cover: change in the annual-average and monthly average cloud cover for each time slice. Wind speed and wind direction: change in the seasonal-average wind speed and wind direction for each time slice. Calm and windy days: change in the annual-average and monthly average number of calm and windy days for each time slice. Strong wind: change in the annual-average, and monthly average ninety-ninth percentile of wind speed for each time slice. Potential Evapotranspiration Deficit (PED): change in the amount of water (in mm) needed to be added as irrigation, or replenished by rainfall, to keep pastures growing at levels that are not constrained by a shortage of water. Change in PED are reported at annual time scale for each time slice. Probability of PED over 200 mm: change in the probability of PED exceeding 200mm per year (as annual average) across each time slice.
Ocean	Annual mean sea level: change in average annual sea level around Southland based on national scale assessment (Coastal Hazards and Climate Change Guidance, MfE 2017) and NZ Sea Rise MBIE endeavour programme (NZ SeaRise Programme). Sea temperature: change in average annual change is sea temperature around Southland based on New Zealand Earth system model and findings from Deep south National Science Challenge project Marine heatwaves and the link with climate extremes (Deep South Challenge)'. Probabilistic assessment being carried out using sea temperature threshold. Change in acidification.
Projected sea-level rise	Sea Level Rise: projected change in average annual sea level around Southland based on national scale assessment (Coastal Hazards and Climate Change Guidance, MfE 2017) and NZ Sea Rise MBIE Endeavour programme as sea-level. Change at decade scales is quite similar across New Zealand, so no large variations are expected. Years when specific increments are reached. Building and replacement costs with SLR and 1% AEP storm. Local variation of sea-level rise: Due to existing data limitation, those changes are limited to an appraisal of vertical land movement (based on available continuous GPS data and publications) and an updated check on the sea-level trend from the Bluff tide-gauge record.

Variable	Description
Hydrology	High flows metrics: change for each time slice in the Mean Annual Flood (representing the change in the mean largest peak flow for each year), Q5 and Q25, which represents river flow exceeded five and twenty-five percent of the time (resp.). Mean Annual Flood typically represents flows that are exceeded less than one percent of the time and have a return period between two and three years. Flushing flows: Annual and seasonal change, per season and time slice, of the average number of events per year exceeding three times the median flow with a gap of 3 days or 5 days between events. Average flows metrics: change for each time slice in the average seasonal river flow metric for each time slice, calculated as annual or seasonal average and annual median as river flows tends to not be normally distributed (as opposed as climate variables). Low flows: change in low flow for each time slice. Low flow metrics are represented by Q75 and Q95, which represents river flow exceeded seventy-five and ninety-five percent of the time, and the 7-day mean annual low flow metrics, calculated as average yearly 7-day rolling minimum discharge over the summer season defined as the period between 1st September and 30th April of each hydrological year. Flow reliability/Water supply reliability: change for each time slice of the average fraction of time during each irrigation season (spanning from 1 September to 30 April) that the river flow is too low and thus irrigation is restricted. To define the river conditions at which abstraction for irrigation becomes restricted, a local minimum flow condition is set based on the proposed National Environmental Standard (NES) for Environmental Flows and Water Levels (Ministry for the Environment 2008) or environmental flow thresholds established by Environment Southland. Water supply reliability varies between 0 and 100 percent and is often between 0.9 and 1.0 for New Zealand rivers. Recharge rates: change for each time slice in annual, seasonal and monthly average net land surface recharge calculated over Southland non-confined aquifers as defined using the Hydrogeological Stack Unit layer (White et al. 2019).
Oceanographic	Projected sea surface temperatures, ocean acidification and marine heat waves including anomalies to historic values will be assessed around Southland. Marine heatwaves are characterised by marine heatwave intensity and annual marine heatwave days.
Water temperature	Water temperature: change in the seasonal and monthly water temperature using linear correlation between air temperature and water temperature where available for each time slice.
Cryospheric	Cryospheric metrics: change for each time slice in annual and seasonal average snow volume, taken as snow water equivalent, snowfall and snow melt calculated using the cryospheric process model embedded within the TopNet model (Clark et al. 2008). Snow line: change for each time slice of the annual and seasonal average elevation of the interface between snow free and snow-covered area. Number of snow days: change for each time slice of the annual and seasonal average number of snow days, characterised as number of days of precipitation concurrent with temperature below the temperature accumulation threshold of 2°C as represented within the degree day model used in TopNet (Clark et al. 2008). Changes reported for 14 catchments.

Variable	Description
Weather events	Drought changes in frequency: change in drought frequency are reported as drought change in frequency above 200 mm threshold of PED. However, large uncertainties exist in the derivation of the downscaled driven climate variables (solar radiation, relative humidity, wind) so results are likely to have a large uncertainty. Change in precipitation events in tabular format for the following locations: Gore, Mataura and Riversdale townships (Mataura catchment); Winton and Lumsden (Oreti catchment); Waihopai Dam (Waihopai River); and New River Estuary.
Spatial extent	Analysis of change for variable of interest will be reported graphically on each Freshwater Management Units (FMU) from ES (i.e., Mataura, Waituna, Aparima, Oreti, Waiau and Fiordland).

3.2 Atmospheric modelling

The CMIP6 model archive was developed to support the IPCC's 6th assessment report (IPCC 2023). Gibson et al. (2024a) evaluated the CMIP6 GCMs for performance over the New Zealand region using model outputs for the period 1980–2015 (referred to as the hindcast period) compared to global observational reanalysis data and based on a large number of climate variables and indicators. Six of the CMIP6 GCMs were selected as best at representing climatic dynamics around New Zealand. These six GCMs span a range of climate model sensitivities and are relatively independent from one another. The six CMIP6 models selected are:

- ACCESS-CM2 (Bi et al. 2020);
- AWI-CM-1.1-MR (Semmler et al. 2020);
- CNRM-CM6-1 (Voldoire et al. 2019);
- EC-Earth3 (Döscher et al. 2022);
- GFDL-ESM4 (Dunne et al. 2020); and
- NorESM2-MM (Seland et al. 2020).

These six CMIP6 GCMs were used to generate climate change projections for the recent historic period plus four emission scenarios: SSP1-2.6 (not used for this report), SSP2-4.5, SSP3-7.0 and SSP5-8.5. Dynamical downscaling to produce the CMIP6 New Zealand climate projections was undertaken using CSIRO's Conformal Cubic Atmospheric Model (CCAM) (Gibson et al. 2024a) to produce hourly climate simulations, for all climate variables of interest, available at 12 km spatial resolution across New Zealand.

While a bias corrected high resolution (5 km spatial resolution) dataset was generated for a limited set of climate variables (precipitation, minimum and maximum temperature and potential evapotranspiration; Campbell et al. 2024), we are proposing to use the lower 12 km resolution products which were created as part of the project (Gibson et al 2024b). This choice of the low spatial resolution dataset is driven by the following:

- Only a limited number of climate variables are available at high resolution which will generate additional uncertainties within the hydrological projections (due to the analytical estimation of climate variables that is required for the hydrological model). For consistency across the requested climate variables, it is recommended to use the low-resolution climate dataset;

- The bias correction of the high-resolution climate dataset was completed using the Virtual Climate Station Network (Tait et al. 2006, Campbell et al. 2024) for daily precipitation and daily maximum/minimum temperature. The use of the CMIP6 bias corrected dataset may require additional bias correction across Southland for each climate variable of interest such as relative humidity, incoming short-wave radiation, atmospheric pressure and wind);
- The bias correction of the high-resolution climate dataset was carried out for each variable independently of other climate variables. Gibson et al. (2024b) recommend using the low-resolution climate projections (12 km) for hydrological assessment;
- The high-resolution bias corrected dataset is available only at daily time step, which will limit its use for assessment of hydro-climate extremes, especially for representation of high flow simulations in the majority of catchments in New Zealand (over 75 percent), which are characterised by a time of concentration of less than 24 hours (Griffiths and McKerchar 2012).

The impact of climate change on climate variables will be based on yearly characterisation over the period 1 January–31 December.

In addition to the provision of spatial change in climate variable, Table 3-2 presents a description of the climate analysis completed at required locations (Table 3-3).

Table 3-2: Variable and methods used to report climate change impact at specific locations.

Variable	Description
Precipitation	Change in number of days with rain above or below threshold: change characterised across time slice in the change in annual average number of days where total precipitation is equal or larger than, or equal or smaller than a defined rainfall threshold.
Temperature	Heatwave: Change in 'heatwave days'. For this analysis a heatwave is defined as at least three consecutive days when the temperature is 25°C or above. The estimation of the distribution of heatwave will be carried out over the year July to June in order not to break up any heatwaves that cross the December-January period. Frost days: change in annual number of days where average daily temperature is 0°C or below. Overnight minimum: represented by change in minimum daily temperature.
Wind	Calm days: change in the annual-average and monthly average number of calm days for each time slice reported at location of interest.

Variable	Description
Weather events	Drought change in frequency: change in drought frequency will be reported as drought change in frequency above specific threshold of PED (to be agreed with ES at the start of the project). However, large uncertainties exist in the derivation of the downscaled driven climate variables (solar radiation, relative humidity, wind) so results are likely to have a large uncertainty. Change in precipitation events in tabular format for the following locations: Gore, Mataura and Riversdale townships (Mataura catchment); Winton and Lumsden (Oreti catchment); Waihopai Dam (Waihopai River); and New River Estuary.

Table 3-3: Location where climate information is required.

Town location	Town location	Town location	Town location
Invercargill	Winton	Tuatapere	Milford Sound
Riverton	Lumsden	Oban	Garston
Mataura	Te Anau	Riversdale	Mossburn
Gore	Otautau	Waikaka	Bluff

3.2.1 Design rainfall

To determine design rainfall totals, we are using the High Intensity Rainfall Design System tool (HIRDS) version 4 developed by NIWA (Carey-Smith et al. 2018). HIRDS provides a map-based interface to enable rainfall estimates to be provided at any location in New Zealand under current and future climate conditions.

As HIRDS has not yet been updated using the New Zealand CMIP6 climate projections, HIRDS projections of rainfall depth are based on the New Zealand CMIP5 climate projections (Ministry for the Environment 2018):

- Change in rainfall depth under future scenario is provided, for each event duration and return period, by HIRDS' CMIP5 derived change in rainfall depth per degree change (Carey-Smith et al. 2018).
- Mid-century and end of century change in temperature is calculated based on the 20-year average change in temperature for each SSP scenario for mid and end-century time period.

Change in rainfall designed precipitation will be provided for the 16 township locations (see Table 3-4).

Table 3-4: Location where design rainfall information is required.

Town location	Town location	Town location
Invercargill	Te Anau	Milford Sound
Riverton	Otautau	Garston
Mataura	Tuatapere	Mossburn
Gore	Oban	Bluff
Winton	Riversdale	
Lumsden	Waikaka	

3.2.2 Possible Maximum Precipitation

Possible Maximum Precipitation (PMP) is generated using a tool recently developed and funded by the New Zealand Dam Safety Hydrology Group (DSHG) chaired by David Payne (Mercury). The Dam Safety Hydrology Group (DSHG), formed from 6 major dam owners in New Zealand (Meridian Energy, Mercury Energy, Contact Energy, Genesis Energy, Manawa Energy and Watercare Services. The tool (Bodeker et al. 2023) provides the ability to calculate PMP estimations calculated over upstream surface water draining catchment, using an automated implementation of the New Zealand's PMP methodology developed by Tomlinson and Thompson (Tomlinson and Thompson 1992, commonly referred as the 'Blue-Book') using two approaches:

- PMP estimations using the original Tomlinson and Thompson (1992) storm catalogue and ancillary data sets;
- PMP estimations using updated storm catalogue, including events over the period 1988–2023, along with updated data sets (such as 100-year dewpoint temperature and rainfall).

The authors indicate that PMP estimations using the updated storm catalogue can be different from those generated using the 'Blue-Book' data (Figure 3-1). In some regions (e.g., Northland, Gisborne, Bay of Plenty, Central North Island, Canterbury) this newer estimate is higher than would be derived using only the Blue Book, and in other regions (e.g., south-west of the North Island and the Southern Alps), lower.

As part of the project, PMP is estimated for upstream catchments associated with 18 locations presented in Table 3-5 for 12-24-48-72 hrs storm durations. To ensure that Probable Maximum Flood (PMF) can be provided for those locations (see Section 3.3.5), those locations are mapped to the closest DN3 Strahler 3 river reach.

Table 3-5: Location where PMP estimation is required. The location represents the outlet of the surface water catchment draining at that location.

PMP location	PMP location	PMP location
Fairlight at Cainard Station	Mokoreta at Mt Alexander	Pourakino Valley at Cascade Road
Five Rivers at Five Rivers Station	Ohai at Wether Hill Station	Garston at Nevis Road
Glenlapa at Round Hill	Waiau River at Clifden	Whitestone Aquifer at Lynwood
Hamilton Burn at Mt Hamilton Road	Waikaia River at Piano Flat	Oban
Hokonui Hills at Lora Station	Wendon Valley at Waikaka	Bluff
McKellars Flat at Gorge Burn	Woodlands at Garvie Road	Invercargill (Waihopai dam)

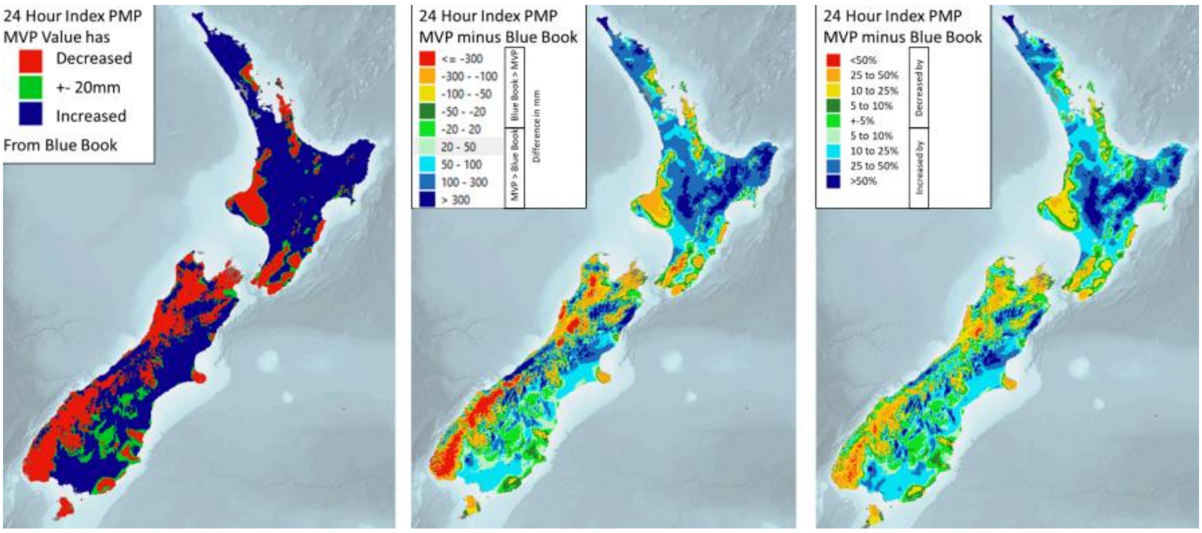

Figure 3-1: Differences in 24-hour indexed PMP. Left panel shows where PMP estimates are within 20 mm (green), where PMP estimates decrease in the MVP (red), and where PMP estimates increase in the MVP (blue). Centre panel shows the absolute change in mm while right panel shows the percentage change. Sourced from Bodeker et al. (2023).

Climate change projections of PMP (magnitude) for different durations are generated using the methodology devised by the DSHG group. Change in rainfall depth is calculated at each location for each emission scenario based on the average change in air temperature in increments of 1 degree ranging from 1°C (current climate warming condition) to 4°C (corresponding to a warming of 3°C by 2100).

3.3 Hydrological modelling

3.3.1 Hydrological model description

The hydrological modelling methods used in this study are adapted from the Deep South National Science Challenge project 'Climate change impact on national hydrology' (DSC 2019, Collins 2020), hereafter referred to as the DSC project. In addition, the modelling methodology follows recommendations provided as part of the Ministry for the Environment project 'Pilot CMIP5-CMIP6 hydrological projections intercomparison study' (Zammit et al. 2024).

In the DSC project the CMIP5 New Zealand climate projections were coupled with TopNet (Clark et al. 2008), NIWA's uncalibrated, nationally parameterised surface water hydrological model. The TopNet model is implemented as part of the New Zealand Modelling (NZWaM) framework (Zammit et al. 2025) and is routinely used for surface water hydrological modelling applications in New Zealand.

As part of the DSC project, the version of TopNet used did not consider water transfers between catchments or other water storages and did not model aquifer water balances, nor did it simulate the impacts of water use activities on hydrological regimes. The Topnet model has been thoroughly validated at the national scale against hydrological records (McMillan et al. 2016) and is similar to the hydrological model used by NIWA for short-term flood forecasting risk across New Zealand (McMillan et al. 2013, Cattoën et al. 2022). It has been generally found to perform better for large, wet catchments, while significantly underestimating high flow responses. The model has been shown to perform well in both low and high flow situations if calibrated, but transferring calibration parameters to ungauged catchments has proven challenging (Singh et al. 2017). As such the uncalibrated model has been considered as useful in undertaking previous national scale climate change impact assessment (Collins 2020).

As part of the project, we implement an updated version of the TopNet model. The updated version of TopNet is suitable for surface water-dominated catchments but does not address the impact of regional groundwater movement on hydrological regimes, or conceptualise water use activities under current and future climate projections (which is outside of the scope of the project). Nevertheless, this version of TopNet is different from the version implemented as part of Collins (2020) and ES's 'CMIP5 Climate change assessment' project (Zammit et al. 2018). A brief description of the differences in TopNet conceptualisation is provided below:

- The current conceptualisation of hydrological processes was improved following enhancements by Griffiths et al. (2021):
  - improved conceptualisation of evapotranspiration processes after Feddes et al. (1978); and
  - improved representation of cryospheric processes achieved by updating TopNet's simple temperature index model parameterisation to represent Mediterranean climate (Conway et al. 2023).
- Parametrisation of watershed scale infiltration excess runoff processes to improve high flow response, implemented as part of the Mā te haumaru ō te wai project.

Spatial information and datasets used to develop TopNet as part of the DSC project included:

- Catchment topography based on a nationally available 8 m Digital Elevation Model (DEM);
- Physiographical data based on the Land Cover Database - version 5 (LCDB5) and Land Resource Inventory (Newsome et al. 2000);
- Soil data based on the Fundamental Soil Layer information (FSL) (Newsome et al. 2000); and
- Hydrological properties based on the River Environment Classification version 2.3 (REC2.3), with land cover characteristics derived from the LCDB5 (Snelder and Biggs 2002) and mapped onto digital river network version 3.1 (DN3v1- high resolution digital river network generated in collaboration with ES as part of the NZWaM project).

Because of TopNet assumptions, soil and land use characteristics within each computational sub-catchment are homogenised. Essentially this means that the soil characteristics and physical properties of different land uses, such as pasture and forest, will be spatially averaged, and the hydrological model outputs will be an approximation of conditions across agricultural land use over the period 1980–2099. Despite the ability of the uncalibrated TopNet model (i.e., a priori parametrised TopNet model across Southland) to reproduce observed hydrological characteristics across New Zealand as a whole (see Booker and Woods 2014, McMillan et al. 2016, Cattoën et al. 2022), model bias correction or calibration is recommended to match catchment-specific hydrological behaviour (Zammit 2019).

3.3.2 Hydrological model parametrisation

TopNet model parameterisation uses the concept of parameter multipliers, because one of the main assumptions of the model implementation is that the spatial distribution of the parameters is determined *a priori* from catchment physiographic information (see Clark et al. 2008).

TopNet requires the determination of seven hydrological parameter multipliers and ten snow-related parameters for each sub-catchment (see Table 3-6). The initial values of the parameter multipliers are set to a value of 1, while snow-related parameters are initialised based on previous study results for the area. The values of the parameter multipliers are adjusted through calibration or bias correction as described below.

Table 3-6: Range of TopNet parameter multipliers used during calibration process. Default values of the TopNet parameters are spatially distributed and provided by national datasets (see Clark et al. 2008 for further information).

Parameter name (internal name)	Parameter description	Calibrated range
Hydrological parameters		
Saturated store sensitivity (topmodf)	Describes exponential decrease of soil hydraulic conductivity with depth	[0.01–2] * default
Drainable soil water (swater1)	Range between saturation and field capacity	[0.05–10] * default
Plant available soil water (swater2)	Range between field capacity and wilting point	[0.05–10] * default

Parameter name (internal name)	Parameter description	Calibrated range
Hydraulic Conductivity at saturation (hydcond0)	Catchment average hydraulic conductivity at saturation at the surface	[0.1–10000]*default
Ch_exp	Clapp-Hornberger c exponent	[0.05–10] * default
Ga-psif	Green-Ampt wetting front suction	[0.05–10] * default
canscap	canopy storage capacity	[0.05–10] * default
canenh	canopy evaporation enhancement factor	[0.05–10] * default
Overland flow velocity (overvel)	Catchment average surface rugosity	[0.1–10]*default
Manning n	Characterises the roughness of each reach	[0.1–10] *default
Atmospheric lapse rate (atmlaps)	Change in temperature with elevation, used to adjust temperatures from climate data sites to basin centroid	[0.7–1.5] * default
Gauge Undercatch (gucatch)	Adjustment for non-representative precipitation	[0.5–1.5] * default
Top-soil Hydraulic Conductivity associated with infiltration excess runoff	catchment-scale infiltration excess runoff factor	[0.1–10] * default
Snow parameters		
threshold for snow accumulation (th_accm)	Temperature threshold for snow accumulation	270.15–275.15 [K]
Threshold for snow melt (th_melt)	Temperature threshold for snow melt	269.15–274.15 [K]
snowddf	Degree-day factor for snow melt	1–10 [mm K^{-1} day^{-1}]
Minddf	Calendar day of the minimum degree-day-factor day	1–366 [days]
Maxddf	Calendar day of the maximum degree-day-factor day	1–366 [days]
snowamp	Seasonal amplitude of degree-day factor for snow melt	0–5 mm K^{-1} day^{-1}]
snowros	Addition in melt factor caused by rain-on-snow events	0–5 mm K^{-1} day^{-1}]
decmelt	Decrease in melt due to higher albedo after fresh snow	0–5 mm K^{-1} day^{-1}]
albdecy	Time decay of snow albedo	1–5 days
cv_snow	Sub-grid variability representing the distribution of the snowpack across the catchment	0–2 [-]

3.3.3 Hydrological variables

As part of the project, impact of climate change is carried out for the following hydrological metrics:

- Mean annual flood (MAF) – The mean of the largest peak flows for each year. For New Zealand rivers, this flow is typically exceeded less than one per cent of the time and has a return period of between two and three years.
- Q5 – The river flow exceeded five per cent of the time. This is typically a high flow metric lower than MAF.
- Average number of Fre3 days – Average number of events per year for which river discharge exceeds three times the median river discharge (Biggs 2000) considering a 3 day and 5 day filter period (Booker 2013).
- Mean river flow – River flow averaged over the period of analysis.
- Mean and median annual flows- Mean and median annual river discharge.
- Seasonal average flows – mean river discharge calculated across hydrological year (i.e., after August 1, until July 31 the following year) during Winter (June-July-August), Spring (September-October-November), Summer (December-January-February) and Autumn (March-April-May) period.
- Mean annual low flow (7dMALF) – The mean of the lowest 7-day average flows in each year of a period of interest.
- Annual Q25 - The annual average river flow that is exceeded twenty-five per cent of the time. This is typically a flow metric providing insights into the average flow during the second quartile of the year.
- Q75 – The river flow exceeded seventy-five per cent of the time. This is typically a low flow metric higher than the 7-day MALF but lower than mean river flow.
- Q95 – The river flow exceeded ninety-five per cent of the time. This is typically a low flow metric used by Environment Southland for riverine water consenting purpose.
- Flow reliability – The fraction of time that the flow is equal to or below the National Environmental Standard (NES) for Environmental Flows and Water Levels (Ministry for the Environment 2008) minimum flow threshold as defined previously, and in this case without accounting for any water takes. Using a simple and nationally consistent method permits national comparisons of the results transparently, even though local circumstances will differ. The minimum flow below which abstraction must cease is thus:

  $$Q_{min} = \begin{cases} 90\%\ of\ MALF\ when\ Q_{mean} \leq 5m^3/s \\ 80\%\ of\ MALF\ when\ Q_{mean} > 5m^3/s \end{cases} \tag{1}$$

  where *MALF* is the mean annual 7-day low flow calculated from the baseline hydrological data.
- Water Supply Reliability - The average fraction of time during each irrigation season (spanning September 1 to April 30 of each hydrological year) that the river flow is too low and thus irrigation is restricted.

Unless specified, the hydrological metric is calculated as an annual average quantity estimated using calendar year aggregation. The 7dMALF and flow reliability hydrological metrics are estimated using hydrological year spanning June 1 to May 31 the following calendar year. Water Supply reliability hydrological metric is estimated over the irrigation season spanning September 1 to April 30 of each hydrological year.

In addition to riverine flow hydrological metrics, recharge rates across non-confined aquifers (see Figure 3-2) are provided as the average annual and seasonal vertical drainage between TopNet's watershed root zone and groundwater zone. As TopNet is a watershed based hydrological model, it is assumed that any watersheds overlying Environment Southland's non confined-aquifer layers (Environment Southland, pers. comm.) is fully contributing to the recharge of the aquifer.

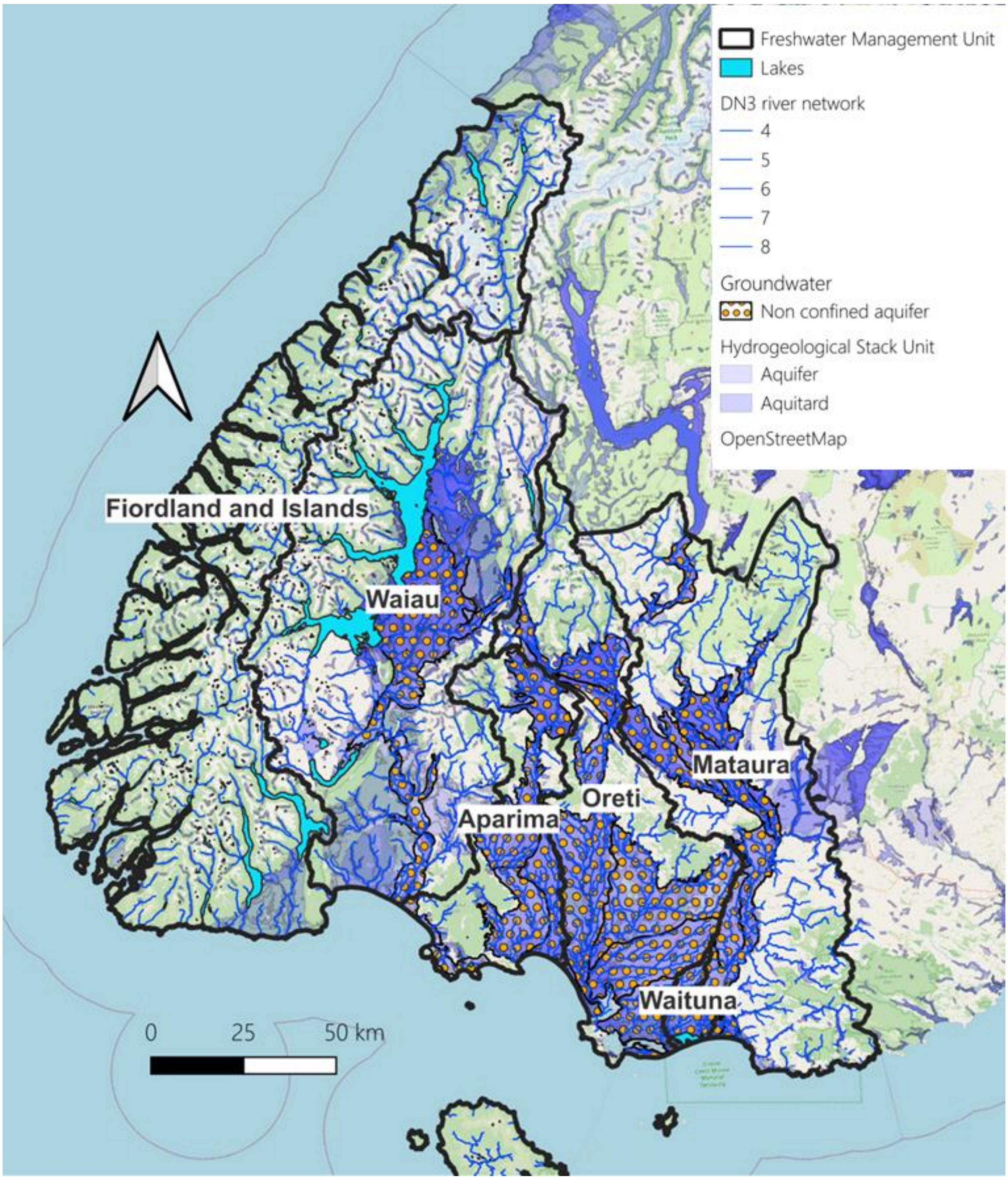

Figure 3-2: Hydrogeological Stack Units and shallow aquifers across the Southland region. Environment Southland's non -confined aquifers presented as orange dot polygon, with national aquifer dataset (blue) and aquitard (blue-green) derived from Hydrogeological Stack Units (White et al. 2019).

3.3.4 Hydrological extremes

Extreme events, such as floods, are often defined to be the maximum value of a quantity over a given period of time. Flood size at a given location is usually characterised by its intensity (peak flow rate or peak flow), its duration (length of the event) and its frequency (annual, seasonal, etc.) (Langbein 1949). The statistical limiting model describing how large a flood will get is referred to as the Extreme Value distribution (EVD). EVDs have similar statistical basis as the Central Limit Theorem (CLT), which specifies that the mean of a large sample has a normal distribution regardless of the underlying distribution. The main difference between EV theory and CLT is that the CLT concerns the behaviour of the mean of random variables, while EV theory concerns the behaviour of the tails of those distributions.

EVDs are of the following three types (Haan 1977), as long as the distribution is well behaved (Gumbel 1958). Extreme value theory, in particular the extremal types theorem (Leadbetter et al. 1983), suggests that the distribution of these maxima should be close to one of the extreme value types:

- EVD Type I (EVD1): Gumbel distribution, that is the most common EVD characterised with no upper or lower limits (usually referred to as unbounded).
- EVD Type II (EVD2): Fréchet distribution, bounded at the lower end, is usually used to model maximum values.
- EVD Type III (EVD3): Weibull distribution, bounded at the upper end, is usually used to assess product reliability to model failure and life data analysis.

The Generalised Extreme Value distribution (GEV, Von Mises 1936, Jenkinson 1955, Coles 2001) is a three-parameter distribution that combines all three extreme value types into a single form. The GEV distribution has been widely used for modelling the distribution of annual maximum flood peaks in at-site and regional settings (Hosking et al. 1985, Smith 1987, Martins and Stedinger 2000). In addition to flood modelling, the GEV distribution is commonly used to model many other natural extreme events (Bauer 1996, Coles 2001). One of the primary advantages of the GEV distribution is its ability to characterise flood magnitude regardless of the underlying flood generating distribution.

The Cumulative Distribution Function (CDF) for the GEV is:

$$F(x;\mu,\sigma,\zeta) = exp\left\{-\left[1+\xi\left(\frac{x-\mu}{\sigma}\right)^{-1/\zeta}\right]\right\} \quad (2)$$

Where σ and 1 + ξ(x-μ)/σ must be greater than zero. μ is referred as the location parameter and indicates where the distribution is located, σ is referred as the scale parameter and determines the width of the distribution, and ξ is referred as the shape parameter that affects the tail of the distribution. The shape parameter defines which distribution the Generalised Extreme Value distribution takes on:

- When the shape parameter ξ is equal to 0, the GEV is EVD Type I (EV1).
- When ξ is greater than 0, the GEV is EVD Type II (EV2).
- When ξ is less than 0, the GEV is EVD Type III (EV3).

Flood frequency analysis (FFA) is used as the basis to devise and implement strategies associated with mitigation and limitation of flood impacts. A key concept is the design flood peak value, typically set in national regulations by specifying an average recurrence interval, or return period, *T*, associated with an annual exceedance probability (AEP) of being exceeded in each year equal to *P = 1/T*. The *T*-year flood, in turn, is estimated based on the analysis of past floods, with the selection of a probability distribution to perform this inference using a sample with size *S* << *T* (Benson 1962) (i.e., *S* is very much less than *T*). In practice, due to the short period of observation record, large T-year flood (e.g., flood magnitude associated with 1000-

year return period) is estimated using a limited sample of historical floods, with S taken between 50 and 70 years.

Flood magnitudes are characterised by their Annual Recurrence Interval (ARI) or Return Period, which represents the average time, in years, between floods of a specific magnitude. Flood magnitudes can also be expressed as, and is defined by, an Annual Exceedance Probability (AEP) which is the probability a flood that size occurs in a given year (e.g., a 100-year ARI flood has a 1% AEP), with the AEP being associated with the GEV CDF.

Hydrological extremes, including estimation of the flood magnitude associated with PMP, are generated at 27 locations presented in Table 3-7 for the following ARI:

- Streamblank flood 1:2.33 year ARI- usually referred as the Mean Annual Flood (MAF);
- Common flood: 1:5 year ARI, 1:10 year ARI, 1:20 year ARI;
- Large floods: 1:50 years ARI, 1:100 year ARI and 1:200 year ARI;
- Extreme floods: PMF associated with PMP.

To ensure that the hydrological extremes are located on the digital river network used, those locations are mapped to the closest DN3 Strahler 3 river reach. It is important to note that the modelling chain bias correction aims to represent mid-range to low flow hydrological regimes. Bakker (2025) identified that the analysis of flood frequency requires bias correcting the CMIP6 modelling chain to match observed flood frequency using the TopNet hydrological model. This additional bias correction (i.e. in addition to the water resource bias correction carried out as part of this project) can be carried out by correcting the volume of precipitation and/or the infiltration excess runoff catchment scale parameters. As a result, the analysis presented in this section is illustrative of the impact of climate change on flood magnitude across the Southland region.

Table 3-7: **Hydrological extreme locations.**

Hydrological extreme location	Hydrological extreme location	Hydrological extreme location
Mataura Catchment to Cattle Flat Township	Waiau at Tutapere Township	Waikaia Catchment
Mataura Catchment to Gore Township	Oreti Catchment to Wallacetown Township	Whitestone Catchment
Mataura Catchment to Wyndham Catchment Township	Winton Stream to Winton Township	Mokoreta Catchment
Waikaia Catchment to Waikaia Township	Aparima River to Otautau Township	Makarewa Catchment
Waikaka Stream to Gore Township	Otautau Stream to Otautau Township	Mararoa Catchment
Oreti Catchment to Lumsden Township	Waihopai River to Invercargill City	Orauea Catchment
Oreti Catchment to Winton Township	Otepuni Stream to Iinvercargill City	Waimea Catchment
Oreti Catchment to Wallacetown Township	Kingswell Creek to Invercargill City	Waikawa Catchment
Winton Stream to Winton Township	Upukerora Catchment	Waituna Catchment
Aparima at Thornbury Bridge	Irthing Catchment	

Due to the relative short length of the annual time series, large uncertainties are expected for the estimation of the magnitude of the peak flow discharge associated with low frequency event (larger than 1:50 year or 1:100-year ARI) and its comparison with FF magnitude derived from observations. This is due mainly to two main reasons. Firstly, large statistical uncertainty is inherent with the characterisation of low frequency event. Secondly, the CMIP6 climate projections, while representing a step change in the reproduction of historical climate extreme indices (Gibson et al. 2024a), have a weak temporal correlation with observed climate that may translate to timing mismatch with observed historical events. As a result, caution has to be taken regarding the calculation of the magnitude of peak flow discharge, their interpretation and their use to inform planning decision process. As some locations of interest are not locations for which streamflow observation is available it is proposed to provide a comparison of the magnitude of the peak flow discharge with the national flood frequency tool (Henderson et al. 2018) for selected ARI. As the flood frequency tool does not provide estimation of change in the magnitude of flood peak discharge under climate change, the intercomparison will be provided only for the hindcast time slice.

Climate change impact on Flood Frequency magnitude (FF) is carried out by fitting GEV distribution, at each location of interest, of the annual maximum discharge time series under future climate projections. The analysis is completed using R (R Core Team 2021) using the ‘extRemes’ package (Gilleland and Katz 2016).

- Due to the relatively short length of each time slice considered (20 years), the analysis will be carried out for each time slice assuming stationarity of hydro-climate processes over the duration of the time slice.
- The annual FF is be completed for each GCM (20 years annual time series) and as pooled GCM (120 years annual time series assuming equiprobability between the six GCMS) to reduce uncertainty of estimations of flood magnitude associated with different annual exceedance probabilities. While results are available for each GCM driven historical simulations, only pooled GCM analysis is presented to reduced uncertainty.
- Analysis of the change in FF will be reported for each location and for each SSP:
  - Magnitude of flood peak discharge for 1 hr and 24 hr duration for each ARI and its associated 95th percentile confidence interval for each GCM and pooled GCM, associated with the estimation of the magnitude of flood peak discharge.
  - Magnitude of flood peak discharge for 24 hr duration for each selected ARI and its associated 95th percentile confidence interval as provided by the national flood frequency tool.

3.3.5 Probable Maximum Flood

As part of the project, at each location of interest PMF is defined as the PMF peak discharge associated with PMP event for each storm duration (Clavert-Gaumont et al. 2017, Rastogi et al. 2017, Sammen et al. 2022).

Estimation of the climate change impacts on Probable Maximum Flood (PMF) requires the combination of estimation of climate change impacts on PMP coupled with hydrological models (Clavert-Gaumont et al. 2017, Rastogi et al. 2017, Sammen et al. 2022). As part of the project, we are following a similar methodology by coupling PMPs (represented using rainfall design event approach for each duration of interest) calculated under current and future climate change scenarios with the hydrological model to calculate PMFs under current and future climate projections. The use of the hydrological model to calculate PMF enables us to simulate the impacts of increased warming on runoff generation processes, combined with a variety of initial conditions (including timing through the year).

The temporal hourly signature of the PMP for 24 hrs storm durations is identified for each GCM as the historical rainfall event generating the largest daily volume of precipitation. As part of the analysis, the following two assumptions are made:

- As PMP events are spatially very large rainfall events, it is assumed that all locations within the same surface water catchment will experience the PMP event over the same time (i.e., the data of the largest event does not change spatially across location within the same surface water catchment).
- Only the rainfall depth is affected by increase warming (i.e., the timing of the PMP storm does not change with warming).

3.3.6 Cryosphere locations

Snow fall volume, snow volume (simulated as snow storage in the TopNet model) and snow melt are generated using the cryospheric module associated with the TopNet hydrological model for 12 mountain ranges located across the Southland region (see Table 3-8). Due to difference in orientation, cryospheric analysis is presented based on the surface water catchment. As a result, a mountain range split between two Freshwater Management Units (FMU) may have different estimations based on their orientation.

Table 3-8: Mountain range information request.

Mountain range	Freshwater Management Unit
Eyre Mountains	Oreti and Mataura
Thomson Mountains	Mararoa and Oreti
Livingston Mountains	Maroroa and Waiau
Takitimu Mountains	Aparima and Waiau
Garvie Mountains	Mataura and Waikaia
Umbrella Mountains	Waikaia
Hunter Mountains	Waiau
Kepler Mountains	Waiau
Murchison Mountains	Waiau
Stuart Mountains	Waiau
Franklin Mountains	Waiau
Earl Mountains	Waiau

The estimation of two additional cryospheric variables is required by Environment Southland for those mountain ranges: number of snow days and snow line defined as follow:

- Snow days are identified as days when snow fall occurs and are identified as rainy days associated with air temperature lower than 1°C.
- Seasonal Snow line is defined as the elevation with an average annual snow cover duration of 90 days. For each catchment, the snow line elevation is determined by finding the annual average snow cover duration within each 100-metre elevation band, then interpolating between the mid-point elevations of the bands with the two closest snow cover durations (i.e. moving upwards, the last band with <90 days and the first band with >90 days). Snow cover duration is determined as the number of days with average fractional snow cover area >50% within each elevation band. The snow line elevation is found separately for each catchment, year and GCM simulation, then averaged for each time period (historical, mid-century, end-of-century).

- Snow volume is calculated as snow storage within the TopNet model and is represented as snow storage in kg/m^2 or meter snow water equivalent.

3.3.7 Modelling chain bias correction methodology

As the GCM simulations are 'free-running' (i.e., based only on initial conditions, and not updated with climate observations) comparisons between present and future hydrological conditions can be made directly (as each GCM is characterised by specific physical assumptions and parameterisation), but this also means that the variability simulated in hydrological hindcasts do not track observational records over the period 1995–2014. Furthermore, biases in the rainfall simulated by the models can combine with biases in the hydrological model to create flow-duration-curves that differ significantly from the observed.

Bias correction and calibration are procedures that involve adjusting a model's parameter values and inputs such that the model's outputs (i.e., simulations) more closely match historical measurements (i.e., observations). For TopNet in particular, despite the ability of the uncalibrated model to reproduce some observed hydrological characteristics across New Zealand as a whole (see Booker and Woods 2012, McMillan et al. 2016, Cattoën et al. 2022), model bias correction or calibration is recommended to match catchment-specific hydrological behaviour (Zammit 2019).

Calibration aims to achieve model simulations that provide the best overall fit to each individual observation, averaged across all observations. Goodness of calibration is often expressed through metrics such as root mean square error or R^2. Calibration is known to be very challenging for GCM-driven hydro-climate modelling, of the type undertaken in this study. As a result, the model outputs exhibit weak temporal correlation with weather observations and as such with hydrological observations. When applied to the hydro-climate modelling chain, Malek et al. (2022) found that common strategies for reducing errors in hydrologic projections can themselves strongly distort projected climate vulnerabilities associated with water resource management. As a result, calibration of GCM-driven hydro-climate simulations is location dependent, difficult to transpose to ungauged locations, computationally intensive and requires observation time series, and hence is overall generally difficult to achieve.

Bias correction aims for the model output to reproduce the long-term statistical characteristics of the observations (e.g., expressed as mean, variance, etc.), rather than their true individual values. Different bias correction techniques are available for both climate and hydrological modelling, including linear scaling, variance scaling, quantile mapping, and trend-preserving quantile mapping, with each method having its own strengths and weaknesses. Regardless of method used, bias correction is generally less computationally intensive than calibration.). Among those methods, quantile mapping has been found to be among the best bias correction techniques for catchment-specific hydro-climate modelling (Willkofer et al. 2018). However, when applying quantile mapping bias correction to climate simulations, Lafon et al. (2012) found it to be sensitive not only to the method used but also the baseline period. In addition, as a catchment-specific method, the application of quantile mapping bias correction is limited at larger scale and cannot be applied for catchments without any hydrological observations.

As part of the project, a bias correction of the rainfall for each GCM driven simulation was carried out, without undertaking a GCM specific calibration for each relevant streamflow observation. Due to its simplicity, the bias correction can be developed for locations at which streamflow time series observations are available, or for locations at which hydrological metrics are available.

Due to its conceptualisation, the bias correction targets long-term average hydrological characteristics, rather than event based hydrological characteristics that would be possible with calibration. It focusses on the following steps and is associated with surface water catchment conceptualisation of hydrological processes:

- *Correction of upstream precipitation volume*, ignoring internal variation in soil moisture and snow storages. For each site we calculate, over the period 1995–2014, the mean annual precipitation (referred as target precipitation hereafter) using mean annual runoff as calculated from the streamflow observation time series and the mean annual actual evapotranspiration. The bias correction of the precipitation is given by the ratio of the GCM-specific mean annual precipitation compared to the target precipitation. The accuracy of the bias correction is depending on the following: i) applicability of surface water conceptualisation to each subbasin, and ii) applicability of the long-term average actual evapotranspiration dataset (Tait et al. 2006, Woods et al. 2006) generated over the period 1966-2006 to the hindcast period (1995–2014).
- *Correction of the annual average upstream snow cover area*, as a proxy for the upstream snow volume. The correction is characterised by the annual average snow cover duration metric and analysis of catchment scale daily average fractional snow cover area derived from NASA's Moderate Resolution Imaging Spectroradiometer Snow Cover Area dataset (MODIS-SCA) (contact: Dr Pascal Sirguey, University of Otago) over the period 2000–2015.The bias correction assumed that the catchment average snow cover duration is primarily controlled by the precipitation volume and snow accumulation temperature threshold within the TopNet model. The cryospheric component of the bias correction aims to improve the representation of snow melt contribution to summer low flow.
- *Correction of the evapotranspiration parametrisation* to better represent low flow hydrological metrics and mean annual runoff across Southland. It is assumed that catchment-scale soil moisture and associated flow generation processes during summer months are mainly controlled by catchment-scale evapotranspiration processes which are characterised by the catchment-scale plant available water parameter within the TopNet model.
- *Correction of infiltration excess runoff parameterisation* to better represent low flow hydrological metrics across Southland. It is assumed that catchment-scale soil moisture and associated flow generation processes during summer months are partially impacted by infiltration excess runoff parametrisation that is characterised by the catchment-scale infiltration excess runoff parameter within the TopNet model. As part of the modelling chain bias correction, this parameter is set to be GCM dependant as it depends on GCM specific sub-daily rainfall intensity distribution that is independent of the hydrological model considered.

The bias correction is carried out at a spatial scale relevant to the climate projections using the following order:

- Bias correction of relevant hydrological characteristics at natural flow sites (i.e., streamflow stations for which impact of human activities is considered negligible) across the full flow range.
- Bias correction of relevant winter hydrological characteristics for flow sites whose flow regime is impacted by human activities, under the assumption that human impacts are minimal during the winter period and that little bias is present in the seasonal cycle of the precipitation (Cambell et al. 2024).
- Bias correction of relevant cryospheric characteristics for large mountainous regions within Southland using MODIS-SCA annual average Snow Cover Duration (SCD) and annual average daily fractional Snow Cover Area (fSCA) for mountain range.
- Bias correction of relevant hydrological metrics for ungauged locations where possible. Those metrics can be based on NZ River Maps hydrological metrics (Booker and Woods 2014, Whitehead et al. 2019). Booker and Woods (2014) dataset is used primarily to generate the upstream precipitation bias correction and (if possible) to provide an estimation of low flow metrics at locations without adequate observations (e.g., locations without streamflow observation, locations where streamflow observation is impacted by human activities, locations where streamflow observations are not long enough to be used as part of the bias correction process).

Figure 3-3 and Table 3-9 present the locations where flow characteristics, selected in collaboration with Environment Southland's Hydrology team, are bias corrected across the full hydrological regimes (annual hydrological regime for natural flow sites) and/or winter flow regime for winter flow sites. Site classed as "winter" site encompasses: i) streamflow observation site that is affected by upstream human activities (e.g., sites located downstream of hydro-electricity generation lakes, or by large surface water and/or groundwater takes activities), ii) streamflow gauging station at site classed as "Natural" impacted by regional groundwater movements, and iii) streamflow gauging stations classed as "Natural" missing at least 50% of daily average streamflow observations over the period 1995–2014.

Table 3-9: Physiographic information for bias corrected streamflow sites.

Freshwater Management Unit	Tideda ID	Site name	Hydrological regime Bias correction
Aparima	78906	Aparima Dunrobin	Natural
	78901	Aparima at Thornburry	Winter
Oreti	78606	Oreti at three Kings	Natural
Mataura	77563	Waikaia at Piano Flat	Natural
	77505	Mataura at Parawa	Winter
	77506	Mataura at Tuturau	Winter
	77519	Mataura at Seaward Downs	Winter
	77527	Mokoreta at McKays Rd	Winter
	77528	Waikaka Stm at Willowbank	Winter
	77561	Waikaia at Mahers Beach Rd	Winter
Waiau	80201	Rowallan Burn at Old Mill	Natural
	79740	Spey at West Arm	Natural
	79708*	Waiau at Queens Reach	Winter
	79737	Mararoa at Cliffs	Winter
Fiordland	83201	Lyvia Ck at Deep Cove	Natural

* Bias correction not carried out for Waiau at Queens Reach as not meeting the threshold of 50% of observations present over the period 1994–2014.

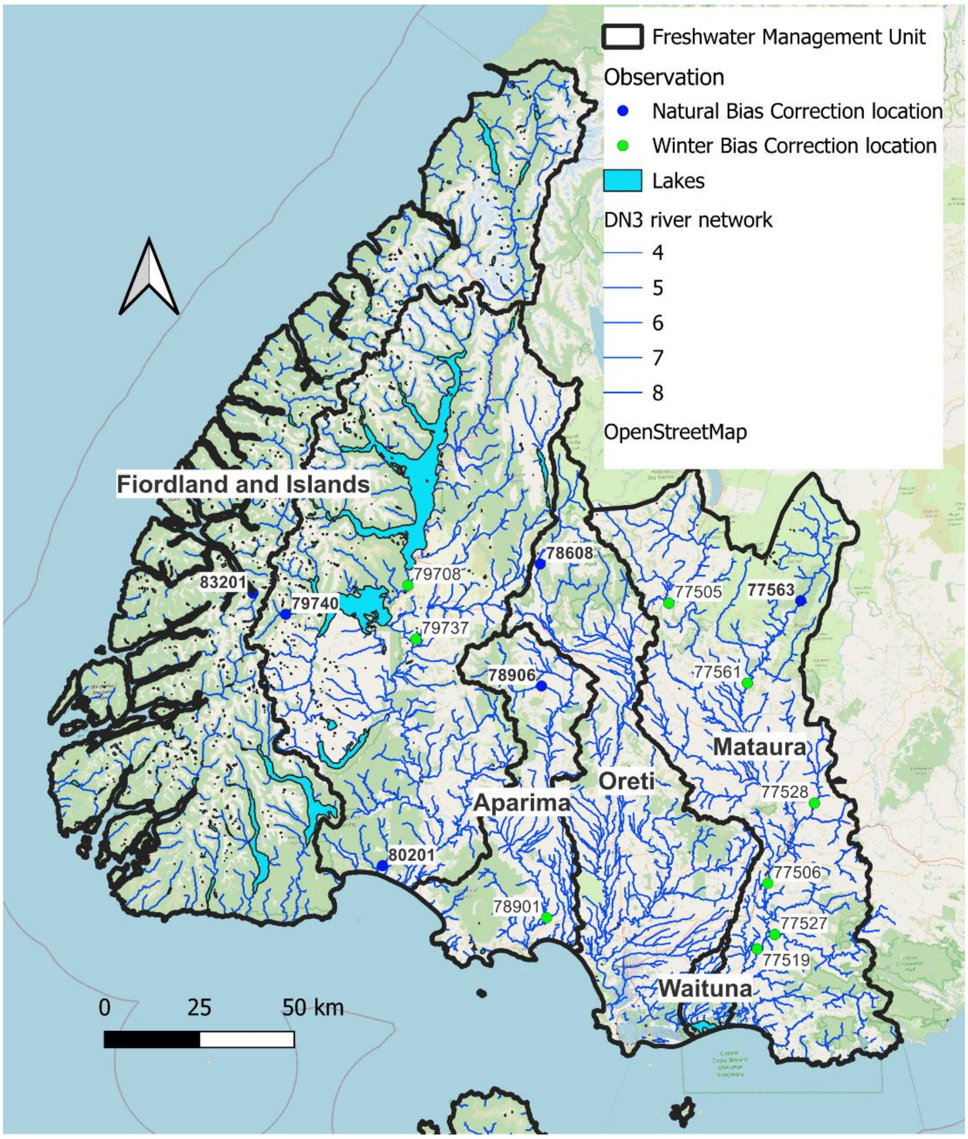

Figure 3-3: Location of the flow sites (represented by their Tideda ID) used as part of the modelling chain bias correction. Blue dots represent locations for which bias correction is carried out across the full hydrological regime ("natural flow sites"), while green dots represent locations for which winter hydrological regimes are bias corrected.

Following discussions with Environment Southland, the Southland region was disaggregated in 100 subbasins on which the bias correction is performed (Figure 3-4) with 32 subbasins located within the Mataura/Waituna FMUs, 8 subbasins located within the Aparima FMU, 22 subbasins located within the Oreti FMU, 25 subbasins within the Waiau FMU and 13 subbasins within the Fiordland FMU.

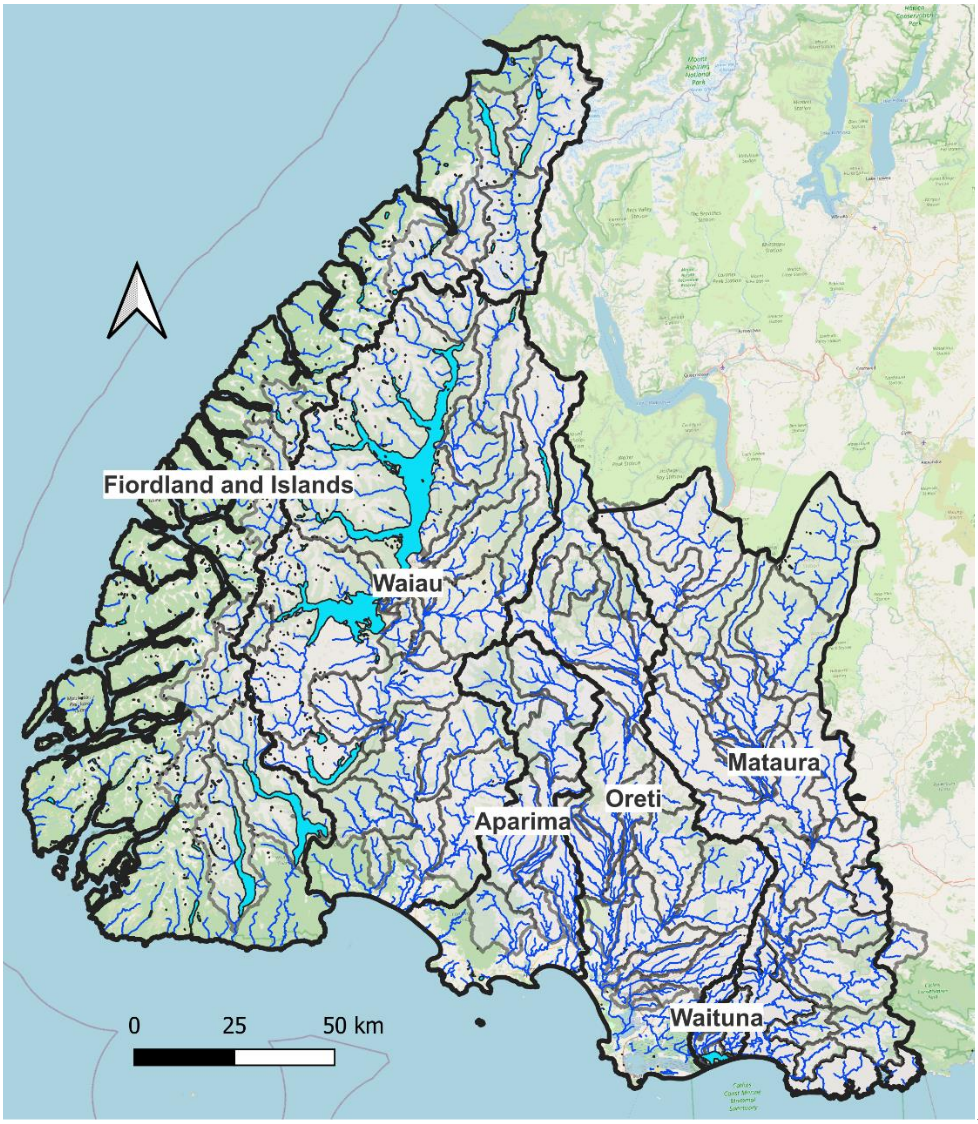

Figure 3-4: Bias corrected watersheds (dark grey outline) within each of Southland's FMU (black outline). Coastal watersheds (not shown on the map) are not bias corrected.

Validation of the bias correction is carried out as follows:

- At the outlet of each subbasin, the accuracy of the bias correction is carried out using Mean flow (QMean) and 7day Mean annual low flow (Q7DMALF). This is the primary objective of the bias correction, as estimation of QMean and Q7DMALF are available either from observation or from NZRiverMaps.
- At streamflow gauged locations, the accuracy of the bias correction is out as per Zammit et al. (2024), using the following nine statistical scores:
  - The accuracy of the calibration process is estimated in terms of the Nash-Sutcliffe efficiency coefficient calculated on the discharge (NS) and on the logarithm of the discharge (NS Log). The NS score represents a measure of the residual variance versus the data variance. A NS score of 1 indicates that the simulation perfectly mimics the observations in time and volume. A negative NS score indicates that the average of the observation is a better

predictor than the modelled flow. The NS score represents the ability of the model to mimic the observations during high flow periods, while the NS Log score represents the ability of the model to mimic the observations during low flow periods.

- KGE (Kling-Gupta Efficiency), KGElf (Kling-Gupta Efficiency for low values) and KGEkm (Kling-Gupta Efficiency with knowable-moment). KGE was developed to provide additional information that the Nash-Sutcliffe efficiency and to facilitate the analysis of the relative importance, in the context of hydrological modelling, of its different components (correlation, bias and variability). KGElf represents a standardised measure of the ability of a model to reproduce low flows (Garcia et al. 2017) and was developed to circumvent some issues associated with the use of NS Log to support water management decisions. KGElf is taken as the average of the Kling-Gupta Efficiency (Gupta et al. 2009) for the discharge and its inverse. KGEkm was developed by Pizarro and Jorquera (2024) to provide a reliable estimation and effective description of high-order statistics from typical hydrological samples and, therefore, reducing uncertainty in their estimation and computation of the KGE. A KGE/KGElf/KGEkm score value of 1 indicates that the simulation perfectly mimics the observations in time and volume. A KGE/KGElf/KGEkm score lower than -0.41 indicates that the average discharge is a better predictor of the observations than the simulation (Kling et al. 2012, Knoben et al. 2019).

- Percentage bias (PBIAS) measures the average tendency of the simulated values to be larger or smaller than their observed ones. The optimal value of PBIAS is 0.0, with low-magnitude values indicating accurate model simulation. Positive values indicate overestimation bias, whereas negative values indicate model underestimation bias.

- Pbiasfdc (PBIASfdc) measures the percent bias in the slope of the mid-segment of the flow duration curve (Yilmaz et al. 2008). The mid-segment is characterised by a percentage of exceedance between 20% and 70%. The optimal value of PBIASfdc is 0.0, with low-magnitude values indicating accurate model distribution of mid-flow range. Positive values indicate overestimation bias, whereas negative values indicate model underestimation bias.

- The Index of Agreement (D) (Willmot 1981) represents a standardized measure of the degree of model prediction error. A value of 1 indicates a perfect match, while a value of 0 indicates no agreement at all between observed and predicted data. Despite D being able to detect additive and proportional differences in the observed and simulated means and variances, it is extremely sensitive to extreme values due to its formulation (Legates and McCabe 1999).

- Volumetric Efficiency (VE) (Criss and Winston 2008) is used to circumvent some problems associated with NS. It varies between 0 and 1 and represents the fraction of water delivered at the proper time.

- At natural and winter flow sites, comparison of the average daily observed and predicted discharge (to identify potential issues in the reproduction of seasonality of the hydrological regime), daily observed and predicted flow duration curve (to identify potential mismatches in the statistical distribution of the flows) and cumulative flow (to identify potential issues related to systematic bias in the bias correction process).

Table 3-10 provides a qualitative description of the quality of the bias correction result based on the value of the statistical scores used.

Table 3-10: Classification of Nash-Sutcliffe, KGE, Pbias, D and VE scores.

Score	Classification
NS/NSLog/KG scores	
NS <0.4	Poor
0.4< NS < 0.6	Adequate
0.6 < NS < 0.8	Good
NS > 0.8	Excellent
PBIAS/ PBIASfdc scores	
|PBIAS|[4] < 10%	Very good
10% < |PBIAS| < 15%	Good
15% < |PBIAS| < 25%	Satisfactory
|PBIAS| > 25%	Poor
D score	
D < 0.5	Poor
0.5 < D < 0.8	Acceptable
D > 0.8	Good to Excellent
VE score	
VE <0.4	Poor
0.4 < V < 0.7	Acceptable
VE > 0.7	Good to Excellent

[4] |PBIAS| represents the absolute value of the PBIAS score.

It is important to note that the validation of the bias correction using statistical scores is dependent on the timing and seasonal behaviour of the GCM driven precipitation. While there is consistency across the GCMs regarding simulation of historical air temperature, this may not be the case regarding precipitation (Campbell et al. 2024). As a result, the use of conventional hydrological statistics to assess the accuracy of the bias corrected modelling chain may be limited at locations with large bias in seasonal cycle of precipitation (further details see Campbell et al. 2024, Figure 3-4).

As part of the bias correction process the following constraints have been imposed to each of the bias corrected parameters

- The value of the correction factor for upstream precipitation volume is limited to a range [0.25-2.5]. The range was determined based on previously completed TopNet calibration for the Aparima-Oreti and Mataura FMUs (Zammit et al. 2019, Zammit et al. 2020, Zammit et al. 2021) that indicated that the correction factor for upstream precipitation volume associated with the use of the observed derived daily gridded Virtual Climate station Network (VCSN) precipitation (Tait et al. 2006) varies between 0.6 and 1.8 across those three FMUs. As part of the investigations, the correction factor for the upstream precipitation is GCM specific.

  Correction factor value larger than 1.2 tends to indicate either a large underestimation in the GCM derived mean annual precipitation to generate long-term average discharge under a surface water conceptualisation (underestimation of the orographic effect on precipitation in catchment with large elevation change due to the spatial resolution of the CCAM model), or that the location where the bias correction is carried out is subject to regional groundwater influence (e.g. spring discharge within the subbasin, reaches within the subbasin can be classed as gaining reaches).

  Correction factor value smaller than 0.8 tends to indicate that either a large overestimation in the GCM derived mean annual precipitation to generate long-term average discharge under a surface water conceptualisation, or that the location where the bias correction is carried out is subject to regional groundwater influence (e.g., reaches within the subbasin can be classed as losing reaches), or could be associated with small basin that were generated due to river network junctions or location of gauging stations.

- The correction of the temperature of accumulation correction is limited to a range of [-5°C, 5°C]. The range of the temperature correction is based on analysis carried out as part of the Deep South National Challenge project "Snow, ice and glaciers in our changing climate" ([Snow, ice and glaciers in our changing climate | Deep South Challenge]) and is GCM specific. This project estimated that the annual average error in air temperature at high elevation could reach up to 5°C to reproduce 250 m resolution gridded derived MODIS-SCA annual average SCD in the Clutha catchment.

  The correction of the temperature aims to: i) minimise the error on the estimation of the annual average SCD by a maximum of 7 days, ii) reproduce the seasonal pattern of MODIS derived annual average SCA, iii) reproduce the timing of seasonal average start of the snow accumulation period as well as the timing of the seasonal average melt of the snowpack, iv) summer month seasonal average SCA as a proxy of percentage of the area covered by permanent snow.

- The correction of the evapotranspiration parametrisation is limited to a range [0.25–2.5]. The range was determined based on previously completed TopNet calibration for the Aparima-Oreti and Mataura FMUs (Zammit et al. 2019, Zammit et al. 2020, Zammit et al. 2021). Currently, the evapotranspiration parameter within the TopNet model is determined based on class 1 land cover classification (Newsome 2000). As such, the correction aims to represent the adaptation of the vegetation to Southland climate and weather patterns, which is GCM specific.

  The correction of the evapotranspiration aims to better represent low flow hydrological metrics while minimising its impact on mean annual runoff across Southland. It is important to note that, under a surface water conceptualisation, low flow estimation is largely controlled by GCM specific seasonal rainfall patterns. As such, the current simple bias correction process may have limited impact on the reproduction of low flow metrics.

- The correction of infiltration excess runoff parameterisation is limited to a range [0.5–9]. This range was determined through experiment as part of the project.

  The correction of the infiltration excess runoff parameterisation is carried out at the FMU scale and is GCM specific (as associated with the sub-daily rainfall distribution that is specific to each GCM). Its optimisation is a trade-off between the ability of the simulations to reflect high flow metrics and hydrological statistics, and the ability of the model to represent low flow metrics based on analysis of Flow Duration Curve (FDC).

As part of the project, TopNet has been run continuously from 1980 to 2099, with a spin-up period 1980–1994 excluded from the final analysis. However, as the hydrological model used is uncalibrated, it is recommended that only the change analysis be used to inform discussions (Collins and Zammit 2016, Collins et al. 2019).

As riverine water stress period occurs over the summer period, climate change impact assessment on annual and seasonal hydrological metrics will be carried out using hydrological year defined as the period between 1 June and 31 May. Climate change impact assessment on annual and seasonal cryospheric metrics are carried out using snow year defined as the period between 1 April and 31 March.

3.4 Stream temperature

3.4.1 Stream temperature observations

Water temperature data, and concomitant river flow data where it was available, were obtained from Canterbury (40 locations), Taranaki (19 locations), and Southland (20 locations) regional councils and from NIWA. Environment Southland water temperature observation meta data is presented in Table 3-11 and mapped on DN3 river network (Figure 3-5).

Table 3-11: Water temperature and flow observation across the Southland region.

Site ID	name	Start Observation	End Observation	Concurrent Flow
77502	Mataura at Wyndham	22-Aug-06	16-Jul-18	No
77504	Mataura at Gore	26-Oct-99	1-Jan-25	Yes
77505	Mataura at Parawa	26-Aug-98	1-May-03	Yes
77506	Mataura at Tuturau	8-Dec-96	29-Apr-09	Yes
77519	Mataura at Seaward Downs	20-Feb-98	22-Mar-17	Yes
77525	Waimea Stream at Mandeville	30-Sep-99	1-Jan-25	Yes
77528	Waikaka Stream at Willowbank	27-Jun-00	10-Jun-19	Yes
77563	Waikaia at Piano Flat	3-Nov-95	14-Feb-25	Yes
77583	Oteramika Stream at Seaward Downs	9-Jan-96	28-Apr-03	Yes
78601	Oreti at Wallacetown	1-Oct-97	1-Nov-13	Yes
78608	Oreti at Three Kings	4-Sep-98	1-Jan-25	Yes
78625	Otapiri Stream at Otapiri Gorge	1-Jul-98	26-Nov-24	Yes
78634	Makarewa at Counsell Road	20-May-99	1-Jan-25	Yes
78636	Oreti at Lumsden Cableway	25-Aug-98	6-Jan-25	Yes
78637	Irthing Stream at Ellis Road	26-Aug-98	1-Jan-25	Yes
78901	Aparima at Thornbury	26-Jul-05	1-Jan-25	Yes
78906	Aparima at Dunrobin	30-Aug-00	1-Jan-25	Yes
78912	Hamilton Burn at Waterloo Road	9-Jul-98	1-Jan-25	Yes
79701	Waiau at Tuatapere	19-May-16	1-Apr-25	Yes
79735	Waiau at Sunnyside	28-Sep-00	1-Jan-25	Yes
789052	Pourakino at Pourakino Valley	2-Jul-04	3-Feb-25	Yes

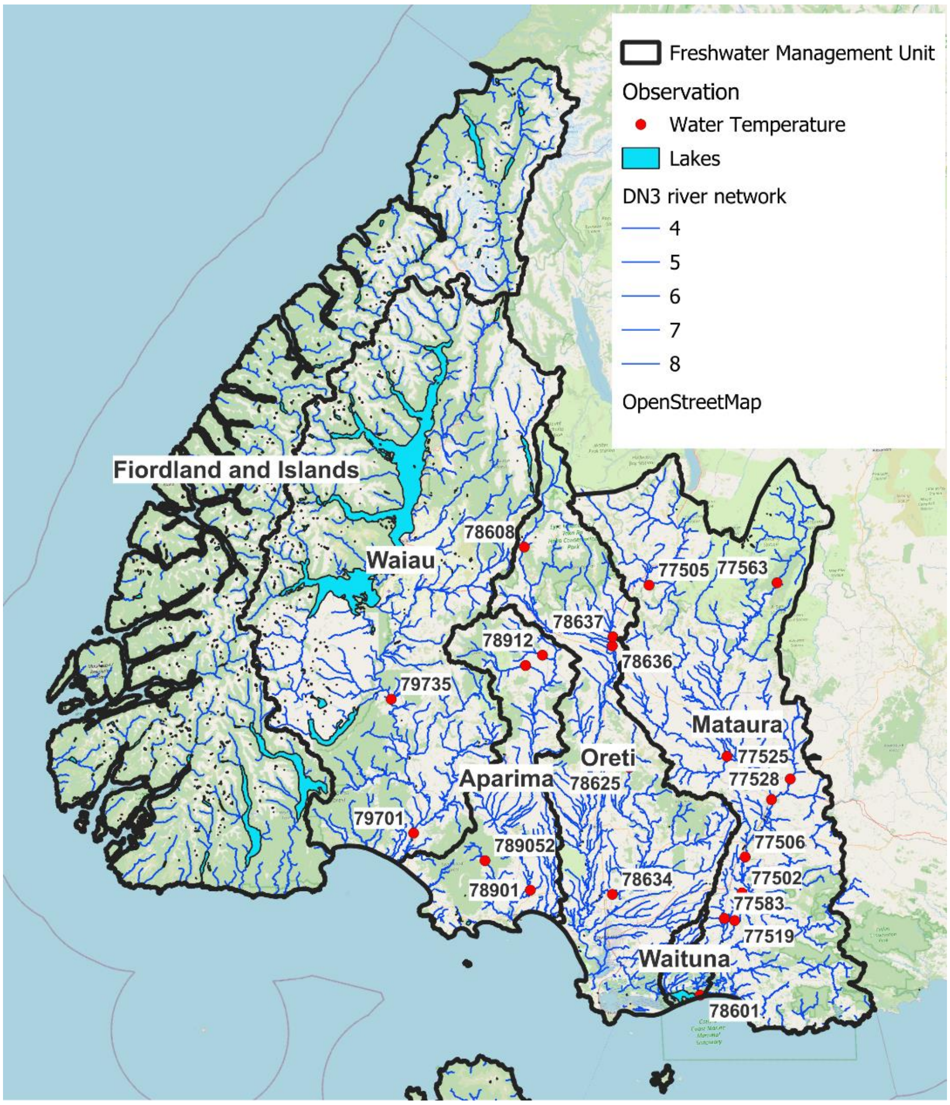

Figure 3-5: Stream water temperature observation location. Water temperature gauges identified using their Tideda ID.

Mean daily water temperature was calculated for each day at each site by averaging all observed water temperature values (e.g., hourly, 15-minutely, irregular) at each site on each calendar day. For each site, mean daily river flow was calculated by averaging all observed river flow values within each calendar day.

Local and average upstream mean daily air temperature time-series were obtained from the Virtual Climate Station Network (Tait et al. 2006) on each day at each site, by averaging maximum daily air temperature and minimum daily air temperature. For each site, the local mean daily air temperature was interpolated by applying a weighted average over the four nearest VCSN grid points. The same interpolation method was applied to obtain the air temperature at the mid-points of all upstream segments. Air temperature over the entire upstream catchment was then calculated by applying an area-weighted average to all upstream air temperatures (but these upstream air temperature data were not used in this analysis).

It is important to note that water temperature is modelled on the DN2.4 river network that is at a coarser resolution than DN3.1 used for hydrological modelling purpose. This is due to the fact that at the time of the project, extrapolation REC classes (Snelder and Biggs 2002) have been generated for DN2.4 digital river network and associated Rec watersheds.

3.4.2 Stream temperature model

The following four stream temperature models have been developed to estimate stream temperature at location of stream temperature and/or flow sites:

- Daily stream temperature as local daily air temperature (control model).
- Daily Stream temperature predicted from local daily air temperature on the same day.
- Daily stream temperature predicted from local daily air temperature on the same day and the difference in air temperature and the average of the preceding three days temperature.
- Daily stream temperature from local daily air temperature and daily river flow.

For validation purpose, air temperature was estimated for the daily minimum and maximum gridded air temperature generated as part of the Virtual Climate Station Network (VCSN, Tait et al. 2006) dataset. The VCSN is a five kilometres resolution daily gridded climate product generated based on daily climate observation directly ingested in NIWA climate database (CliDB).

Validation across all sites (regardless of region) shows that at the daily resolution the addition of flow improved predictive performance but also introduced a bias. The addition of antecedent air temperature improved predictive performance more than the addition of flow. Regional analysis indicated that for Southland, the model that incorporated some antecedent information about air temperature performed best across the 20 sites.

The parameters of the stream temperature model for Southland were modelled as a function of landscape-scale watershed characteristics in order to produce a landscape-scale model that could predict water temperature at ungauged locations. The landscape-scale predictor variables chosen, described hereafter, have some physical basis for influencing the relationship between river water temperature and air temperature (and flow where relevant):

- LocHab: an ordinal estimate of habitat type (influence water depth and therefore water volume).
- segSolarRadSum: an estimate of summer solar radiation (influence energy input to the river water from the sun directly).
- segAveElev: local elevation (influences slope and therefore hydraulics, but also the range or air temperatures, possible link to influence of snow).
- TURB_Median: an estimate of water clarity defined as turbidity (influences albedo of water and therefore transfer of solar radiation to heat, may also have a link to hydraulic conditions).

As part of the climate change impact assessment on stream temperature, we assume that landscape predictors are independent of future warming conditions.

3.5 Ocean modelling

Oceanic data for this report is based on hydrodynamic modelling using New Zealand Earth System Model (NZESM). This Coupled Model Inter Comparison Project -class model uses the Unified Model to simulate the atmosphere, NEMO for the ocean, Medus for the ocean-biogeochemistry and CICE for sea-ice. NZESM has been tuned for New Zealand applications by refining the oceanic model mesh from 1° to 0.2° around New Zealand, allowing for a better representation of fine scale oceanic processes (e.g., coastal currents as per Behrens et al. 2020).

A set of 3 historical simulations have been conducted covering the period from 1940 to 2014, which were initialized from simulations of United Kingdom Earth System Model. Three future scenarios branched off from the historical simulation in 2015, following the SSP 2.6, SSP 4.5 and SSP 7.0, Greenhouse gas emissions specifications. These future climate change scenarios cover the period 2015 to 2100. For this report sea surface temperatures and pH have been extracted and analyzed. Marine heatwave metrices (MHW intensities and MHW days) have been computed using 5-daily SST temperature fields using the widely used definition as per Hobday et al. (2016).

3.6 Sea-level analysis

3.6.1 Gauge Data

Two tide-gauge records from Port of Bluff and Stead Street Bridge (Invercargill) were used in this report to summarise mean sea-level trends and extreme storm-tide levels (Figure 3-6).

Figure 3-6: Locations of the two tide gauges used in this report. Stead Street is operated by Environment Southland, Port of Bluff by South Port.

The >30-year tide-gauge data from the Port of Bluff (Southport) was obtained from Land Information NZ (LINZ) sea-level archive and quality-controlled for any major glitches and gaps focusing on extreme storm-tide events. The Bluff sea-level recorder runs from 12 December 1982 and has a recording duration of 32-years until the end of 2024. Initially, the sampling interval was 1-hourly until 2010, then it changed to 5 minutes until 2011, and since then it has been sampling at 1-minute intervals. The complete sea level record includes 81 gaps larger than 24 hours, with the longest gap spanning 1661 days from 1 July 1989. The full QA'd Port of Bluff sea-level record in Figure 3-7 shows no apparent issues with datum or offset.

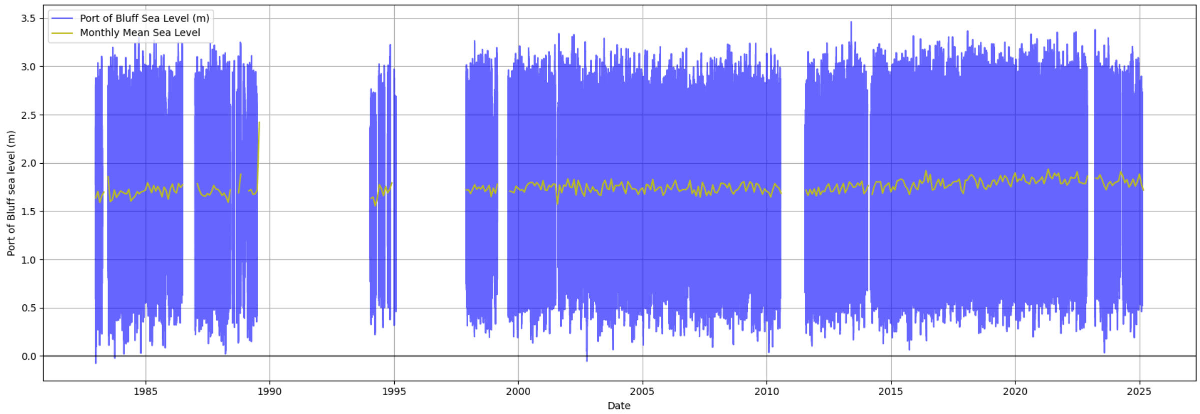

Figure 3-7: Quality assured Bluff sea-level data. Blue line sea level (m) relative to BVD-55 where gauge zero is 1.611 m. Black line represents gauge zero relative to local vertical datum. Yellow line monthly mean sea-level.

The Stead Street sea-level record, obtained from Environment Southland, spans approximately 32 years, beginning with the installation of the gauge on 24 March 1992 and continuing to the present. Over this period, several changes in sampling frequency have occurred. Initially, sea-level measurements were recorded at 15-minute intervals until March 1999, when the sampling frequency increased to 10-minute intervals. In February 2000, the recording interval reverted to 15 minutes before being further refined to 5-minute intervals in September 2000. From August 2010, the gauge initially recorded data at 1-minute intervals; however, this was later reduced to 10-minute intervals from 20 June 2013. On 9 October 2019, the sampling frequency was increased again to 5-minute intervals, which remains the current recording resolution.

Despite these changes in sampling frequency, the record is generally continuous. The most significant gap in the record lasts 37 days, beginning on 5 October 1993. Figure 3-8 presents the complete quality-assured (QA'd) Stead Street sea-level record. There are no apparent offset changes evident in the record, nor are there any documented changes to the gauge setup that would indicate systematic shifts in the measured sea level..

The Stead Street tidal record is strongly influenced by river flow, leading to a tidal signal that is noticeably skewed. This freshwater input reduces the overall tidal range at Stead Street compared to the more marine-dominated conditions observed at the Port of Bluff. As a result, the difference between high and low tide is less pronounced. Additionally, the tidal residual at Stead Street exhibits distinct peaks, which are primarily driven by freshwater flood events. These peaks reflect the impact of river discharge on local water levels, further complicating the tidal dynamics in this area.

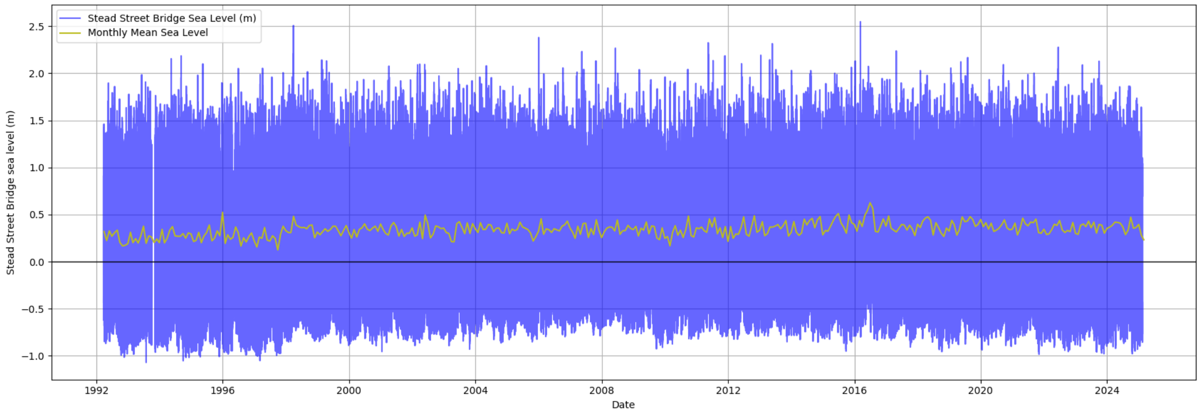

Figure 3-8: Quality assured Stead Street sea-level data. Blue line sea level (m) relative to BVD-55. Black line represents gauge zero relative to local vertical datum. Yellow line monthly mean sea-level.

3.6.2 Tides, sea level rise, change in extreme storm tide

Tidal elevations were predicted using tidal harmonic analysis, which was applied to each sea-level record.

Tidal harmonic analysis was undertaken using Versatile Harmonic Tidal Analysis (Foreman et al. 2009), applying a signal-to-noise ratio (the ratio of the tidal variance to the non-tidal variance) of 10.

Timeseries of tidal predictions were created by fitting the tidal harmonics and predicting the tides on an annual basis. The tidal predictions are more accurate when resolved on an annual basis, since the tidal harmonic constituents change subtly from year to year. Long-term average tidal harmonic constituents (e.g., M_2, N_2 and S_2 in Table 7-1) were calculated by applying the harmonic analysis to the whole record (epoch-averaged constituents).

High-tide exceedance curves for the present day were generated for both the Bluff and Stead Street records. These curves quantify the distribution of all high tides based on tide predictions spanning several decades (excluding weather and climate influences). They were then be used to assess how the frequency of high tides exceeding specific thresholds changes under three example sea-level rise (SLR) scenarios: 0.2 m, 0.4 m, and 0.8 m.

All sea-level records exhibit a rising trend in MSL. Tides and surges all ride on top of the MSL. To analyse changes in storm tides and surge through time, the underlying sea-level rise trend must first be removed. There are two main causes of rising MSL through time:

1. Absolute (or eustatic) rise in ocean levels around New Zealand.
2. Relative (or local) sea-level rise, which is the net rise from absolute, regional-sea offsets and local vertical land movement, measured relative to the local landmass.

The sea level gauges sit on the land surface, so they are recording relative sea-level rise (RSLR). It is RSLR that is observable through time in Figure 3-7 and Figure 3-8.

To detrend the datasets, linear fits to the relative annual MSL were calculated using a least-squares analysis (Table 3-12). These linear fits were subtracted from the non-tidal residuals before undertaking further analyses.

We note that the trend for Stead Street, calculated in Table 3-13, as not used in the analysis. This is because Stead Street is influenced by freshwater inflows, and the observed trend is as likely to reflect increased storminess and freshwater inflows as it is mean sea-level changes.

Table 3-12: Linear fits for sea level detrend.

Sea Level record	Linear detrend (mm/year)	Zero axis crossing date for linear fit detrending (MVD)
Bluff	1.78	7 December 1921
Stead Street	2.18	28 June 1848

Earth Science New Zealand assessed the local variations in sea-level to apply New Zealand-wide sea-level rise (SLR) scenarios to 2200. These scenarios are based on the most recent global SLR projections issued by the IPCC in 2021 (Fox-Kemper et al. 2021). These projections use updated shared socio-economic pathways (SSPs), which incorporate socioeconomic assumptions and emissions trajectories. The updated SSPs are outlined in the **MfE Coastal Hazards and Climate Change Guidance** (Ministry for the Environment 2024). The scenarios cover a wide range of plausible societal and climatic futures, from a 1.5°C 'best-case' low-emissions scenario (SSP1-2.6) to a high-emissions scenario (SSP5-8.5) with over 4°C of warming by 2100 (Chen et al. 2021). Even under the low-emissions scenario (SSP1-2.6), average SLR around Aotearoa could exceed 1 m shortly after 2200 (Table 6).

Table 3-13: Estimated years when absolute sea-level rise (SLR) thresholds may be reached based on recommended projections for a central location in Aotearoa New. Adapted from the MfE Coastal Hazards and Climate Change Guidance.

SLR (metres)	Year achieved for SSP5-8.5 H+ (83rd percentile)	Year achieved for SSP5-8.5 (median)	Year achieved for SSP3-7.0 (median)	Year achieved for SSP2-4.5 (median)	Year achieved for SSP1-2.6 (median)
0.2	2035	2040	2045	2045	2050
0.3	2050	2055	2060	2060	2070
0.4	2055	2065	2070	2080	2090
0.5	2065	2075	2080	2090	2110
0.6	2070	2080	2090	2100	2130
0.7	2080	2090	2100	2115	2150
0.8	2085	2100	2110	2130	2180
0.9	2090	2105	2115	2140	2200
1.0	2095	2115	2125	2155	>2200
1.2	2105	2130	2140	2185	>2200
1.4	2115	2145	2160	>2200	>2200
1.6	2130	2160	2175	>2200	>2200
1.8	2140	2180	2200	>2200	>2200
2.0	2150	2195	>2200	>2200	>2200

The records were analysed for annual exceedance probabilities (AEP) of storm-tide levels (relative to Bluff Vertical Datum-1955) up to potentially 1% AEP. In parallel, extreme storm-tide level results were also available from a project completed by NIWA for Invercargill City Council on a storm-tide analysis of the Stead Street gauge in Invercargill. These updated AEP storm-tide levels are then available to the councils for improving design levels for coastal stopbanks, roads, stormwater and drainage systems and assessing coastal flood risks.

High-tide exceedance curves for present day were produced from the Bluff record and Stead Street (Invercargill), which can be used to quantify the distribution of all high tides from tide predictions over many decades (excluding weather and climate effects) and then assess the change in these high-tide markers being exceeded by high tides with three example values of SLR (0.2, 0.4 and 0.8 m).

3.6.3 Storm tide extreme value analysis

Storm-tide is defined as the sea-level peak reached during a storm event, from a combination of **tide + storm surge** (Figure 3-9)**.** It is the storm-tide that is primarily measured by sea-level gauges such as the Port of Bluff and Stead Street gauges analysed here.

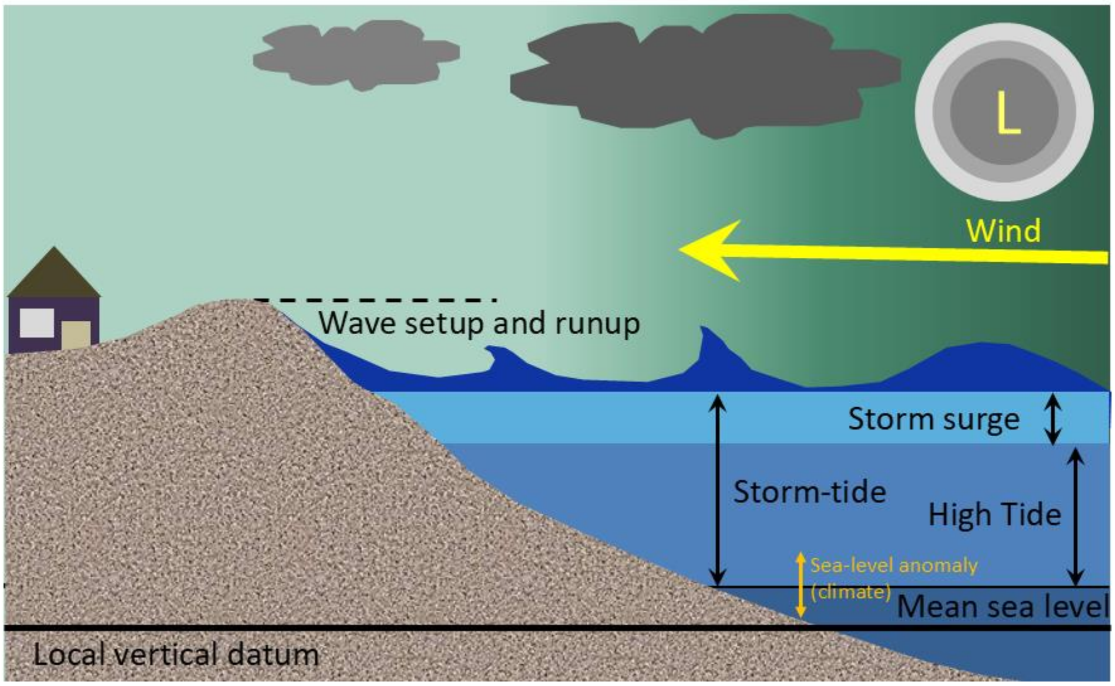

Figure 3-9: Components that contribute to storm-tide and wave overtopping. L= low-pressure weather system.

We subsampled all sea-level data to 1 h intervals after first applying a 15 min running average to minimise the effect of far infragravity and tsunami waves. The data were linearly detrended to a zero-mean relative to local vertical datum for each gauge, to remove the effects of historical SLR from the sea-level distribution and create a quasi-stationary time series required for extreme value analysis (Coles 2001). Therefore, here we define the term **"storm tide"** as the total water level above BVD-55 from the sum of tide plus storm surge.

The extreme value analysis was applied to high water levels, with mean sea-level adjustments incorporated where appropriate. The time series was declustered to isolate statistically independent peak events—defined as local maxima separated by at least three days and with a storm-tide level. These peaks were then analysed using a univariate approach, primarily fitting a Generalised Pareto Distribution (GPD) to values exceeding the 95th percentile threshold. This approach enables the estimation of return levels (with 95% confidence intervals) for the storm-tide itself, without requiring assumptions about the dependence structure between separate processes like surge and tide. By focusing on the storm-tide as an integrated variable, the analysis provides a direct assessment of the water levels most relevant to coastal inundation and stop bank performance under extreme conditions.

For the Bluff site, the GPD fit resulted in a location parameter of 1.44, scale of 0.145, and shape of –0.316.

In parallel with the GPD method, a Generalised Extreme Value (GEV) distribution was fitted to the annual maxima series at each site. The GEV approach uses block maxima (in this case, one annual maximum per year) and provides an independent validation of the return level estimates derived from the GPD method. Both methods are commonly used for extreme value modelling, and they generally yield similar results. For short records, the GPD makes use of more of the data so can provide better confidence in the fits and reliably extrapolate up to 10 times the record length, at the expense of requiring subjective choice of sampling thresholds. The GEV fitted to annual maxima overcomes the need to choose a threshold but is wasteful of data because it only uses the largest event within the calendar year so is best used on long records—it can reliably extrapolate to return periods of 3–5 times the record length.

At the Stead Street site, the GPD fit did not perform well, likely due to the limited number or quality of peak events above the threshold. As a result, a GEV analysis was applied instead, using annual maximum storm-tide levels. This alternative provided a more stable and representative fit for estimating extreme return levels at this location. The Stead Street GEV fit had a location parameter of 1.98, scale of –0.028, and shape of 0.098. By employing both GPD and GEV models, the analysis ensures robustness and flexibility in accounting for site-specific data limitations and behaviours.

Once fitted, the extreme-value models can be checked by plotting them against the measured maxima, assuming that the maxima follow a double-exponential (Gumbel) distribution (Gringorten 1963). The maxima should lie within the confidence intervals of the fitted model.

There are two commonly used ways to express the likelihood of occurrence of an extreme sea level:

- Annual exceedance probability (AEP) – The probability of a given (usually high) sea level being equalled or exceeded in elevation, in any calendar year. AEP can be specified as a fraction of 1 (e.g., 0.01) or a percentage (e.g., 1%).
- Average recurrence interval (ARI) – The average time interval (ideally averaged over a long time period and many "events") that is expected to elapse between recurrences of an infrequent event of a given large magnitude (or larger). A large infrequent event would be expected to be equalled or exceeded in elevation, once, on average, every "ARI" years. The term return period is often substituted for ARI.

3.7 Coastal risk exposure

Coastal flood risk in this study is the frequency and magnitude of property exposure to flood waters. We analyse flood risk using the RiskScape multi-hazard model framework (Paulik et al. 2023a), which is a modular and configurable system designed to evaluate coastal flooding exposure and impacts from extreme sea levels (ESLs) and sea level rise (SLR) under current and projected climate conditions.

3.7.1 Hazard maps

Spatiotemporal maps of episodic flooding from extreme sea-levels (ESLs) for present-day and higher sea levels were obtained from Paulik et al. (2023b). ESL flooding maps represent a 100-year annual recurrence interval (ARI) event i.e., 1% annual exceedance probability (AEP). The mapping process involved extracting digital elevation model (DEM) raster cells situated below ESL and elevations using a static inundation mapping technique (Stephens et al. 2021). A comprehensive national DEM for coastal regions (Paulik et al. 2020, 2021), up to an elevation of 20 m above present-day mean sea levels, was established by amalgamating LIDAR DEMs resampled to a 10-meter resolution and employing a fully convolutional neural network (FCN) model to rectify vertical biases in the Shuttle Radar Topography Mission (SRTM) data (Meadows and Wilson 2021). Flood depth calculations were performed by computing the disparity between ESL water surface heights and land elevations for DEM grid cells. To ensure accuracy, only grid cells with a hydrologic connection to coastlines were considered, thereby minimizing the risk of overestimating inundation extents. Additionally, topographic protection structures like levees were identified from aerial imagery and incorporated into the analysis, albeit without detailed design-level information. Consequently, land protection was assumed up to ESLs corresponding to a 100-year recurrence interval at present-day mean sea level (MSL). This was consistent with statutory flood hazard risk management directed by the New Zealand Coastal Policy Statement (Department of Conservation 2010), which requires regional and local authorities to avoid increasing the risk of social, environmental and economic harm from coastal hazards, over at least a future 100-year timeframe. Future ESL flooding maps were generated incrementally for MSL increments ranging from 0.1 to 2 m, regardless of future projections of RSL rise.

New Zealand RSL projections for Shared Socioeconomic Pathways (SSPs) were obtained from probabilistic RSL projections estimated using the Framework for Assessing Changes to Sea-level (FACTS) from the IPCC Assessment Report 6 (Levy et al. 2023; Naish et al. 2024). These projections, incorporating VLM effects, were based on interferometric synthetic aperture radar (InSAR) data calibrated with campaign and continuous Global Navigation Satellite System (GNSS) measurements. In this study, we investigate land movement effects on flood risk timing and magnitude by selecting RSL projections with and without local VLM for SSP 1 to 5 over 100 years (2020–2120) relative to a 1995–2014 baseline period (0 m at year ~ 2005). RSL projections from Naish et al. (2024) were obtained for 7435 coastal sites from https://searise.takiwa.co/. Medium confidence climatic processes for each SSP scenario were represented by RSL curves, which were spatially joined to flood-exposed properties to evaluate the future timing of expected exposure to flood waters.

3.7.2 Exposure maps

Southland property information represented in the national district valuation roll (DVR) was obtained from Environment Southland via South Territorial Authorities. The DVR contains a representation of the boundaries of all known properties in Southland. Property features (i.e., vector polygons) are represented as land parcels and contain identifier information about the property unit and other property perspectives. Information on property ‘zoning’, ‘actual use’, ‘capital value’, ‘improvement value’ and ‘land value’ is reported in this study along with property location based on the corresponding territorial authority.

3.7.3 Exposure calculation

Southland property risk to coastal flooding was evaluated using RiskScape software (Paulik et al. 2023a). RiskScape supports a flexible modelling engine for multi-hazard risk analysis that is centred on a well-established conceptual framework for risk quantification:

$$R = f(H_i, E) \tag{3}$$

where, risk (R) is a function (f) of the consequences from hazard (H) event i exposing a property (E) to flood waters. Property exposure to flood waters ($\mathrm{F_E}$) is determined when:

$$F_E = \begin{cases} 1, \ WD < 0.05 \text{ m} \\ 0, \ WD \geq 0.05 \text{ m} \end{cases} \tag{4}$$

where, water depth above ground (WD) exceeds a limiting depth of 0.05 m. The limiting depth value accounts for the potential uncertainty of water presence at the flood map grid cell, with water depths >0.05 m indicating a high likelihood of flooding. Southland properties corresponding to $F_E = 1$ are then enumerated at regional and territorial authority levels for each ESL and RSL scenario.

4 Present-day and future climate of Southland

Southland is both the most southern and western part of New Zealand and generally the first to be influenced by weather systems moving onto the country from the west or south. It is well exposed to these systems, although western parts of Fiordland are sheltered from the south and the area east of the western ranges is partially sheltered from the north and northwest. The Southland region is in the latitudes of prevailing westerly winds, and areas around Foveaux Strait frequently experience strong winds, but the winds are lighter inland. Winter is typically the least windy time of year, as well as the driest time of year for many areas.

The western ranges, with annual rainfall exceeding 6,000 mm in some parts, are among the rainiest places on Earth. In contrast, the drier eastern lowlands and hills receive annual rainfall totals predominantly between 500–1,000 mm. Dry spells of more than two weeks occur occasionally. Temperatures are on average lower than over the rest of the country, with frosts and snowfalls occurring comparatively frequently each year. On average, Southland receives less sunshine than the remainder of the country. For more information about Southland's present-day climate other than the information presented in this report, see Macara (2013).

The future climate of Southland will be influenced by a combination of the effects of anthropogenic climate change, plus the natural interannual and interdecadal variability resulting from phenomena such as the El Niño-Southern Oscillation (ENSO), the Interdecadal Pacific Oscillation (IPO), and the Southern Annular Mode (SAM).

5 Climate change impact by industry sectors

5.1 Council Infrastructure

5.1.1 Roads and highway infrastructure

One of the main considerations for roading is a significant decline in frost days, with a potential decline of up to 40 fewer frost days per year by 2090. While this may reduce some winter maintenance costs, higher temperatures and more frequent heatwaves could cause problems for road surfaces, including softening and damage to bitumen.

Southland's transport network is also vulnerable to flooding and storm damage. Projected increases in heavy rainfall, including more very wet days and higher extreme daily and five-day totals, and flood magnitude will raise the risk of bridge and road washouts, particularly where infrastructure is close to rivers such as the Mataura, Oreti, and Aparima.

In addition, coastal roads will face increasing challenges from sea-level rise combined with storm surge, particularly in low-lying urban areas such as Bluff and Riverton, and local flooding. Maintenance and adaptation planning will be needed to manage the growing risks to both inland and coastal transport corridors.

5.1.2 Potable water supply

Much of Southland's potable surface water is sourced from the region's four main rainfall fed rivers: the Waiau, Aparima, Oreti, and Mataura. Increases in the mean annual rainfall and streamflow may make riverine supply more reliable. However, flows into **meltwater-fed lakes** such as **Manapōuri and Te Anau**, which may become less reliable as climate change drives higher temperatures and snowpack losses.

Due to projected reduction in annual average land surface recharge (mainly in the lower Oreti and Mataura FMUs), potential drinking groundwater source in those areas may be increased stress under future warming. While land surface recharge is projected to increase across all shallow aquifers during winter, large decreases are projected during summer months (and a lesser extent spring) potentially exacerbating recently on-going water scarcity in those regions.

At the same time **higher temperatures, more frequent heatwaves, and urban growth** could increase demand for drinking water. And more intense rainfall could raise the risk of contamination of potable water stores. The potential quality and quantity of potable water is impacted by Southland's future climate.

5.1.3 Stormwater and wastewater

Stormwater and wastewater systems in Southland are particularly vulnerable to climate change because their discharge points are typically located at the **lowest elevations of populated areas**. Projected increases in **extreme rainfall intensity and multi-day rainfall totals** could readily exceed current design capacities, leading to system overflows and localised flooding. This risk is especially relevant for towns like **Invercargill, Gore, and Riverton**, where infrastructure is close to rivers and estuaries.

In **low-lying coastal areas**, where groundwater levels are closely linked to the sea, **sea-level rise** could further reduce the performance of both stormwater and wastewater systems. Rising seas combined with storm surges could increase the likelihood of backflow and reduced drainage capacity.

In urban areas climate change could impact the handling of sewage sludge with increased maximum temperatures combined with increase in green-waste volume (due to increase in favourable growing conditions).

Southland's wastewater systems are more likely to face **persistent increases in inflows and infiltration**, compounding the challenge of managing higher peak loads during extreme rainfall events.

Table 5-1: Infrastructure summary table.

Climate Change Driver	Risks	Opportunities
Decline in frost days (up to –40 by 2090)	Reduced frost heave but potential shift in road maintenance priorities	Lower winter maintenance costs; safer winter travel conditions
Higher temperatures & more heatwaves	Softening and rutting of roading materials; heat damage to roads	Adoption of more heat-resilient road materials and design standards
Heavier rainfall & extreme events	Bridge and road washouts (Mataura, Oreti, Aparima valleys); rural access disrupted	Strengthens case for upgrading culverts, bridges, and flood-resilient designs
Sea-level rise & storm surge (coastal areas e.g. Bluff, Riverton)	Overtopping, erosion, and permanent inundation of coastal roads	Coastal adaptation planning, potential realignment or retreat of vulnerable routes
Increased annual rainfall & streamflow	Possible flooding impacts on water supply infrastructure	Improved reliability of supply from rainfall-fed rivers (Waiau, Aparima, Oreti, Mataura)
Snowpack loss (meltwater-fed lakes Manapōuri, Te Anau)	Less reliable inflows affecting drinking water security	Incentive for diversified water storage and supply sources
Hotter summers & urban growth	Rising potable water demand and pressure on supply networks	Promotes water conservation, smart metering, and demand management
More intense rainfall	Contamination of potable water and storage facilities	Investment in upgraded treatment and protection of source water
Extreme rainfall intensity & multi-day totals	Stormwater/wastewater system overflows; localised flooding in towns (e.g. Invercargill, Gore, Riverton)	Incentivises green infrastructure (wetlands, permeable surfaces) to reduce runoff

Climate Change Driver	Risks	Opportunities
Sea-level rise & groundwater interaction	Reduced stormwater/wastewater capacity; saline intrusion and backflow	Opportunity to redesign outfalls, install backflow prevention, and integrate coastal resilience
Persistent inflows & infiltration to wastewater	Greater treatment volumes and costs; bypass risks in wet weather	Sewer renewal programmes and stormwater–wastewater separation
Higher maximum temperatures & more vegetation growth	Increased sewage sludge and green waste volumes to manage	Potential to capture biogas/compost from increased organic waste streams

5.2 Agriculture

Agriculture is central to Southland's economy, particularly dairy, sheep and beef farming. While many of the climate impacts listed above can indirectly affect these industries, it is worth describing how climate change could directly impact this sector given its importance in the region.

5.2.1 Impacts of warming temperatures on pasture and crops

Southland is projected to experience increases in annual rainfall; nevertheless, projections of **potential evapotranspiration deficit (PED)** suggest the region could face greater water stress in the future. Drier soils may create challenges for farming systems that depend on reliable pasture growth. Even under scenarios where annual rainfall increases, higher temperatures and evapotranspiration rates mean that water demand for irrigation could rise, placing pressure on available water resources.

A plant's demand for water, and its sensitivity to water stress, varies across its growth cycle. The **timing of drier conditions** is therefore critical: a late-summer drying, after most growth is complete, is far less damaging than a late winter or early spring drought that prevents crops and pastures from reaching their full productive potential.

At the same time, not all impacts of warming are necessarily negative. Higher temperatures will bring an increase in **growing degree days (GDD)**, supporting faster pasture growth and potentially enabling new crop opportunities. The projected **decrease in frost days** may also be beneficial to agriculture by reducing frost damage to sensitive plants and extending the growing season. These positive shifts need to be weighed against the risks of higher PED and increased water stress.

Elevated atmospheric CO_2 may partially mitigate increased PED under limited water supply. The **CO_2 fertilisation effect** reduces water loss through transpiration and improves crop water-use efficiency. Faster growth stimulated by higher CO_2 can also allow some plants to mature earlier, helping them avoid exposure to late-season drying.

Ultimately, the balance of these trade-offs will depend on the crop or pasture system in question. **Bespoke crop and pasture modelling** will be needed to fully assess how increased

GDD, reduced frost days, and higher PED interact to shape productivity and resilience in Southland's farming systems.

5.2.2 Impacts of flooding on pasture and crops

Heavy rainfall events are projected to become more intense in Southland, increasing the risk of land erosion, landslides, and slips. These processes are particularly significant in areas used for agriculture, horticulture, and forestry, where exposed soils and steep terrain make landscapes more vulnerable. Erosion not only reduces the productive capacity of the land but also has downstream effects through sedimentation of rivers and streams. Sediment loading reduces water quality, alters aquatic habitats, and contributes to greater flood risk by raising riverbeds and reducing channel capacity.

For farmers and land managers, these processes translate into **significant economic stress**. Major erosion events can strip away fertile topsoil, reduce pasture productivity, and damage crops, fences, and farm infrastructure. The long-term recovery from such events can be costly, requiring re-grassing, soil stabilisation, and sometimes changes in land use. Horticultural enterprises are particularly vulnerable, as even short-term waterlogging of soils can damage crops, while sediment deposition reduces soil fertility and structure.

In addition to erosion, **pluvial flooding** (caused by extreme rainfall overwhelming drainage capacity) will increasingly affect low-lying farmland. Crops and pastures may be inundated for longer periods, leading to plant stress, reduced yields, and, in severe cases, total loss of production. Livestock systems may also be disrupted, with grazing land unavailable and animal health compromised by wet conditions.

Coastal flooding, driven by a combination of extreme rainfall, storm surge, and sea-level rise, presents another growing risk for Southland's agricultural sector. Coastal farmland is at risk of periodic inundation and saltwater intrusion, which can reduce soil productivity and, over time, make some areas unsuitable for traditional farming practices.

Together, these impacts highlight the dual challenge of managing both the **direct physical damage** caused by heavy rainfall events and the **longer-term productivity losses** that follow. Adaptation will likely require investment in improved drainage systems, soil conservation measures such as riparian planting and erosion control, and, in some areas, shifts to more flood- and salt-tolerant crops or alternative land uses. Catchment-scale approaches to land and water management will become increasingly important in minimising the cascading effects of erosion and sedimentation on both farms and downstream communities.

Table 5-2: Agriculture summary table.

Climate Change Driver	Risks	Opportunities
Warming temperatures & higher PED	Drier soils, greater irrigation demand, pressure on water resources	Improved water-use efficiency; investment in on-farm water storage
Timing of dryness (late winter/early spring)	Reduced pasture growth and crop yields	Adjusted planting/grazing schedules; better drought forecasting
Increased growing degree days (GDD)	Faster pasture turnover may stress farm systems if unmanaged	Longer growing season; potential for new crop opportunities

Climate Change Driver	Risks	Opportunities
Decline in frost days	Fewer natural pest controls, adverse impacts to plants/animals adjusted to frost	Reduced frost damage to crops; extended growing season
Elevated CO_2 (fertilisation effect)	Uneven benefits across species; does not eliminate evaporative drought risk	Improved water-use efficiency and faster plant growth
More intense rainfall & flooding	Soil erosion, landslides, slips, and sedimentation reducing land productivity	Justifies soil conservation practices and riparian planting
Erosion & sedimentation	Reduced water quality, altered aquatic habitats, raised riverbeds increasing flood risk	Catchment-scale land management to protect both farms and downstream communities
Pluvial flooding (overland runoff)	Waterlogging of soils, crop loss, livestock health impacts	Improved farm drainage and flood-adapted land-use planning
Coastal flooding & saltwater intrusion	Loss of productive coastal farmland	Transition to salt-tolerant crops; strategic land-use change or retreat

5.3 Forestry

The Otago–Southland wood supply region is dominated by ***Pinus radiata*** **(63%)**, **Douglas-fir (*Pseudotsuga menziesii*; 26%)**, and **eucalypts (7%)**.

5.3.1 Growth and disease

Warming temperatures and a longer growing season are expected to improve ***Pinus radiata* growth** in cooler parts of Southland. However, these gains may be offset by increased disease pressure. **Dothistroma needle blight**, which thrives in moderate temperatures, is projected to become more severe in Southland under climate change, reducing potential productivity gains.

For **Douglas-fir**, the outlook is less positive. The severity of **Swiss needle cast** (*Phaeocryptopus gaeumannii*), the most widespread disease affecting this species, is strongly correlated with warmer winter temperatures. As winters become milder, outbreaks are expected to intensify, reducing Douglas-fir productivity across the region.

5.3.2 Wildfire risk

Climate change is likely to **increase wildfire risk** in Southland's forests. Key contributing factors include:

- Warmer weather, drier soils, and windier conditions that make fuels easier to ignite.
- A longer fire season, starting earlier in spring and extending later into autumn.
- More thunderstorms and lightning, increasing ignition events as possible.

Afforestation with exotic species such as ***Pinus radiata***, a key climate change mitigation strategy, may further heighten fire risk. Exotic conifer and eucalypt plantations burn more intensely than pasture, shrubland, or most native forests. Among plantation species, ***Pinus radiata*** is particularly fire-sensitive, while **Douglas-fir and eucalypts** are more fire-adapted. In contrast, many indigenous New Zealand plants are relatively non-flammable, offering potential value in fire-resilient landscape planning.

Table 5-3: Foresty summary table.

Climate Change Driver	Risks	Opportunities
Warming temperatures & longer growing season	Productivity gains may be offset by disease outbreaks	Enhanced growth potential for *Pinus radiata* in cooler parts of Southland
Dothistroma needle blight (*Pinus radiata*)	More severe infections in Southland under warmer, wetter conditions	Investment in resistant genotypes, improved silviculture, and targeted spraying
Swiss needle cast (Douglas-fir)	Increased severity as winters warm, reducing Douglas-fir productivity	Potential to shift planting zones, trial alternative species better suited to warmer climates
Warmer, drier soils, and windier conditions	Fuels dry faster, making forests more flammable	Stronger fire management planning, fuel reduction, and landscape-scale firebreaks
Longer fire season	Greater firefighting demand and costs; higher risk of large-scale burns	Justifies investment in year-round fire readiness and rapid-response capacity
More thunderstorms and lightning	Increased ignition events	Advanced monitoring, early-warning systems, and fire suppression technology
Afforestation with exotic species (e.g. *Pinus radiata*, eucalypts)	Exotic plantations more flammable than native shrubland/forest; higher fire hazard	Strategic land-use planning: mix of exotics with less-flammable native plantings for resilience
Species flammability differences	*Pinus radiata* highly fire-sensitive, Douglas-fir and eucalypts more fire-adapted	Potential to select species mixes that balance growth, economic value, and fire resilience

5.4 Terrestrial biosecurity and infectious diseases

Climate change is widely regarded as one of the greatest challenges facing indigenous ecosystems in the coming century. For Southland, where the economy relies heavily on efficient primary production systems, the risk of exotic pests and diseases affecting agriculture and horticulture must be carefully managed.

5.4.1 Terrestrial biosecurity

A warmer climate could bring new biosecurity challenges by enabling the establishment of pest animals, weeds, and diseases that are currently limited by Southland's cooler conditions. The greatest concern lies with species that already arrive seasonally but fail to establish, along with those already recognised as high-risk invaders.

Although climate change will affect organisms and ecosystems in many ways, temperature is expected to be the most important driver of pest invasions, influenced by rainfall, humidity, and carbon dioxide levels. Regional winds and currents may also play a role by increasing the likelihood of new pests reaching New Zealand and establishing locally.

Some of the world's most damaging invasive ant species, such as the big-headed ant (*Pheidole megacephala*) and the Argentine ant (*Linepithema humile*), are already present in New Zealand. Continued warming could encourage their spread into Southland. Similarly, fruit flies remain a major threat to horticulture, with central and eastern Southland projected to become increasingly suitable for their establishment later this century, though the region will still be less suitable than much of the North Island.

The arrival of new pest plants and the greater invasiveness of existing weeds is also a concern. As frosts become less frequent and more insect pollinators are able to survive warmer winters, a wider range of weed species will be able to establish and compete with native plants. Farmers and growers in Southland may increasingly adopt plant species currently grown further north, while ornamental plants that escape cultivation may naturalise and become invasive under warmer conditions.

It is important to note that while many biosecurity threats may arrive from outside New Zealand, others are already present within the country. These "sleeper pests" remain at low levels under current conditions but may spread and flourish under a warmer climate. Examples include invasive weeds as well as invertebrates such as the migratory locust (*Locusta migratoria*) and the tropical armyworm (*Spodoptera litura*).

5.4.2 Infectious diseases

The Southland region may face increasing pressure from novel diseases and their vectors as the climate warms. Twelve mosquito species were present before human settlement, and since then four exotic species capable of spreading disease have become established. More than 30 additional exotic mosquito species have been intercepted at national entry points, highlighting the ongoing risk of new arrivals.

Some pathogens carried by ticks and mosquitoes are currently limited in New Zealand by cooler temperatures — for example, *Theileria orientalis* (tick-borne), West Nile virus, and bovine ephemeral fever virus. However, under warmer conditions these diseases can spread rapidly, with the potential for explosive outbreaks once suitable climate thresholds are crossed.

Table 5-4: Summary of terrestrial biosecurity and infectious diseases.

Climate Change Driver	Risks	Opportunities
Warmer climate & fewer frosts	Greater survival and spread of exotic pests, weeds, and diseases	Strengthened biosecurity; investment in monitoring and eradication programmes
Spread of existing pests	Damage to crops, pastures, and horticulture systems	Justifies targeted pest management and development of resistant crop varieties
New invasive weeds & ornamental escapees	Competition with native plants and productive species; increased management costs	Adoption of alternative crops/land uses; expansion of weed biocontrol programmes
Sleeper pests already in NZ (e.g. locusts, armyworms)	Potential outbreaks triggered by warmer conditions	Proactive risk assessments and adaptive farm biosecurity planning
Expansion of insect pollinators under warmer winters	Facilitation of invasive weed spread and establishment	Improved ecosystem monitoring and adaptive management of landscapes
Novel disease vectors (mosquitoes, ticks)	Establishment of new species capable of transmitting disease	Stronger surveillance and vector control measures
Pathogens (e.g. *Theileria orientalis*, West Nile virus, bovine ephemeral fever)	Risk of outbreaks affecting livestock and human health	Vaccines, treatments, and farmer preparedness programmes

5.5 Fishing and aquaculture

5.5.1 Aquatic Biosecurity

The primary entry pathway for aquatic biosecurity risks into New Zealand remains **international shipping**, through organisms carried in ballast water or attached to ship hulls. Climate change is expected to increase these risks: warmer ocean temperatures and shifting currents may allow marine species — including pests and pathogens not previously seen in New Zealand waters — to arrive, survive, and establish. Around New Zealand, sea temperatures are projected to rise (especially to the west), while seawater pH will continue to decrease. Both factors will influence which organisms can thrive in local ecosystems.

Long-term changes in marine conditions, particularly warmer water, may create new ecological compatibilities and alter host–pathogen relationships. Warmer seas and freshwater systems are known to increase host susceptibility to disease, raising the likelihood of new aquatic diseases emerging in regions such as Southland. This could affect both wild species and aquaculture operations.

5.5.2 Freshwater biosecurity

In freshwater systems, higher water temperatures could favour the expansion of **warm-water pest fish** such as koi carp, goldfish, tench, rudd, and catfish. These species degrade water quality and reduce indigenous biodiversity. Warmer conditions may also facilitate the establishment of tropical aquarium fish that are accidentally or deliberately released into the wild.

Similarly, invasive **warm-climate aquatic plants** such as water hyacinth (*Eichhornia crassipes*) and water fern (*Salvinia molesta*) could expand under rising temperatures, further threatening freshwater habitats.

5.5.3 Emerging risks

As with terrestrial biosecurity, many organisms already present in New Zealand but not currently considered pests could become problematic under a changing climate. Warmer waters and altered ecological balances may provide the trigger for these “sleeper species” to spread and cause new impacts on ecosystems, water quality, and aquaculture.

Table 5-5: Summary Fishing and aquaculture.

Climate Change Driver	Risks	Opportunities
International shipping (ballast water & hull fouling)	Introduction of new marine pests and pathogens	Strengthened port biosecurity, ballast water treatment, and hull management
Warmer sea temperatures	Expansion of invasive marine species; higher disease susceptibility in wild species and aquaculture	Diversification into species that thrive in warmer waters; selective breeding for resilience
Shifting ocean currents	New pathways for exotic species to reach Southland	Enhanced forecasting and regional monitoring of marine biosecurity risks
Ocean acidification (lower seawater pH)	Stress on shellfish and calcifying organisms; altered species competitiveness	Adaptation in aquaculture practices, investment in resilient species and farming systems
Altered host–pathogen interactions	Increased likelihood of aquatic disease outbreaks affecting fisheries and aquaculture	Proactive surveillance, early-warning systems, and improved veterinary responses
Higher freshwater temperatures	Spread of warm-water pest fish (koi carp, catfish, rudd, etc.); degraded water quality and biodiversity loss	Stronger freshwater biosecurity controls; targeted eradication and management programmes
Invasive aquatic plants (e.g., water hyacinth, water fern)	Habitat loss, reduced water quality, competition with native species	Early detection, containment strategies, and use of biocontrol agents where feasible

5.6 Tourism

Changes in **snow cover** are expected to significantly impact the ski industry in Southland and Central Otago. Shorter and less reliable ski seasons will create challenges for operators. However, tourism numbers from Australia may increase as snow cover in Australia declines rapidly, making New Zealand a more attractive option for winter holidays. Beyond skiing, New Zealand's alpine and glacier landscapes may become more appealing as their relative rarity increases globally.

In general, **warmer and drier conditions** are likely to benefit Southland's tourism sector by lengthening the summer season and supporting activities such as hiking, cycling, and nature-based tourism. On the other hand, **wetter conditions, flooding, and extreme weather events** could disrupt visitor access, damage infrastructure, and reduce the attractiveness of outdoor destinations.

A significant portion of Southland's tourism relies on **rivers and lakes**, including activities such as fishing, jet boating, rafting, kayaking, and scenic cruising. Projected changes in river flow – with more water in winter and spring but drier conditions in summer – will directly affect these operations year-round, altering both safety and the reliability of experiences.

Table 5-6: Summary snow cover.

Climate Change Driver	Risks	Opportunities
Declining snow cover	Shorter, less reliable ski seasons; reduced viability for winter sports	Increased Australian visitor numbers as snow declines there; NZ alpine/glacier landscapes gain global appeal
Warmer, drier conditions	Risk of heat stress during peak summer; potential pressure on water resources	Longer summer season; boost to hiking, cycling, camping, and nature-based tourism
Wetter conditions & extreme weather events	Flooding, slips, and storms may damage infrastructure and disrupt visitor access	Incentive to strengthen infrastructure resilience and diversify tourism products
Changes in river flow (wetter winters/springs, drier summers)	Unreliable conditions for fishing, rafting, jet boating, kayaking, and scenic cruising	Potential to extend river-based tourism in winter/spring when flows are higher
Alpine & glacier retreat	Loss of iconic features over time, reducing attraction for some visitors	Increased global rarity may enhance scenic value in the medium term

5.7 Renewable Energy Sector

Climate change is expected to alter hydro-electricity generation in Southland through changes in river inflows, lake storage, and drought risk. Large hydro-generation schemes located in Southland western catchments (Fiordland, western Waiau) may see more stable or increased inflows (mainly through increase in precipitation), while smaller and local hydro schemes located in eastern and inland FMUs (Waikaia, Mataura, Oreti) may face increased variability through higher drought risk and reduced reliability. Two main risks are identified for the large hydro-generation lakes: i) increased inflow variability due to projected decline of snowpack with warming (reduced spring/summer snowmelt), ii) and increased flood and spill risk due to increased heavy rainfall events forcing spill from hydro-generation lakes.

Climate change is expected to have mixed impact on wind electricity generation across Southland. While little change is expected in term of wind patterns, strong seasonal changes are projected through time and warming for wind speed (decrease in summer and autumn, increase in winter and spring), frequency of windy days and strong wind magnitude (decrease in summer, increase for remaining season). As a result, wind generation could potentially increase under future warming, but changes in wind intensity, and increased seasonal variability could impact electricity supply balance when paired with hydro-lakes inflows.

6 Summary

The global climate system is changing and will continue to change for decades to come. These changes will have implications not only for New Zealand's climate and weather systems but also for freshwater availability for downstream users and for hazard exposure (inland and coastal). Due to the nature of climate change, trends will vary across the country, over the course of the century, and among scenarios of climate change. This report addresses potential impacts of climate change on a range of components of climate, hydrology, marine and coastal processes across Southland using downscaled Global Climate Model (GCM) outputs from 1971–2099 under different global warming scenarios. The combination of six GCMs and three emission scenarios allows us to consider a plausible range of future trajectories of greenhouse gas emissions and climatic responses.

It is not possible to attribute the modelled differences between two time periods (in this report, mid-century and end of century) solely to climate change, as natural climate variability is also present and may add to, or subtract from, the climate change effect. The resulting potential impacts of climate change are presented through averaging of the six model projections, which does reduce the underlying natural variability to some extent. With these caveats in mind, the potential effects of changing climate over this century across the Southland region and Freshwater Management Units (FMUs) are summarised as follows across the different domains for each metric considered:

Climate domain

The projected mean temperature changes across the Southland region increase with time and emissions scenario. Future annual average warming spans a wide range: 0.9–1.4°C by 2050, and 1.8–3.6°C by 2090, largely dependent on scenario. Seasonally, most warming occurs in summer by mid-century and during autumn-winter and summer by the end of century. Changes in maximum and minimum temperature follow similar patterns, with the largest change in the maximum temperature projected during summer (up to 3.9 °C for SSP5-8.5 by the end of century), while the largest change in the minimum temperature is projected during winter (up to 3.6°C for SSP5-8.5 by the end of century). Diurnal range (i.e., difference between daily minimum and maximum temperature) is expected to increase with time and emission scenario, with the largest increase projected for summer and autumn.

Changes in extreme temperatures mirror the average annual signal. The average number of hot days is expected to rise from 3–5 days by 2050 to 6-19 days by 2090. Heatwave days (consecutive days >25°C) are projected to increase — particularly at higher elevations — across all scenarios by up to 8 days by 2050 and up to 36 days by 2090. Conversely, frost days are expected to decline by up to 20 days by mid-century, and by 25–45 days by the end of the century. Growing degree days will increase with time and emissions scenario, with the largest seasonal rise during summer.

Projected changes in rainfall show a marked seasonality and variability across the Southland region. Annual rainfall is expected to increase slightly by mid-century (5%), rising to 8–13% by the end of the century, with larger increases in western and eastern areas. One exception is the Garvie Mountains region (headwaters of the Waikaia River) where annual rainfall is projected to decrease by up to 10% across all scenarios. Seasonally the largest increases are projected during winter and spring (up to 40%), while summer and autumn precipitation is more variable with increase/decrease (between -20% and + 20%) depending on elevation, location, time and emission scenarios.

By mid-century, the number of wet days may increase slightly (up to 1 day) but decrease by up to 2.5 days by the end of the century. Spatially, increases in wet days occur mainly in the headwaters of the three Waiau hydro-electric generation lakes and coastal Southland. Decreases are concentrated in the eastern Waiau and north of the Hokonui syncline.

The number days where the total precipitation exceeds 10 mm is projected to increase across Southland at all time slices and emission scenarios, except for some areas in the lower Fiordland (Cameron/Princess Mountain ranges) and upper Mataura (Waikaia catchment) where small decreases (up to eight days) are expected for both mid-century and end of century.

The number of very wet days (i.e., days where the total precipitation exceeds 25 mm) is projected to increase across Southland at all time slices and emission scenarios, except for a small area in the eastern Waiau catchment (northern part of the Livingstone/Thomson Mountain range) and lower Fiordland (Cameron/Princess Mountain ranges) where a small decrease is expected by mid-century.

The number of heavy wet days (i.e., days where the total precipitation exceeds 50 mm) is also projected to increase across Southland under all scenarios, with the largest increase in northern Fiordland (north of the Cameron-Princess-Kaherekoau Mountain ranges). This increase is smaller than that for very wet days.

Projected changes in maximum annual 1-day rainfall (Rx1d) rise with time and emissions scenario, reflecting changes in very wet and heavy wet days. Rx1d is expected to increase by an average of 11 mm by mid-century and 23 mm by the end of the century, with local increases of up to 80 mm in the Waiau headwaters.

By mid-century, decreases in annual maximum 5-day rainfall (Rx5d) are projected for the Thomson and Ailsa Mountain ranges (up to 25 mm) while other areas, particularly Fiordland, may see increases of up to 50 mm. At the end of the century, Rx5d is projected to increase across the region, with Fiordland experiencing rises of up to 75–100 mm.

By mid-century the number of dry days may decrease slightly (up to 1 day), mainly in low-elevation areas, except in the three Waiau hydro-electric generation lakes. The number of dry days is expected to increase by up to 2 days by century's end, particularly north of the Hokonui syncline, while Waiau headwaters may see decreases.

Relative humidity is projected to slightly decrease with time and emission scenarios by up to 1%.

Solar radiation is projected to rise slightly (up to 1 $W.m^{-2}$) except under SSP 2-4.5 where a decrease is expected. Seasonally, solar radiation is expected to increase in summer and autumn (by up to 7 $W.m^{-2}$ in summer by end of century) and decrease in winter and spring (by up to 4 $W.m^{-2}$ in winter by end of century). Spatially, the largest decrease occurs in Fiordland's Darran Range, while the largest increase is across Mataura and Oreti Freshwater Management Units (FMUs).

Cloud cover is expected to decrease slightly (by up to 2%), mainly from October to April, with some minor increases expected for July/August. Fiordland headwaters may see increases mid-century, while Mataura–Oreti and eastern Waiau experience the largest decreases by the end of the century.

We note that work is ongoing on the wind speed projections, therefore caution is required in using the following results:

Annual average wind speed may rise slightly (up to 4%) across Southland under all scenarios. Seasonally, wind speed is expected to increase in winter and spring (up to 12% in winter under SSP5-8.5) and decrease for summer and autumn (up to 5 % under SSP3-7.0) by the end of the century. Spatially, annual average wind speed may decrease in Fiordland (north of the Franklin Range) and north-east Mataura (Old Man Range), while the largest increases occur north of the Hokonui syncline for the Mataura and Oreti, for most of the Aparima and most of the Waiau River downstream of Manapouri and south of Takitimu mountain ranges.

The number of calm days (i.e., mean wind speed <1.5 $m.s^{-1}$) is expected to remain relatively unchanged, with minor seasonal shifts: slight decreases in June–September across scenarios and slight increases by the end of the century for February–March.

Windy days (i.e., mean wind speed >10 $m\ s^{-1}$) are expected to increase (up to 10 days annually) especially during June–October (up to 3 days per month by the end of the century). Changes in windy days are spatially variable; northern Fiordland may experience reductions (up to 20 days/yr), while most of the Mataura-Oreti-Aparima and Waituna may experience increases (up to 20 days/yr). Strong wind changes (99^{th} percentile) are likely to follow similar patterns.

Southland's prevailing wind direction - typically spanning the westerly quarter (from northwest through to southwest) - is projected to remain unchanged through time and warming.

Changes in meteorological drought (assessed using Potential Evaporation Deficit or PED) indicate Southland will experience increased drought: limited under SSP2-4.5 (up to 7 mm) and greater under higher emission scenarios (up to 23 mm). Spatially, the headwaters of the Waikaia River may see the largest increase (up to 150 mm under SSP5-8.5), with smaller changes elsewhere. Conversely, the probability of annual PED accumulation >200 mm reflects similar trends: the Waikaia and upper Mataura may exceed 200 mm by at least 50%, while coastal Oreti and Waituna catchments may see decreases (up to 20%).

Changes in rainfall intensity correlate with temperature changes: mid-century rainfall intensity increases range from 4.8% (2-year ARI, 1-hour event) to 18% (100-year ARI, 1-hour event); by century's end the range is 16% to 46% for the same events.

Probable Maximum Precipitation (PMP) is projected to change linearly with mean annual air temperature at a rate of 8.34% per °C, regardless of scenario or storm duration.

Cryosphere domain

Changes in snow volume reflect changes in precipitation and temperature. Decreases in snow volume are expected to reach at least 50% across Southland mountain ranges by mid-century for SSP3-7.0 and SSP5-8.5, with smaller reductions in the lower-elevation range of the Waiau and the Garvie mountain range under SSP2-4.5. By the end of the century and regardless of the level of warming, snow volume is projected to decrease by at least 50% across Southland.

Changes in the number of snow days are projected to remain largely unchanged by the 2050s across emission scenarios, although the northern headwater mountain ranges are projected to experience a reduction of up to 30 snow days. Similar spatial patterns are projected for the end of the century with greater reductions in snow days under higher-emission scenarios. The Garvie Mountains (Mataura) are projected to experience snow day reductions of at least 90 days.

Snow line is expected to rise across Southland mountain ranges over time and under all emission scenarios. By mid-century, the snow line is expected to increase by up to 300 m across Southland regardless of scenario. By the end of the century, snow line is expected to increase by approximately 400–670 m depending on the level of warming. While snow may persist on the highest mountain ranges, it is projected that lower mountain ranges will have fewer than 90 days of snow cover by the end of the century.

Hydrology domain

The bias-corrected hydrological modelling chain is judged appropriate for use in this project. The hydrological bias correction was undertaken using a simple method that aims to match the long-term average riverine and cryospheric metrics across Southland rivers. This method allowed the bias-corrected model to adequately reproduce the mean flow metrics for all natural flow catchments. The bias corrected models also matched the mid- to lower-end values of the high flow metrics (95^{th} percentile of non-exceedance) and the mid- to high-end values of the low flow metrics (25^{th} percentile of the annual flow time series) for the natural flow catchments. However, the bias corrected models were less able to reproduce the more extreme flows, i.e., the lowest values for the low flow metrics, or the highest values for the high flow metrics. This indicates that further work is required to ensure the adequacy of the bias correction methods for these more extreme flow conditions.

Mean annual flood is projected to increase across nearly all Southland River reaches (at least 95%), with the greatest increases by the end of the century under SSP5-8.5 (up to 42%). While the magnitude of mean annual flood is projected to decrease for a small pocket within each FMU, flood magnitude is projected to decrease for the headwaters of the Waikaia catchment (Mataura FMU).

Changes in the magnitude of the Q5 (i.e., river flow magnitude exceeded five percent of the time) and of the Q25 (i.e., river flow magnitude exceeded twenty-five percent of the time) reflect changes in mean annual flood albeit with a reduced range of magnitude (up to 20% for changes in Q5 and up to 9% for changes in Q25), suggesting a possible shift in the distribution of high flows under warming conditions.

Changes in the average annual number of Fre3 events are projected to vary across Southland rivers depending on location, time and emission scenario. By the end of the century under SSP5-8.5, the number of annual Fre3 events is projected to increase by between one and two events across all FMUs, except the Waiau FMU for which the increase is limited to one additional event. Spatially, decreases are projected for up to 11% of river reaches across Southland with decreases of up to five events in some areas of the Aparima and Mataura FMUs.

Changes in the magnitude of mean discharge reflect changes in average annual precipitation. The magnitude of the mean flow is projected to increase across nearly all Southland river reaches (at least 90%), with the greatest increases by the end of the century under SSP5-8.5 greenhouse gas concentration scenarios (average increase of 15% across Southland rivers).

However, the magnitude of mean flow is projected to decrease for several reaches within each FMUs (up to 16% in the Waiau FMU and 23% in the Mataura FMU).

Changes in annual flows were considered in terms of changes in mean annual flows (more sensitive to extremes) and median annual flows. Across most Southland rivers, mean annual flow is projected to increase, with the greatest increases by the end of the century under SSP5-8.5 (average increase of 15% across Southland rivers). A notable difference occurs in the lower Mataura River, whereby the end of the century under very high greenhouse gas concentration scenarios, median annual flows are projected to increase by 10%, while mean annual flows are projected to increase only by up to five percent. The magnitude of mean annual flow is projected to decrease in some areas within each FMU (e.g., up to 17% of the reaches in the Mataura and Waiau FMUs).

The magnitude of seasonal average flows is projected to increase across nearly all Southland rivers in every season except summer, with the greatest increases by the end of the century under SSP5-8.5 (average increase of at least 30% across Southland rivers). In contrast, the direction of change in summer flows is projected to vary with time and scenario, while the magnitude of change is limited to a range of -5 to 5%. Regardless of time and scenario, summer average flows are predicted to decrease in the Waiau and Fiordland FMUs, and across the Mataura FMU by the end of century under SSP3-7.0 and mid-century under SSP5-8.5. Summer flows across the Aparima and Oreti FMUs are projected to increase regardless of time and scenarios. Overall, changes in seasonal average flows suggest a possible larger shift in Southland rivers' hydrological regimes under warming conditions, with spatial differences identified across FMUs.

Changes in the magnitude of the Q75 (i.e., river flow magnitude exceeded seventy-five percent of the time) and Q95 (i.e., river flow magnitude exceeded ninety-five percent of the time), shows a different spatial pattern to the mean flow. Both metrics are projected to decrease by the end of the century under high (SSP3-7.0) and very high (SSP5-8.5) greenhouse gas scenarios, with average decrease up to 6% across Southland rivers. However, while Q75 is projected to increase across the Fiordland FMU, and to a lesser extent the Aparima and Waituna FMUs, Q95 is projected to increase only for the Fiordland FMU. Overall, changes in low flow metrics associated to changes in seasonal average flows suggest a possible larger shift in Southland rivers' hydrological regimes under warming conditions, with spatial differences identified across FMUs.

Changes in the magnitude of mean annual 7-day low flow vary across Southland rivers in both direction and magnitude. At FMU scale, the largest decrease in mean annual 7-day low flow is predicted across the Waiau FMU by mid-century under SSP3-7.0 (-11.1%), while the largest increase is predicted across the Aparima FMU by the end of the century under SSP2-4.5 (+ 50%). Spatially, mean annual 7-day low flow is expected to decrease by at least 23% for most reaches located in the Oreti and Mataura main stems by mid-century century under SSP3-7.0. Mean annual 7-day low flow is projected to decrease in parts of every FMU (up to more than 60% across all FMUs, except for the Waituna FMU). These changes point to significant shifts in low flow hydrological regimes across Southland rivers under emission scenarios.

Changes in the magnitude of flow reliability vary across Southland rivers in both direction and magnitude. At FMU scale, the largest decrease in flow reliability is predicted across the Aparima, Waituna and Mataura FMUs by end of century under SSP2-4.5 (up to at least -0.05), while the largest increase is predicted across the Waiau FMU by the end of the century under all emission scenarios (+ 0.02). Spatially, flow reliability is projected to decrease across at least 69% of the reaches in every FMU, with the largest decrease projected in the Aparima-Oreti and Mataura main stems (up to -0.08) by the end of century under SSP5-8.5. Changes in flow reliability point to river water being less consistently available to meet environmental or water-use needs under future warming, reflecting a shift towards greater variability and uncertainty in water regimes.

Changes in the magnitude of water supply reliability vary across Southland rivers in both direction and magnitude. At FMU scale, the largest decrease in water supply reliability is predicted across the Aparima, Waituna and Mataura FMUs by end of century under SSP2-4.5 (up to at least -0.05), while the largest increase is predicted across the Waiau FMU by the end of the century under SSP2-4.5 and SSP5-8.5 (+ 0.01). Spatially, water supply reliability is projected to decrease for most FMU river main stems (up to -0.1) by end of century under SSP5-8.5. Changes in water supply reliability extend changes in flow reliability and indicate river water will be less consistently available to meet water demand during the irrigation season, likely increasing the frequency of water-use shortages under future warming.

Changes in the magnitude of land surface recharge vary across Southland shallow aquifers in both direction and magnitude. At annual and FMU scale, land surface recharge is projected to decrease across the Oreti FMU with warming (up to -6%), with the largest increase in land surface recharge predicted across all other FMUs under all emission scenarios (up to 12.6% for the Aparima by end of century under SSP5-8.5). At seasonal scale, land surface recharge is projected to increase during autumn and winter across all FMUs through time and emission scenarios, except for small pockets within the Oreti and Waiau FMUs. In summer, however, land surface recharge is projected to decrease with warming across most shallow aquifers in the Oreti (except the Five Rivers aquifer), Waituna and lower Mataura FMUs (by up to 50% for some aquifers under SSP5-8.5). In spring, decreases in land surface recharge are limited to parts of the lower Oreti and Waiau aquifers, specifically under SSP5-8.5. Overall, decreases in spring and summer land surface recharge points to intensifying risks of hydrological and agricultural drought, water scarcity (e.g., access to groundwater resource for drinking water supply in the lower Mataura), and ecosystem stress across the Oreti-Waituna and lower Mataura FMUs under future warming.

Magnitude of the maximum daily and hourly annual peak discharge is generally projected to increase with time, warming and return period across Southland rivers, noting that for some locations the largest rises occur under SSP3-7.0 rather than the SSP5-8.5 scenario. These results mirror the changes in precipitation extremes.

Magnitude in peak daily discharge associated with 12hr and 24 hr duration PMP (referred as 12hr PMF and 24 hr PMF) is projected to increase with warming. Under a 1°C warming scenario, most FMUs, except the Mataura FMU, are projected to experience an increase between 10 and 20% in the magnitude of the PMFs (independently of the event duration). Spatial differences in changes to the 12hr and 24 hr PMF magnitude are more pronounced for warming greater than 2°C, with the largest changes predicted for the Mataura FMU because of the timing of the PMP event (winter PMP). While changes in PMF across different storm durations generally follow the same pattern as rainfall depth changes predicted by the HIRDS tool, changes in PMF are predicted to not increase with storm duration at some locations. This may be due to the methodology used to predict climate-change adjustments to PMP rainfall depths across duration. Further investigations are needed to understand these changes and better characterise the impacts of rarer rainfall events.

Stream temperature is projected to increase uniformly across Southland rivers with time and warming, at both annual and seasonal time scale. Spatially, the largest increases are projected under SSP5-8.5 at the outlets of the main river stems (up 4°C), while the lowest increases are projected in the upper reaches of FMUs (up to 1°C). These changes directly reflect changes in air temperature at both annual and seasonal time scale and signal greater stress on freshwater ecosystems.

Intercomparison of the CMIP5-CMIP6 changes in hydrological metric projection

Intercomparison of the magnitude of the projected change in three hydrological metrics (mean annual flood, annual and seasonal average discharge and water supply reliability) was carried out between the CMIP5 derived assessment and this assessment. As per the atmospheric analysis, the intercomparison was completed for two scenarios producing similar levels of CO_2 by the end of the century. The scenarios considered were RCP4.5 (CMIP5) versus SSP2-4.5 (CMIP6) and RCP8.5 (CMIP5) versus SSP3-7.0 (CMIP6). While the CMIP5 assessment was developed using a non-bias-corrected modelling chain, the CMIP6 assessment uses a bias-corrected modelling chain. The bias correction methodology does not affect the intercomparison of changes in river flow metrics. The key messages relevant to the Southland region are:

- Change in the magnitude of mean annual flood under CMIP6 are generally smaller than those projected under CMIP5. However, the spatial distribution of these changes differs, especially under the lower-emission scenario, mean annual flood projected to decrease up to 50% under RCP4.5 (CMIP5) for the upper Oreti FMU by the end of the century, while increases of up to 20% are projected under SSP2-4.5 (CMIP6).
- Changes in annual average discharge remain similar under the lower-emission scenario for the Mataura FMU. However, while little change is projected for the remaining Southland main rivers under RCP 4.5 (CMIP5), increases of up to 20% are projected under SSP2-4.5 (CMIP6). Under higher emission scenarios, similar magnitudes of change are projected under RCP8.5 (CMIP5) and SSP5-8.5 (CMIP6), except for the Mataura FMU, where little change or a decrease is projected under CMIP6.
- While differences in the magnitude of change are noted for seasonal average flow, with smaller changes projected under CMIP6, differences in spatial distribution and direction of change are identified depending on the emission scenario, as follows:
  - Winter flows are projected to decrease for the Waikawa catchment under CMIP6, while winter flows are projected to increase under CMIP5. Winter flows are projected to increase in lower Mataura and Oreti FMU under SSP2-4.5 (CMIP6), with little change projected under RCP4.5 (CMIP5) for those locations.

  - Changes in summer discharge are consistent between the two assessments for the Mataura FMU. However, differences in the direction of the change are projected for the Aparima, Oreti and Waiau FMUs particularly under higher-emission scenarios.
  - Changes in spring discharge are consistent between the two assessments with differences only in the magnitude of change.
  - Changes in autumn discharge are not spatially consistent between the two assessments, particularly for the Aparima, Oreti and Mataura FMUs, with increases in autumn discharge projected under RCP4.5 (CMIP5), while decreases are projected under SSP2-4.5 (CMIP6).
- The magnitude of absolute change in water supply reliability is consistent across both assessments. However, differences in the direction of the change are identified for both emissions scenarios, with water supply reliability projected to decrease across most rivers under RCP8.5 (CMIP5) while increasing under SSP5-8.5 (except for the lower Oreti FMU).

Sea-level domain

Tidal analysis showed differences in the tidal characteristics at Bluff and Stead Street, with Bluff experiencing a larger tidal range. High-tide exceedance curves were used to identify how often high tides might reach critical elevations, and these were then adjusted to reflect projected sea-level rise. Additionally, storm-tide levels—caused by the combined effects of tide, storm surge, and wave setup—were statistically modelled for both locations. Bluff was found to be more exposed to extreme sea levels under storm-tide conditions, which are expected to become more frequent as sea levels rise.

Historical observations from Bluff and Stead Street show a clear and ongoing rise in mean seas level, consistent with trends observed elsewhere in New Zealand. While year to year variability remains high, recent decaded show higher average sea levels than earlier periods, increasing the frequency of coastal flooding during high tides and storm events.

Future projections, based on IPCC AR6 scenarios and data from the NZSeaRise programme, indicate that sea levels will continue to rise under all emission pathways. These projections include estimates of when critical sea-level thresholds (e.g., 0.2 m, 0.6 m, and 1.0 m above 2020 levels) are likely to be reached, both with and without adjustments for local vertical land movement—information that is essential for future risk and infrastructure planning.

This sea-level analysis highlights the increasing exposure of coastal communities and infrastructure to flooding and erosion hazards. It provides important information to inform planning, infrastructure investment, and risk management decisions across Southland. The modelling shows that Southland property exposure to coastal flooding (with a 100-year ARI) will increase anywhere from 20% to 76% over time, under different emission scenarios and if land subsidence over the next 100 years. The results underline the need for proactive adaptation measures that consider not only global sea-level rise but also local land movement and extreme events.

Marine domain

Sea surface temperatures (SSTs) are projected to increase over time and under all emission scenarios due to anthropogenic greenhouse gas emissions. By the end of the century, SSTs are projected to increase by up to 1°C under the low-emission stabilisation scenario (SSP1-2.6) and by more than 3°C under SSP3-7.0.

Linked to SST change, marine heatwaves are expected to increase over time and under all emission scenarios. By the end of the century, marine heatwave intensity is projected to increase by up to 0.5°C under SSP1-2.6 and exceed 2 °C under SSP3-7.0. Conversely, the frequency of marine heatwaves is expected to increase from 15–30 days per year (under current climate conditions) to 120–200 days per year under SSP1-2.6 and up to 365 days per year under SSP3-7.0.

The below table is provided in this Executive Summary to assist readers interested in the key findings for each Freshwater Management Unit (FMU).

7 Glossary of abbreviations and terms

Anthropogenic	Originating in human activity
ARI	Average recurrence interval. The average time interval (averaged over a very long time period and many "events") that is expected to elapse between recurrences of an infrequent event
Bias-corrected	The process of using observations (e.g. data from existing climate stations, or streamflow stations) to calibrate climate model or climate-hydro modelling chain simulations. This ensures climate model or hydrological output resemble a given location's climate and weather/ hydrological characteristics appropriately
CCAM	Conformal Cubic Atmospheric Model, which uses dynamical downscaling to simulate climate and weather at finer resolutions than GCMs. Developed by CSIRO in Australia
Climate variable	An element of the climate that is liable to vary or change, e.g. temperature, rainfall
CMIP5	Fifth Coupled Model Intercomparison Project. This project involved a number of experiments with coupled atmosphere-ocean global climate models, most of which were reported on in the IPCC Fifth Assessment Report, Working Group I
CMIP6	Sixth Coupled Model Intercomparison Project. This project involved a number of experiments with coupled atmosphere-ocean global climate models, most of which were reported on in the IPCC Sixth Assessment Report, Working Group I
Downscaling	Deriving local climate information from larger-scale model or observational data. Two main methods exist – statistical and dynamical. Statistical methods develop statistical relationships between large-scale atmospheric variables (e.g., circulation and moisture variations) and local climate variables (e.g., rainfall variations). Dynamical methods use the output of a regional climate/weather model driven by a larger-scale global model
Ensemble	Referring here to the collection of six GCMs that were selected for downscaling and bias-correction in New Zealand
GCMs	Global Climate Models
IPCC	Intergovernmental Panel on Climate Change. This body was established in 1988 by the World Meteorological Organisation (WMO) and the United Nations Environment Programme (UNEP) to objectively assess scientific, technical and socioeconomic information relevant to understanding the scientific basis of risk of human-induced climate change, its potential impacts and options for adaptation and mitigation. One of its major outputs is regular assessments of the climate system and our knowledge of climate change, known as Assessment Reports

Radiative forcing	The difference between the energy entering our atmosphere and the energy leaving it. If the number is greater than zero, our atmospheric temperature can increase
RCPs	Representative Concentration Pathways. These comprise scenarios used in CMIP5 modelling. They describe four possible climate futures represented by the concentration of greenhouse gases in the atmosphere, all of which are considered possible depending on how much greenhouse gases are emitted in the years to come. The four RCPs, RCP2.6, RCP4.5, RCP6, and RCP8.5, are named after a possible range of radiative forcing values in the year 2100 relative to pre-industrial values (+2.6, +4.5, +6.0, and +8.5 W m2, respectively)
Seven-station series	This refers to seven long-term temperature records used to assess New Zealand's warming on the century time-scale. The sites are located in Auckland, Wellington, Masterton, Nelson, Hokitika, Lincoln, and Dunedin
SSPs	Shared Socioeconomic Pathways. These comprise scenarios of projected socioeconomic global changes up to 2100, used in CMIP6 modelling
SST	Sea-surface temperature. The temperature of the uppermost layer of the ocean

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